STOCHASTIC METHODS FOR FINANCE
Lecture Notes

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Contents

1 Discrete Time Finance **4**
1.1 Introduction to Finance . . . 4
1.2 Forward Contracts . . . 4
1.2.1 Example: Hedging FX Risk . . . 5
1.3 Arbitrage . . . 5
1.3.1 Pricing the Forward Contract . . . 5
1.4 Future Contracts . . . 6
1.5 Options: Call and Put . . . 6
1.5.1 Payoff and Profit of Options . . . 7
1.6 Pricing a Contract: The Binomial Tree . . . 7
1.6.1 The One-Period Binomial Model . . . 8
1.6.2 Imposing No Arbitrage . . . 8
1.6.3 Risk-Neutral Valuation . . . 8
1.7 Pricing, Hedging, Dividends . . . 9
1.7.1 Recap: Pricing and Hedging . . . 9
1.7.2 From Theory to Practice: Volatility . . . 9
1.7.3 The Martingale Measure $\mathbb{Q}$ . . . 11
1.7.4 Dividends: A Practical Example . . . 11
1.7.5 Put-Call Parity . . . 12
1.8 Market Completeness . . . 13
1.8.1 Sanity Check . . . 13
1.8.2 Market Completeness . . . 13
1.8.3 Lattice Tree: Recombining . . . 14
1.8.4 Pricing and Hedging on a Recombining Tree . . . 14
1.8.5 Self-Financing Strategy . . . 15
1.9 Turning Point: From Binomial to Geometric BM . . . 16
1.9.1 Continuous Time: Brownian Motion . . . 17
1.10 From Binomial to Black-Scholes and American Options . . . 18
1.10.1 American Options and Stopping Times . . . 19
1.10.2 Pricing an American Call Option with No Dividends (Toy Example) . . . 20
1.10.3 The General Case (Optimal Stopping) . . . 21
1.11 American Options: Optimal Stopping and Snell Envelope . . . 21
1.11.1 The Optimal Value Process . . . 21
1.11.2 The Snell Envelope . . . 22
1.11.3 Exercise: Pricing an American Put Option . . . 22
1.12 More General Models: Multi-Asset and Absence of Arbitrage . . . 23
1.12.1 The Multi-Asset Model (1-period) . . . 23
1.12.2 Portfolio Strategies and Arbitrage . . . 23
1.12.3 Farkas' Lemma and Arbitrage Pricing . . . 24
1.12.4 The Fundamental Theorems of Asset Pricing (FTAP) . . . 24

2 Continuous Time Finance **25**
2.1 From Discrete to Continuous Time: Trees, Martingales, and Itô Calculus . . . 25
2.1.1 Risk-Neutral Pricing and Information Structures . . . 25
2.1.2 Binomial Trees: Lattice vs. Non-Recombining . . . 25
2.1.3 Standard Brownian Motion and Stochastic Integrals . . . 26

2.1.4 Itô's Formula and Geometric Brownian Motion . . . 26
2.2 The Feynman-Kac Formula: Connecting PDEs and Stochastic Processes . . . 27
2.2.1 Feynman-Kac: Version 1 (The Elementary Case) . . . 27
2.2.2 Feynman-Kac Formula: Version 2 (The Discounting Term) . . . 27
2.2.3 Feynman-Kac Formula: Version 3 (The Source Term) . . . 28
2.2.4 Example: Solving a PDE with Feynman-Kac . . . 28
2.3 The Black-Scholes Market Model (1973) . . . 28
2.4 The Black-Scholes Model: Portfolio Dynamics and the PDE . . . 29
2.4.1 The Problem with the "Naive" Limit in Continuous Time . . . 29
2.4.2 The Correct Formulation (Self-Financing Portfolio) . . . 29
2.4.3 Deriving the Black-Scholes PDE . . . 30
2.4.4 Risk-Neutral Evaluation (1st Fundamental Theorem) . . . 30
2.5 The Martingale Approach and The Greeks . . . 31
2.5.1 Imposing Q-Martingality (The Girsanov Theorem) . . . 31
2.5.2 Pricing under $\mathbb{Q}$ . . . 31
2.5.3 The Greeks (Sensitivities) . . . 32
2.6 Put Options, Portfolio Immunization, and Digital Options . . . 33
2.6.1 Pricing and Greeks of a Put Option . . . 33
2.6.2 Portfolio Immunization (Delta, Gamma, Vega Neutrality) . . . 35
2.6.3 The Black-Scholes PDE and the Sign of Gamma . . . 35
2.6.4 Digital Options (or Binary) . . . 36
2.7 Option Strategies, Dividends, and Implied Volatility . . . 36
2.7.1 Static Option Strategies . . . 36
2.7.2 Decomposing Complex Payoffs . . . 38
2.7.3 Black-Scholes with Continuous Proportional Dividends . . . 39
2.7.4 Implied Volatility and The Smile Effect . . . 39
2.8 Multidimensional Market Model and Correlation . . . 40
2.8.1 Pricing a Multi-Asset Payoff: The Exchange Option . . . 41
2.8.2 Intuition Behind Margrabe's Formula . . . 42
2.9 The Change of Numéraire Technique . . . 42
2.9.1 Application 1: The Exchange Option . . . 43
2.9.2 Application 2: Stochastic Interest Rates . . . 43
2.10 Itô's Quotient Rule and the Forward Price . . . 44
2.11 Black-Scholes Call Price with Stochastic Rates . . . 44
2.11.1 Computing the Forward Measure Part ($\mathbb{Q}^T$) . . . 45
2.11.2 Computing the Stock Measure Part ($\mathbb{Q}^S$) . . . 45
2.12 The Vasiček Interest Rate Model (1977) . . . 45
2.12.1 Bond Pricing and Volatility . . . 46
2.13 Black Formula (1976): Options on Futures . . . 46
2.14 (FOREX) Exchange Rates and Currency Conversion . . . 47
2.14.1 Pricing Options on Foreign Assets . . . 48
2.14.2 Market Note: Volatility Smile in Forex . . . 49
2.15 Mathematical Foundations of Barrier Options . . . 49
2.15.1 Hitting Times and Stopping Times . . . 49
2.15.2 The Reflection Principle . . . 49
2.15.3 Moment Generating Function of the Hitting Time (via Localization) . . . 50
2.15.4 Brownian Motion with Drift and Barrier Options . . . 51
2.15.5 Uniform Integrability (UI) and Stopping Times . . . 52
2.16 Björk's Unified Approach for Barrier Options . . . 53
2.16.1 Building Blocks: Vanilla Contracts . . . 54
2.16.2 Up-and-Out (UO) Contracts . . . 54
2.16.3 Pricing Down-and-Out (DO) Contracts . . . 55
2.16.4 In-Out Parity . . . 56
2.17 Lookback Options . . . 56
2.17.1 Pricing a Lookback Put . . . 57
2.18 Pricing When Volatility is Stochastic . . . 57
2.18.1 The Fourier Transform Approach . . . 57
2.18.2 Fourier Transforms of Standard Payoffs . . . 57

2.19 The Heston Stochastic Volatility Model (1993) . . . 59
2.19.1 Feller Condition and Squared Bessel Processes . . . 60
2.19.2 The Characteristic Function PDE (Feynman-Kac) . . . 60
2.19.3 Solving the Heston Characteristic Function . . . 60
2.19.4 Alternative Approach: Matrix Exponentials . . . 61
2.19.5 Multi-Factor Stochastic Volatility (2-Heston) . . . 62
2.20 Limitations of the Multi-Factor Heston Model . . . 62
2.21 Generating Dependence while Preserving Affinity . . . 63
2.22 The Wishart Process . . . 63
2.22.1 Static Wishart Distribution . . . 64
2.22.2 The Laplace Transform of the Wishart Process . . . 65
2.22.3 Extension to Real Dimensions (α) . . . 65
2.22.4 Adding Mean Reversion: The Affine Wishart Model . . . 65
2.22.5 Wishart's characteristics function . . . 66
2.22.6 Feynman-Kac with State-Dependent Discounting . . . 67
2.23 The Multi-Factor Wishart Volatility Model . . . 68
2.23.1 Matrix Brownian Motion Correlation . . . 68
2.23.2 Log-Price Dynamics and Joint Generator . . . 68

Chapter 1

Discrete Time Finance

1.1 Introduction to Finance

Derivative

A **derivative** is a contract involving two parties who agree to a future transaction based on an **underlying asset**.

Typical underlying assets include:

- Stock
- Index (e.g., S&P 500)
- Interest Rate
- Foreign Exchange Rate
- Price of Hogs, Wheat, or Electricity

The simplest possible derivatives are **Futures** and **Forward** contracts.
Derivatives contracts are traded in two main ways:

1. **Derivatives Exchange:** Entities trade contracts with prices established by the exchange (standardized).
2. **Over-The-Counter (OTC):** Large financial institutions or investment banks trade directly.

1.2 Forward Contracts

A forward contract is an agreement to **BUY (Long Position)** or **SELL (Short Position)** an underlying asset whose value at time t is denoted by S_t.

- **Long Forward:** I have to buy S at time T at the fixed price K.
- **Short Forward:** I have to sell S at time T at the fixed price K.

Strike Price

The price K is called the **Strike Price** or **Delivery Price**. It is the agreed-upon price decided at time $(t = 0)$ which the underlying asset will be bought or sold at maturity $(t = T)$.

Forward contracts are typically OTC contracts.

1.2.1 Example: Hedging FX Risk

Suppose today is May 21, 2020, and a treasurer will have to pay 1 million GBP on Nov 21, 2020. To hedge against FX risk, the treasurer can enter into a **Long Forward** contract.

Assuming the 6-month forward exchange rate is quoted at 1.2230 USD/GBP, the treasurer locks in the purchase of 1 million GBP at 1.2230 USD. The counterparty (e.g., a bank) has a Short position and a binding commitment to sell 1 million GBP for 1.2230 USD.

- If the exchange rate raises to 1.3 USD/GBP at maturity, the contract's value to the treasurer is: $1,300,000 - 1,223,000 = 77,000$ USD.
- If the exchange rate goes down to 1.2 USD/GBP, the contract's value is: $1,200,000 - 1,223,000 = -23,000$ USD.

Payoff

Payoff it's the value of the contract at maturity, which depends on the underlying asset's price at time T:

- Long: $S_T - K$
- Short: $K - S_T$

1.3 Arbitrage

How do we determine K? By imposing **NO ARBITRAGE**. Arbitrage is defined as a "Free lunch with vanishing risk!".

Arbitrage

In mathematical terms, a process $V_0, V_1, \ldots, V_T$ representing the value of a portfolio constitutes an arbitrage if:

1. $V_0 = 0$ (Initial not random cost for out strategy)
2. $V_T \geq 0$ a.s. (At maturity, the value is a non-negative random variable)
3. $P(V_T > 0) > 0$ (There is a positive probability of making a profit)

In a fair financial market, arbitrage opportunities should be avoided.

1.3.1 Pricing the Forward Contract

To avoid arbitrage, we construct a portfolio combining:

1. Take a **Long** position at time $t = 0$ on the forward contract, so I didn't have to buy anything, only sign the contract. At maturity, this position will be worth $S_T - K$.
2. Take a **Short** position on the underlying asset S at time $t = 0$. At maturity, this position will be worth $-S_T$.
3. Use the money obtained from the short position to invest in a **Risk-less Market** (e.g., a bank account) that grows at a risk-free rate R. At maturity, this position will be worth Ke^{RT}.

Movements	$t = 0$	$t = T$
Long Forward	0	$S_T - K$
Short Underlying	$+S_0$	$-S_T$
Risk-free Investment	$-Ke^{-RT}$	$+K$
Resulting Portfolio	0	0

There are two classical ways of computing interest on $[0, T]$ for the risk-free investment at time $t = 0$:

1. **Simple:** $C_0 \longrightarrow C_T = C_0 + RT$
2. **Compound:** $C_0 \longrightarrow C_T = C_0 \cdot e^{RT}$ (where R is the interest rate over 1 year)

In order to get K euros at time T for our portfolio, we need to invest $K \cdot e^{-RT}$ today. Therefore, to have a portfolio value $V_0 = 0$ at inception, we must have:

$$0 + S_0 - Ke^{-RT} = 0$$

Which gives the fair strike price:

$$K = S_0 \cdot e^{RT}$$

1.4 Future Contracts

A **future contract** is an agreement between two parties to buy/sell an asset at a specified time in the future for a specified price.

Q: What's the difference with forward contracts? Futures are traded on an exchange and have standardized features.

Example: Assume that on September 1st, 2022, the quotation for the **December Future** on gold is quoted at \$1,750. This is the price, without commissions, at which traders agree to buy or sell gold for December delivery.

If you check the contract specifications on the Japan Exchange (JPX), you find:

- **Contract Months:** nearest 6 months
- **Last Trading Day:** 3rd business day preceding the delivery day
- **Contract Unit:** 1 kg / contract

Two large exchanges where contract futures are traded are CBOE (Chicago Board Options Exchange) and CME Group.

1.5 Options: Call and Put

Today we will focus on vanilla (or European) options that depend only on the underlying asset price at maturity (S_T). There are two types of positions: Long and Short.

Option

- **Call Option:** gives the holder the opportunity (not the obligation) to **BUY** the underlying at time T at the strike price $K > 0$.
- **Put Option:** gives the holder the opportunity to **SELL** the underlying at time T at the strike price $K > 0$.

Curiosity: What if an option is not European? Do we also have American options? Yes! They can be exercised at any intermediate day between $t = 0$ and $t = T$.

The key ingredients of a European Option are:

1. Underlying Asset: S (has to be traded!)
2. Maturity: $T > 0$
3. Strike Price: $K > 0$
4. Initial Time (Signature): $t = 0$

*N.B. The holder is <u>not obliged</u> to exercise the option (in forwards and futures there is an obligation!). Having this opportunity comes at a cost called **Premium** (c), which is the price of the option at time $t = 0$.*

1.5.1 Payoff and Profit of Options

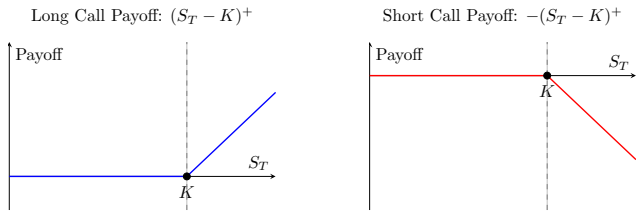
Let S_T be the spot price at maturity and K the strike price.

Call Option:

- **Long Payoff:** $\max(S_T - K, 0) = (S_T - K)^+$
- **Short Payoff:** $-(S_T - K)^+$

If the payoff is positive, the holder will exercise the option. If the payoff is negative, the holder will let the option expire worthless.

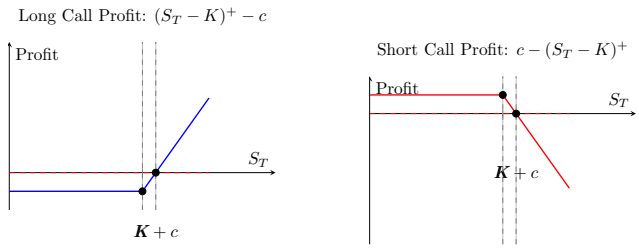
Payoff Diagrams for Call Options:



The net profit has to take into account the initial cost (premium), which we call $c > 0$.

- **Long Profit:** $(S_T - K)^+ - c$
- **Short Profit:** $c - (S_T - K)^+$

Profit Diagrams for Call Options (assuming $c = 10$):



Put Option:

- **Long Payoff:** $\max(K - S_T, 0) = (K - S_T)^+$
- **Short Payoff:** $-(K - S_T)^+$

1.6 Pricing a Contract: The Binomial Tree

How do we price an option? We will see the simplest possible math model for the evolution of a stock: the binomial model (random walk). Key aspects of this model are:

- No Arbitrage
- Discrete time and space (useful also to price American options)
- Risk-Neutral Valuation

1.6.1 The One-Period Binomial Model

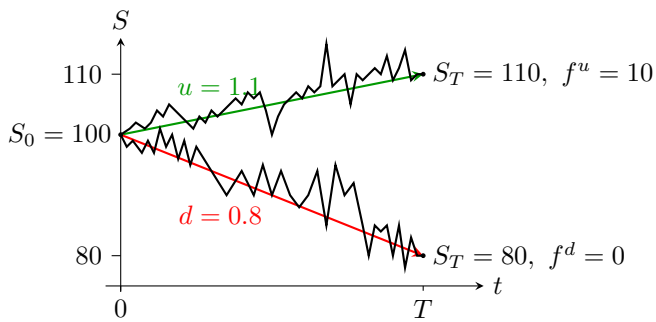
At $t = 0$, the stock price is $S_0 = S$. At $t = T$ (tomorrow), the price can either go:

- **UP:** $S_T = S \cdot u$ (with payoff f^u)
- **DOWN:** $S_T = S \cdot d$ (with payoff f^d)

where $d < u$. S_T is a random variable, seen as of today.

Example: Let $S_0 = 100$ and $K = 100$.

- Up state ($u = 1.1$): $S_T = 110$. Call Payoff $= 10$
- Down state ($d = 0.8$): $S_T = 80$. Call Payoff $= 0$



Terminology based on K:

- $K = S_0$: **At** the money
- $K < S_0$: **In** the money
- $K > S_0$: **Out** of the money

1.6.2 Imposing No Arbitrage

To find the fair price of the option at $t = 0$, we introduce a portfolio strategy $h = (\alpha, \beta) \in \mathbb{R}^2$ where α represents the number of shares of the underlying and β represents the money in the bank account.

- Today cost for creating the portfolio (at $t = 0$): $V_0^h = \alpha S_0 + \beta \cdot 1$
- Tomorrow value of the portfolio (at $t = T$): $V_T^h = \alpha S_T + \beta e^{RT}$

where $R > 0$ is the spot interest rate. We impose that the strategy is a **replicating portfolio**, meaning it replicates exactly the payoff of the option in both states:

$$\text{Portf. Payoff} = \text{Opt. Payoff} \implies \begin{cases} \alpha S \cdot u + \beta e^{RT} = f^u \\ \alpha S \cdot d + \beta e^{RT} = f^d \end{cases} \implies \begin{cases} \alpha = \frac{f^u - f^d}{Su - Sd} = \frac{\Delta \text{Payoff}}{\Delta S} = \Delta \quad [\textbf{delta edging}] \\ \beta = e^{-RT}\left(f^d - \alpha S d\right) = e^{-RT}\left(f^u - \alpha S u\right) \end{cases}$$

By imposing **Absence of Arbitrage Opportunities (AAO)**, the price of the option today P_0 must be exactly the initial cost of the replicating portfolio V_0^h:

$$P_0 = V_0^h = \alpha S + \beta$$

1.6.3 Risk-Neutral Valuation

Substituting α and β into the price formula and rearranging, we get:

$$P_0 = e^{-RT}\left[q f^u + (1-q) f^d\right] \quad \text{where} \quad q = \frac{e^{RT} - d}{u - d}$$

This equation means the price is computed as a **discounted expected value** of the future payoff under a different probability measure $\mathbb{Q}$ (Risk-Neutral probability), not the subjective real-world probability $\mathbb{P}$.

$$P_0 = \underbrace{e^{-RT}}_{\text{Discount}} \cdot \underbrace{\mathbb{E}^{\mathbb{Q}}[H_T]}_{\text{Expected value}}$$

For q to be a valid probability measure, we must have $0 < q < 1$, which requires:

$$d < e^{RT} < u$$

If this is not true (e.g., $e^{RT} \leq d$), there is an evident arbitrage opportunity (one could borrow money at rate R, buy the stock, and guarantee a profit since even the worst-case stock return d is greater than the cost of borrowing).

1.7 Pricing, Hedging, Dividends

1.7.1 Recap: Pricing and Hedging

First of all, let us sum up what we saw until now.

Pricing (Long position): For example small investor wants to buy a call option.

1. Identify the payoff: H_T (which is a random variable).
2. Introduce the risk-neutral probability measure $\mathbb{Q}$, where the probability of an up-movement is:

$$q = \frac{e^{RT} - d}{u - d}$$

3. Compute the price:

$$P_0 = e^{-RT}\mathbb{E}^{\mathbb{Q}}[H_T]$$

Hedging (Short position): Bank or financial institution view. Sell the option, receive P_0, but we need to have a strategy to be able to possibly provide the underlying at time T.

1. Solve the pricing to find P_0.
2. Find the hedging portfolio $h = (\alpha, \beta)$, so how many shares of the underlying we need to buy/sell and how much money we need to invest in the risk-free asset:

$$\begin{cases} \alpha = \Delta = \frac{f^u - f^d}{Su - Sd} = \frac{\Delta \text{Payoff}}{\Delta S} \\ \beta = P_0 - \Delta S_0 \end{cases}$$

Example: Let us consider the binomial tree and an At-The-Money Call option ($K = 100$ €, $S_0 = 100$ €). Assume $R = 0.01$ (over 1 year), $T = 1$, $u = 1.1$, $d = 0.9$. We immediately find:

$$q = \frac{e^{0.01 \cdot 1} - 0.9}{1.1 - 0.9} = \frac{1.01 - 0.9}{0.2} = 0.55 \implies P_0 = e^{-0.01}(0.55 \cdot 10 \text{ €} + 0.45 \cdot 0 \text{ €}) = 5.45 \text{ €}$$

So for the Long position, we have to pay 5.45 euros to buy the option ($c = 5.45$ or $P_0 = 5.45$). For the Short position, we receive 5.45 euros but we need to hedge our position by investing in the underlying and in the risk-free asset. Now the delta hedging strategy is:

$$\begin{cases} \Delta = \frac{10-0}{110-90} = \frac{10}{20} = 0.5 \quad \text{(Bank need to buy 0.5 shares of the underlying to hedge what sell to you)} \\ \beta = 5.45 \text{ €} - (0.5 \cdot 100 \text{ €}) = -44.55 \text{ €} \quad \text{(Bank need to invest 44.55 euros in the risk-free asset)} \end{cases}$$

1.7.2 From Theory to Practice: Volatility

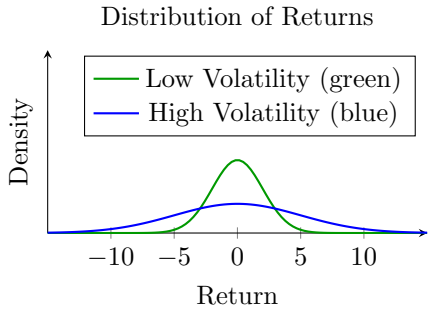
If we want to use a binomial tree in a real-life situation, we need to know u, d, and compute q. How do we find u and d? Which information can we extract from market data?

Volatility

The fluctuation of a stock is measured by its **Volatility (σ)**, defined as the standard deviation of percentage returns of the stock on a Δt-time window:

$$\text{Variance of return} = \sigma^2 \Delta t$$

Graphical Comparison: High vs Low Volatility



Q1: How do I define the return of a stock? Taking S_0 as the price at time $t = 0$ and S_T as the price at time $t = T$, subtracting the initial price from the final price and dividing by the initial price gives us the percentage return:

$$R = \frac{S_T - S_0}{S_0} \qquad t \in [0, T]$$

Q2: How do I derive the standard deviation on real observations? Consider a time series of closing prices $S_0, S_1, S_2 \ldots, S_n$ at daily observations $t_0, t_1, t_2 \ldots t_n$. Time series of returns:

$$R_0 = \frac{S_1 - S_0}{S_0}, R_1 = \frac{S_2 - S_1}{S_1}, R_2 = \frac{S_3 - S_2}{S_2}, \ldots, R_T = \frac{S_T - S_{T-1}}{S_{T-1}} \qquad \forall t \in [0, T]$$

We then estimate the standard deviation of this time series. Recalling the variance formula $\mathrm{Var}(R) = \mathbb{E}(R^2) - \mathbb{E}(R)^2$, we compute the root of the variance to get the estimated daily standard deviation:

$$\hat{\sigma}_{DAILY} = \sqrt{\mathrm{Var}(R)}$$

Q3: How do I project the standard deviation on a different time window? We use the number of working days in 1 year (typically 252):

$$\hat{\sigma}_{YEARLY} = \hat{\sigma}_{DAILY} \cdot \sqrt{252}$$

Thanks to this we can estimate the volatility of a stock using historical data. However, we can also estimate the volatility relating u and d to the volatility of the stock returns, which is the key ingredient for the binomial tree. So in binomial tree we have two possible realizations for the return over a small interval Δt:

$$R := \begin{cases} R_u = \frac{Su - S}{S} = u - 1 \text{ (with q prob.)} \\ R_d = \frac{Sd - S}{S} = d - 1 \text{ (with 1-q prob.)} \end{cases} \implies \mathbb{E}(R) = q(u-1) + (1-q)(d-1) = q(u-d) + d - 1$$

Now from q formula, we have

$$q = \frac{e^{R\Delta t} - d}{u - d} \implies q(u-d) = e^{R\Delta t} - d \implies \mathbb{E}(R) = (e^{R\Delta t} - d) + d - 1 = e^{R\Delta t} - 1$$

(*Note: same calculation can be done for* $\mathbb{E}(R^2)$.)

The variance of the model returns is $\mathrm{Var}(R) = \mathbb{E}(R^2) - \mathbb{E}(R)^2$, which leads to:

$$\mathrm{Var}(R) = \sigma^2 \Delta t = e^{2R\Delta t}(u + d) - ud - e^{2R\Delta t}$$

As we can see that equation cannot be solved for u and d separately because is an equation with 2 unknowns. We need to impose an additional condition:

$$u \cdot d = 1$$

That condition is called **recombining tree** and it means that if the price goes up and then down, we end up in the same state as if it goes down and then up. Thanks to this we can approximate solution (using Taylor expansion $e^x = 1 + x + \frac{x^2}{2} + \ldots$) gives:

$$\begin{cases} u = e^{\sigma\sqrt{\Delta t}} \\ d = e^{-\sigma\sqrt{\Delta t}} \end{cases}$$

Thanks to this, we can compute q and price the option.

1.7.3 The Martingale Measure $\mathbb{Q}$

We saw that $\mathbb{Q}$ is called the Risk-Neutral Probability Measure, and it is also a **Martingale Measure**. Let's compute the expected value of S_T under $\mathbb{Q}$:

$$\mathbb{E}^{\mathbb{Q}}[S_T] = Su \cdot q + Sd \cdot (1-q) = Su\frac{e^{RT}-d}{u-d} + Sd\frac{u-e^{RT}}{u-d} = Se^{RT}$$

Equivalently, we have:

$$\mathbb{E}^{\mathbb{Q}}\left[\frac{S_T}{e^{RT}}\right] = S_0$$

Defining the discounted stochastic process $\tilde{S}_t := \frac{S_t}{e^{Rt}}$, we discover that:

$$\mathbb{E}^{\mathbb{Q}}\left[\tilde{S}_T \mid \mathcal{F}_0\right] = \tilde{S}_0$$

So we discovered that **discounted prices are $\mathbb{Q}$-martingales** because the mean of the discounted prices is equal to the initial discounted price.

1.7.4 Dividends: A Practical Example

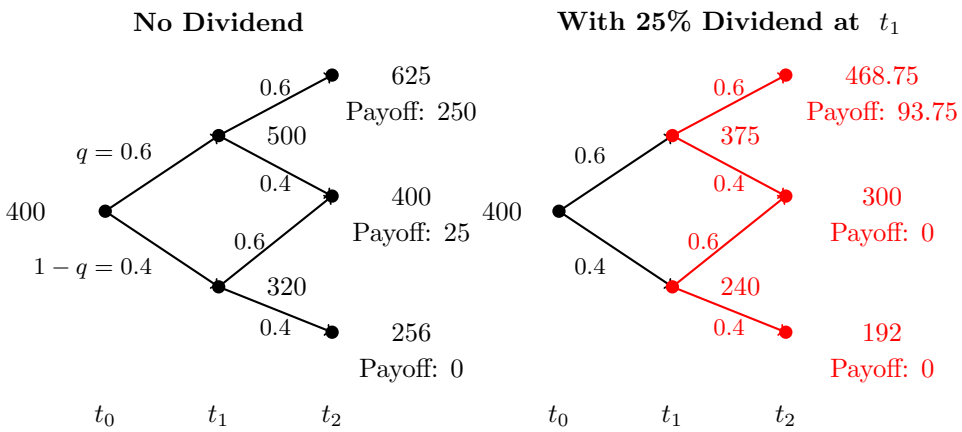
Let us see an example with 3 time instants: t_0, t_1, t_2. Assume the company of stock S pays out **25% of its value at** t_1 as a dividend. Shareholders are happy, but option holders do not receive dividends and only care about the stock value at maturity (European Option).

Let us define the parameters for the tree:

- $S_0 = 400$ as the initial stock price at t_0.
- $u = 1.25$ and $d = 0.8$ as the up and down factors for the stock price at t_1.
- $e^{R \cdot 1} = 1.07$ as the risk-free growth factor for 1 year (from t_0 to t_1).
- $q = \frac{1.07-0.8}{1.25-0.8} = 0.6$ as the risk-neutral probability of an up-movement.
- Strike Price $K = 375$

Remark on Recombining Trees: When you have a fixed cash drop (e.g., a dividend of D euros), the tree does not remain recombining and it becomes very delicate to handle (you have to separate the firm value from the drop value, see Hull Section 21.3). However, if the drop is a percentage (dividend yield, like 25%), the tree remains recombining!

Let us doing graphical tree and compute the option price with and without the dividend to see the impact.



1. **NO-DIVIDEND Case:** Without dividends, the tree evolves normally:

$$t_0 : S_0 = 400 \implies t_1 : \begin{cases} Su = 500 \\ Sd = 320 \end{cases} \implies t_2 : \begin{cases} Su^2 = 625 & (NodeA) \\ Sud = 400 & (NodeB) \\ Sd^2 = 256 & (NodeC) \end{cases}$$

The payoffs for the Call Option $(S_2 - 375)^+$ at t_2 are:

$$\begin{cases} (625-375)^+ = 250 \quad (NodeA) \\ (400-375)^+ = 25 \quad (NodeB) \\ (256-375)^+ = 0 \quad (NodeC) \end{cases}$$

The fair price at time 0 is:

$$\begin{aligned} P_0^{Call} &= e^{-R\cdot 2}\mathbb{E}^{\mathbb{Q}}[(S_2-375)^+] = (1.07)^{-2}[q^2 \cdot 250 + 2q(1-q)\cdot 25] \\ &= 0.8734 \cdot [0.36 \cdot 250 + 0.48 \cdot 25] = 89.09 \end{aligned}$$

2. **WITH DIVIDEND Case (25% drop at t_1):** With the dividend, the stock loses 25% of its value right at t_1.

$$t_0 : S_0 = 400 \implies t_1 : \begin{cases} 375 & \text{(after 25\% drop from 500)} \\ 240 & \text{(after 25\% drop from 320)} \end{cases} \implies t_2 : \begin{cases} 375 \cdot 1.25 = 468.75 \quad (NodeA) \\ 375 \cdot 0.8 = 300 \quad (NodeB) \\ 240 \cdot 0.8 = 192 \quad (NodeC) \end{cases}$$

The payoffs for the Call Option $(S_2 - 375)^+$ at t_2 are now:

$$\begin{cases} (468.75-375)^+ = 93.75 \quad (NodeA) \\ (300-375)^+ = 0 \quad (NodeB) \\ (192-375)^+ = 0 \quad (NodeC) \end{cases}$$

The fair price at time 0 is:

$$P_0^{Call} = e^{-R\cdot 2}\mathbb{E}^{\mathbb{Q}}[(S_2-375)^+] = (1.07)^{-2}[q^2 \cdot 93.75] = 0.8734 \cdot [0.36 \cdot 93.75] = 29.48$$

Conclusion: Notice the considerable reduction in the fair price with respect to the no-dividend scenario.

1.7.5 Put-Call Parity

We are now ready for the **Put-Call Parity**. The useful ingredients are:

- Payoff Call: $(S_T - K)^+$
- Payoff Put: $(K - S_T)^+$

Using the mathematical identity $X = X^+ - X^-$, we put this all together:

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K$$

Taking the conditional expectation at $t = 0$ under $\mathbb{Q}$ and recalling that $P_0 = e^{-RT}\mathbb{E}^{\mathbb{Q}}[\text{Payoff}_T]$:

$$P_0^{Call} - P_0^{Put} = e^{-RT}\mathbb{E}^{\mathbb{Q}}[S_T] - e^{-RT}K$$

Since $e^{-RT}\mathbb{E}^{\mathbb{Q}}[S_T] = S_0$, we find the fundamental relationship:

Put-Call Parity

The price of a European Put option can be derived from the price of the corresponding Call option (and vice versa) using the following formula:

$$P_0^{Put} = P_0^{Call} - S_0 + e^{-RT}K$$

So if we know the price of a European Call option, we can derive the price of Put without needing to use the binomial tree or any other model.

1.8 Market Completeness

1.8.1 Sanity Check

Before talking about completeness, let us notice that a sanity check on market prices can be performed without any mathematical model, independently of the underlying asset. Sanity check with respect to what? **Absence of arbitrage!**

Indeed, it can be easily shown that we should have, at any intermediate time $0 < t < T$:

$$P_t^{Call} \geq S_t - Ke^{-R(T-t)}$$

This comes from the trivial inequality for the Call payoff: $(S_T - K)^+ \geq S_T - K$.

Analogously, one should have:

$$P_t^{Put} \geq Ke^{-R(T-t)} - S_t$$

Remark: Whenever these inequalities are not satisfied, a static arbitrage opportunity exists (no rebalancing of the portfolio is needed!).

1.8.2 Market Completeness

A financial mathematician should always know, when framing a mathematical model, whether the market is **complete** or **incomplete**. We will work under the following realistic assumptions:

1. Any investment strategy $h = (\alpha, \beta) \in \mathbb{R}^2$ is admissible, namely short positions and fractional holdings are allowed.
2. No bid-ask spread: strictly speaking, buying price is equal to selling price.
3. No transaction costs.
4. Perfect liquid market: it is always possible to trade, buy, or sell any quantity of assets (and derivatives) in the market.

Discrete time models are built on a **Filtered Probability Space**:

$$(\Omega, \mathcal{F}, \mathbb{P}, (\mathcal{F}_{t_n})_{t_n \in [t_0, T]})$$

with a finite sample space such that $\mathbb{P}(\{\omega_i\}) > 0 \quad \forall i$.

Contingent Claims (or Financial Derivative)

A stochastic variable $H \sim \mathcal{F}_T$ written on S and with maturity T. Here $\mathcal{F}_T$ is the natural filtration associated with S.

Replicating Portfolio

We say that there exists a portfolio h such that $V_T^h = H$ (P-a.s.). In this case, we say that h is a *hedging portfolio*.

Complete Market

A market is **COMPLETE** if every claim H admits a replicating portfolio.

Completeness has a crucial consequence on pricing:

Proposition: If H derivative is reachable with a portfolio $h \in \mathbb{R}^2$ and the market is arbitrage-free, then:

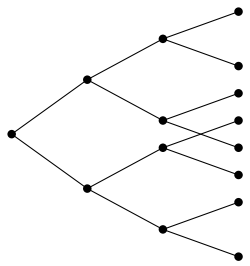
$$\text{Price}_0(H) = V_0^h$$

So the price of a derivative is equal to the cost of the initial replicating portfolio. Moreover, the binomial model with $d < e^{RT} < u$ (or $1 + R$) is both arbitrage-free and complete.

1.8.3 Lattice Tree: Recombining

Can we do something better with respect to the binomial model?

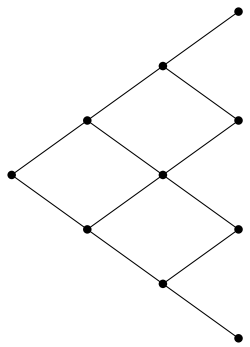
- Can we use multinomial trees? Yes! But they lead to incomplete models! How do I price derivatives then?
- Can we add more assets in the market? Yes! But what if we want more possible final values for the underlying?
- More time steps? Yes, but what's the issue here?



8 distinct states at $t = 3$

The issue here is clearly that with many time steps, a non-recombining tree generates about 2^N terminal states (and many intermediate nodes), so both pricing and hedging become computationally expensive and difficult to interpret, also for large model and advanced computer can be dangerous.

So, what is the solution? A **Lattice Tree**! The assumption is **Recombining**: $Down + Up = Up + Down$.

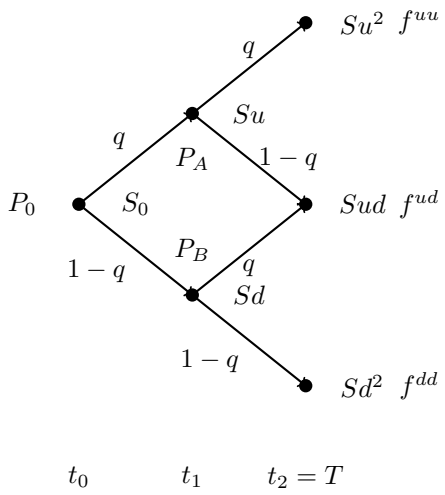


4 distinct states at $t = 3$

Is this assumption realistic? Is the market behavior symmetric? Obviously not, but it is a good approximation for vanilla options and barrier options. We should be careful when using it to price path-dependent options.

1.8.4 Pricing and Hedging on a Recombining Tree

Let us consider a probability space with 4 trajectories: $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ $\mathcal{F}_0 = \{\emptyset, \Omega\}$ $\mathcal{F}_1 = \{\emptyset, \Omega, \{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}\}$ $\mathcal{F}_2 = \mathcal{P}(\Omega)$ Let us work with a 2-step lattice ($T = 2$), with probability of an up movement equal to $q \in (0, 1)$ under measure $\mathbb{Q}$.



Pricing (Proceed backwardly): We first consider what happens in the time interval $[t_1, t_2]$:

$$P_A = e^{-RT/2} \cdot \mathbb{E}^{\mathbb{Q}}[H_T|\mathcal{F}_A] = e^{-RT/2}[f^{uu} \cdot q + f^{ud}(1-q)] \quad \text{or} \quad P_B = e^{-RT/2}[f^{du} \cdot q + f^{dd}(1-q)]$$

Then we work in $[t_0, t_1]$:

$$P_0 = e^{-RT/2}[q \cdot P_A + (1-q)P_B]$$

Due to recombination ($f^{ud} = f^{du}$), this expands to:

$$P_0 = e^{-RT}[q^2 f^{uu} + 2q(1-q)f^{ud} + (1-q)^2 f^{dd}] = e^{-RT}\mathbb{E}^{\mathbb{Q}}[H_T]$$

With that we prove that is possible to price the derivative at time 0 using the risk-neutral expectation of the payoff at maturity *directly without needing to compute the intermediate prices* P_A and P_B.

Hedging (Proceed forward): We need to find the investment strategy, i.e., an admissible sequence (α_1, β_1) on $[0, 1]$ and (α_2, β_2) on $[1, 2]$. The strategy must be **Predictable**, which means $(\alpha_t, \beta_t) \sim \mathcal{F}_{t-1}$ (decided at time $t-1$ and kept until t).

$$\Delta = \frac{\Delta\text{Payoff}}{\Delta\text{Underlying}}$$

For the first step ($t = 0 \to t = 1$):

$$\begin{cases} \alpha_1 = \Delta_0 = \frac{P_A - P_B}{Su - Sd} \\ \beta_1 = P_0 - \Delta_0 S_0 \end{cases}$$

For the second step ($t = 1 \to t = 2$), we have two possible market states:

$$\text{Market UP:} \begin{cases} \alpha_2^A = \Delta_A = \frac{f^{uu} - f^{ud}}{Su^2 - Sud} \\ \beta_2^A = \beta_A e^{-RT/2}(P_A - \Delta_A Su) \end{cases} \qquad \text{Market DOWN:} \begin{cases} \alpha_2^B = \Delta_B = \frac{f^{du} - f^{dd}}{Sud - Sd^2} \\ \beta_2^B = \beta_B e^{-RT/2}(P_B - \Delta_B Sd) \end{cases}$$

So here *we need to visit all the nodes in the tree* to compute the strategy, which is more computationally expensive than pricing.

1.8.5 Self-Financing Strategy

What did we use to compute the strategy in the bank account? We used the **Self-Financing** property. In math models for finance, an investment strategy is self-financing if there are:

- No exogenous infusions of money.
- No exogenous withdrawal of money.

So, for every ω and at every time t_n,

Self-Financing Strategy

The change in the value of the portfolio is only due to the change in the value of the assets in the portfolio, and not due to any external cash flow:

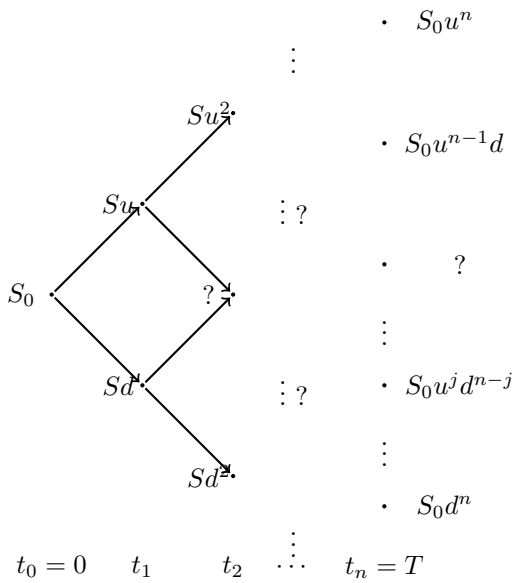
$$\alpha_n S_n + \beta_n \cdot B_n = \alpha_{n+1} S_n + \beta_{n+1} \cdot B_n$$

where both sides give exactly the value of the portfolio at time t_n.

So in reality, the self-financing condition is not always satisfied (e.g., when you pay dividends or when you rebalance your portfolio), but it is a good approximation for vanilla options and barrier options. We should be careful when using it to price path-dependent options.

1.9 Turning Point: From Binomial to Geometric BM

What happens to a binomial model if we let the number of time steps $n \to \infty$? Let us consider a tree with $T = n$ instead of $T = 2$.



In an intermediate step, to have j up-movements and $(n-j)$ down-movements, the underlying asset takes the value:

$$S_n = S_0 u^j d^{n-j}$$

The probability under the risk-neutral measure $\mathbb{Q}$ of reaching this specific node is given by the Binomial Theorem:

$$\mathbb{Q}(S_n = S_0 u^j d^{n-j}) = \binom{n}{j} q^j (1-q)^{n-j}$$

where $\binom{n}{j} = \frac{n!}{j!(n-j)!}$ represents the number of ways (combinations) to reach that node.

Pricing an Option: To compute the price of a European Call option at time 0 in this n-step tree, we must sum over all possible final nodes:

$$P_0^{Call} = e^{-RT} \mathbb{E}^{\mathbb{Q}} [H_T] = e^{-RT} \sum_{j=0}^{n} \binom{n}{j} q^j (1-q)^{n-j} (S_0 u^j d^{n-j} - K)^+$$

The problem is that as $n \to \infty$, the number of nodes explodes, making computational pricing extremely heavy.

1.9.1 Continuous Time: Brownian Motion

When the mesh of the binomial model becomes infinitely dense, it converges to a **Geometric Brownian Motion (GBM)**.

Brownian Motion

A stochastic process $W = (W_t)_{t\in[0,T]}$ is a Brownian Motion if:

1. $W_0 = 0$.
2. It has **independent increments**: if $0 \le s \le t \le u$, then $(W_u - W_t)$ and $(W_t - W_s)$ are independent random variables.
3. The increments are normally distributed: $(W_t - W_s) \sim \mathcal{N}(0, t-s)$.
4. W has continuous trajectories.

GBM as Limit of Binomial Tree

A Geometric Brownian Motion is a stochastic process that takes <u>always positive</u> values. Its dynamics are described by a Stochastic Differential Equation (SDE):

$$\frac{dX_t}{X_t} = \alpha dt + \beta dW_t$$

where α is the drift and β is the volatility (stochastic noise).

Using **Itô Calculus** and Itô's Lemma, the solution to this SDE is:

$$X_t = X_0 e^{\left(\alpha - \frac{\beta^2}{2}\right)t + \beta W_t}$$

Taking the natural logarithm, we find that $\ln(X_t)$ is a Gaussian random variable:

$$\ln X_t \sim \mathcal{N}\left(\mathbb{E}[\ln X_t] = \ln X_0 + \left(\alpha - \frac{\beta^2}{2}\right)t, \quad \sigma^2 = \beta^2 t\right)$$

Proof. Let us define an underlying process $S^{(N)}$ on $[0,T]$, split into N equal sub-intervals. The evolution of price is:

$$S_n^{(N)} = S_{n-1}^{(N)} Z_n^{(N)}$$

where $Z_n^{(N)} \in \{u, d\}$ is the jump u or d at time t_n.

1. Martingale Property: In our binomial model with N time steps, the discounted price is a $\mathbb{Q}$-martingale:

$$\mathbb{E}^{\mathbb{Q}}[S_{n+1}^{(N)} e^{-R(n+1)\frac{T}{N}} | \mathcal{F}_n] = S_n^{(N)} e^{-Rn\frac{T}{N}}$$

2. Convergence in Law via Characteristic Function: We study the limit of the characteristic function of the logarithm of the price as $N \to \infty$. Thanks to that we pass from multiplicative jumps to additive increments, which are easier to handle. The final price is the product of all intermediate jumps:

$$S_T^{(N)} = S_0 \cdot Z_1 \cdot Z_2 \cdot \dots \cdot Z_N = S_0 \prod_{j=1}^{N} Z_j^{(N)} \implies \ln(S_T^{(N)}) = \ln(S_0) + \sum_{j=1}^{N} \ln(Z_j^{(N)})$$

Now we need to fix the problem of having parameters that can explodes as $N \to \infty$. We set from BM:

$$u = e^{\sigma\sqrt{T/N}} \quad \text{and} \quad d = e^{-\sigma\sqrt{T/N}}$$

Now define the characteristic function as:

$$\Phi_{\ln S_T^{(N)}}(\lambda) = (S_0)^{i\lambda} \cdot \left(\mathbb{E}^{\mathbb{Q}}\left[e^{i\lambda \ln Z^{(N)}}\right]\right)^N$$

that derive from the fact that all the step are independent so we can consider one step elevated to N. Applying Taylor expansions, and replacing $q^{(N)}$, we obtain an expression of the form:

$$\Phi_{\ln S_T^{(N)}}(\lambda) \approx (S_0)^{i\lambda}\left[1+\frac{1}{N}\left(i\lambda T\left(R-\frac{\sigma^2}{2}\right)-\frac{1}{2}\lambda^2\sigma^2 T\right)\right]^N$$

3. The Limit: Using the fundamental limit $\lim_{N\to\infty}(1+\frac{x}{N})^N = e^x$, we find:

$$\lim_{N\to\infty}\Phi_{\ln S_T^{(N)}}(\lambda) = (S_0)^{i\lambda}e^{i\lambda T\left(R-\frac{\sigma^2}{2}\right)-\frac{1}{2}\lambda^2\sigma^2 T}$$

This matches exactly the characteristic function of a log-normal distribution, proving that the continuous limit of the binomial tree is:

$$S_T = S_0 e^{\left(R-\frac{\sigma^2}{2}\right)T+\sigma Z_T}$$

where $Z_T \sim \mathcal{N}(0,T)$. This means S_t follows a Geometric Brownian Motion under $\mathbb{Q}$:

$$\frac{dS_t}{S_t} = Rdt + \sigma dW_t^{\mathbb{Q}}$$

□

1.10 From Binomial to Black-Scholes and American Options

What is the Black-Scholes formula? It is the most well-known formula to price Call (and Put) options on a non-dividend paying stock, when the stock follows a Geometric Brownian Motion.

We check what happens when the number of steps $n \to +\infty$ in our binomial tree. The time to maturity T is divided into n intervals, hence each step is of length $\Delta t = T/n$. The price of a European Call option in the n-step binomial model is given by:

$$P_0^{Call} = e^{-RT}\sum_{j=0}^{n}\binom{n}{j}q^j(1-q)^{n-j}(S_0u^jd^{n-j}-K)^+$$

Notice that many terms in the sum above are zero. It suffices to sum only the non-null terms, which are those where the final stock price is greater than the strike price:

$$S_0u^jd^{n-j} > K$$

By taking the natural logarithm and substituting $u = e^{\sigma\sqrt{T/n}}$ and $d = e^{-\sigma\sqrt{T/n}}$, we find the minimum number of up-movements ($\hat{j}$) required for the option to finish in-the-money:

$$\hat{j} := \left\lfloor \frac{n}{2} - \frac{\ln(S_0/K)}{2\sigma\sqrt{T/n}} \right\rfloor$$

We can split the pricing formula into two parts and rewrite it using probabilities:

$$P_0^{Call} = S_0\sum_{j=\hat{j}}^{n}\binom{n}{j}\tilde{q}^j(1-\tilde{q})^{n-j} - Ke^{-RT}\sum_{j=\hat{j}}^{n}\binom{n}{j}q^j(1-q)^{n-j}$$

where $\tilde{q} = que^{-R(T/n)}$. This can be expressed using a Binomial Random Variable $B(n,p)$:

$$P_0^{Call} = S_0\mathbb{Q}(B(n,\tilde{q}) \geq \hat{j}) - Ke^{-RT}\mathbb{Q}(B(n,q) \geq \hat{j})$$

The Central Limit Theorem (CLT): To find the final limit as $n \to +\infty$, we use the Central Limit Theorem. The CLT states that given independent and identically distributed (i.i.d.) random variables X_i with mean μ and variance σ^2:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{n\to\infty} \mathcal{N}(0,1)$$

Applying the CLT to our binomial distribution, we know that $B(n,q) \sim \mathcal{N}(nq, \sqrt{nq(1-q)})$. Taking the limit of the probabilities in our pricing equation, we finally obtain the **Black-Scholes Formula**:

$$P_0^{Call} = S_0\mathcal{N}(d_1) - Ke^{-RT}\mathcal{N}(d_2)$$

where $\mathcal{N}(\cdot)$ is the cumulative distribution function of a Standard Normal variable, and:

$$d_1 = \frac{\ln(S_0/K) + (R + \sigma^2/2)T}{\sigma\sqrt{T}} \qquad d_2 = d_1 - \sigma\sqrt{T}$$

1.10.1 American Options and Stopping Times

Let us briefly list the types of options we are interested in:

- **European:** can be exercised only at maturity T. Payoff: $H_T = f(S_T)$.
- **Bermudan:** many possible exercise dates $t_1, t_2, \dots, T$. Payoff: $H_{t_n} = g(S_{t_n})$.
- **American:** exercise is possible at any intermediate date $t \in [0, T]$.

In order to mathematically treat American options, we need the notion of a **Stopping Time**, which is a special Random Time.

Example: Random Time Consider a sequence of random variables taking values in $\{-1, 1\}$ like a game. Assuming they are Independent and Identically Distributed (i.i.d.):

$$Y_i = \begin{cases} 1 & (1/2) \\ -1 & (1/2) \end{cases} \quad \forall i$$

We define $X_0 = 0$ and introduce the discrete time stochastic process X:

$$X_n := \sum_{i=1}^{n} Y_i$$

We can introduce a specific variable τ_{20} that represents the first time X reaches the value 20 and i want to stop the game:

$$\tau_{20} := \inf\{n \in \mathbb{N} : X_n = 20\}$$

This is clearly a **Random Variable** taking values in $\{t_1, t_2, \dots, t_n, \dots\}$, hence it is a **Random Time**.

Notice that if τ_{20} is bounded, then:

$$X_{\tau_{20}} = 20$$

Why is τ_{20} a random variable? Because its value depends on the specific scenario (path) ω:

$$\tau_{20}(\omega) = \inf\{n : X_n(\omega) = 20\}$$

Random Time

A random time τ taking values in $\{t_1, t_2, \dots, t_n, \dots\}$ is a Stopping Time with respect to the filtration $(\mathcal{F}_n)_n$ if the event:

$$\{\tau \leq t_n\} \in \mathcal{F}_n$$

In plain words: The random time τ is NON-ANTICIPATIVE. At any time t_n, we can determine whether τ occurred or not using only present information.

Example: What is NOT a Stopping Time Is the following random time a stopping time? Consider, from the previous example:

$$\tau_{20} := \sup\{n \in \mathbb{N} : X_n = 20\}$$

This is **NOT** a stopping time! Why? The event $\{\tau_{20} \leq t_n\}$ also depends on future events, i cannot determine the last time my game reaches 20 without looking at the future.

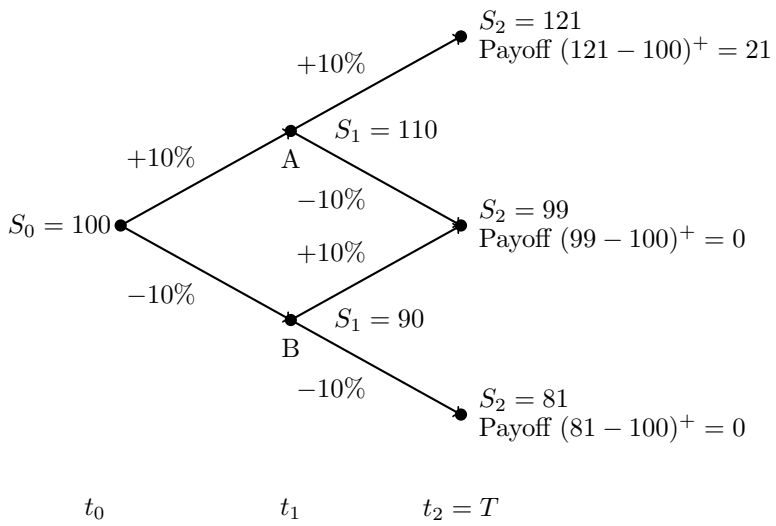
Example: American Call Option Payoff How is the payoff of an American Call option defined?

$$\text{PAYOFF IN CASE OF EXERCISE AT } \tau = (S_\tau - K)^+$$

So, a fundamental question arises: How do I choose τ in an optimal way?

1.10.2 Pricing an American Call Option with No Dividends (Toy Example)

Consider a 2-step binomial model with $S_0 = 100$, $K = 100$, $u = 1.10$, $d = 0.90$, $R = 0$, and $q = 1/2$. How do we compute the price of an American Call? We proceed backwardly:



1. **At $t = 2$ (Maturity):** The option expires. We simply compute the intrinsic payoff $(S_T - K)^+$.
2. **At $t = 1$:** In each node, the holder must compare two values:

$$\begin{cases} \text{Exercise Value (Immediate): } (S_1 - K)^+ \\ \text{Continuation Value (Waiting): } e^{-R\Delta t}\mathbb{E}^{\mathbb{Q}}[\text{Payoff}_{t=2}|\mathcal{F}_1] \end{cases}$$

   In the upper node A ($S_1 = 110$):

$$P_A^{US} = \begin{cases} (110 - 100)^+ = 10 \\ e^{-R\Delta t}\mathbb{E}^{\mathbb{Q}}[\text{Payoff}_{t=2}|\mathcal{F}_A] = e^{-0}[21 \cdot 1/2 + 0 \cdot 1/2] = 10.5 \end{cases}$$

   Since the continuation value (10.5) is greater than the immediate exercise value (10), it is **optimal to WAIT** until the maturity.
   In the lower node B ($S_1 = 90$):

$$P_B^{US} = \begin{cases} (90 - 100)^+ = 0 \\ e^{-R\Delta t}\mathbb{E}^{\mathbb{Q}}[\text{Payoff}_{t=2}|\mathcal{F}_B] = e^{-0}[0 \cdot 1/2 + 0 \cdot 1/2] = 0 \end{cases}$$

   Here, both values are equal, so it does not matter whether we exercise or wait.
3. **At $t = 0$:**

$$P_0^{US} = \begin{cases} (100 - 100)^+ = 0 \\ e^{-R\Delta t}\mathbb{E}^{\mathbb{Q}}[P_1^{US}] = e^{-0}[10.5 \cdot 1/2 + 0 \cdot 1/2] = 5.25 \end{cases}$$

Conclusion: We discovered that it is never optimal to exercise the American Call early. The price coincides perfectly with the European price!

No-Dividends Assumption

In any model with NO-DIVIDENDS we have:

$$P_0^{US}(\text{Call}) = P_0^{EU}(\text{Call})$$

Proof. We notice that the price of a European Call option at time t is given by the discounted expected payoff at maturity:

$$\begin{cases} P_t^{EU}(\text{Call}) \geq S_t - \underbrace{e^{-R(T-t)}}_{<1} K > S_t - K \\ P_t^{EU}(\text{Call}) \geq 0 \end{cases} \implies \underbrace{P_t^{EU}(\text{Call})}_{\text{waiting value}} \geq \underbrace{(S_t - K)^+}_{\text{exercise value}} \quad \forall t \in [0, T]$$

As a conclusion, when I compare the waiting value and the exercise value, **waiting wins!** If it is always optimal to wait, then the US option reduces to a European one. □

1.10.3 The General Case (Optimal Stopping)

Our goal is to compute the optimal time $\hat{\tau}$ to exercise the option to maximize the expected payoff:

$$\sup_{0 \leq \tau \leq T} \mathbb{E}^{\mathbb{Q}}[Z_\tau] = \mathbb{E}^{\mathbb{Q}}[Z_{\hat{\tau}}]$$

where τ is a stopping time and Z is a stochastic process that represent the sequence of payoffs.

Trivial Cases:

- If $(Z_n)_n$ is a **Sub-Martingale** ($Z_n \leq \mathbb{E}[Z_{n+1}|\mathcal{F}_n]$), then late stopping is optimal

$$\hat{\tau} = T$$

- If $(Z_n)_n$ is a **Super-Martingale** ($Z_n \geq \mathbb{E}[Z_{n+1}|\mathcal{F}_n]$), then immediate stopping is optimal

$$\hat{\tau} = 0$$

- If $(Z_n)_n$ is a **Martingale**, then all stopping times are optimal

$$\hat{\tau} \in \{0, 1, \ldots, T\}$$

1.11 American Options: Optimal Stopping and Snell Envelope

1.11.1 The Optimal Value Process

Recall that τ is a stopping time and $(Z_n)_{n \leq T}$ is a sequence of payoffs. We define two fundamental objects to determine the optimal exercise strategy:

1. **The Value Process (J_n):** The expected payoff if we stop at a specific time τ.

$$J_n(\tau) := \mathbb{E}[Z_\tau|\mathcal{F}_n] \quad \text{for } n \leq \tau \leq T$$

2. **The Optimal Value Process (V_n):** The best possible expected payoff across all possible future stopping times.

$$V_n := \text{ess sup}_{n \leq \tau \leq T} \mathbb{E}[Z_\tau|\mathcal{F}_n]$$

It is natural to think that at any time n, we only have two choices:

- **STOP:** We exercise the option and get the immediate payoff: $V_n = Z_n$.
- **WAIT:** We hold the option, and its value is the discounted expectation of tomorrow's optimal value: $V_n = \mathbb{E}^{\mathbb{Q}}[V_{n+1}|\mathcal{F}_n]$.

Optimal Value Process

The optimal value process V_n satisfies the following recursive equation:

- $V_{Tn} = Z_{Tn}$
- $V_{tn} = V_n = \max\{Z_n, \mathbb{E}^{\mathbb{Q}}[V_{n+1}|\mathcal{F}_n]\}$

This equation tells us to compare two quantities: the immediate exercise value (Z_n) and the continuation value ($\mathbb{E}^{\mathbb{Q}}[V_{n+1}|\mathcal{F}_n]$).

- If $\hat{\tau}_n = n$, then $V_n = Z_n \implies$ **I STOP**.
- If $\hat{\tau}_n > n$, then $V_n = \mathbb{E}^{\mathbb{Q}}[V_{n+1}|\mathcal{F}_n] > Z_n \implies$ **I WAIT**.

Optimal Stopping Rule

The optimal stopping time $\hat{\tau}$ is the first time the payoff matches the optimal value process:

$$\hat{\tau} := \min\{n \geq 0 : V_n = Z_n\}$$

1.11.2 The Snell Envelope

We introduce a useful mathematical tool to price American options.

Snell Envelope

Let X and Y be discrete-time stochastic processes. We say that X **dominates** Y if $X_n \geq Y_n$ almost surely (P-a.s.). If $Y \in L^1$ (i.e., $\mathbb{E}[|Y_n|] < \infty$), then the **Snell Envelope** of Y is defined as the smallest super-martingale dominating Y.

Theorem: The optimal value process V is exactly the Snell Envelope of the payoff process Z.

Proof. By construction, $V_n \geq Z_n$, so it dominates the payoff. Since $V_n \geq \mathbb{E}^{\mathbb{Q}}[V_{n+1}|\mathcal{F}_n]$, it is a $\mathbb{Q}$-super-martingale. To prove it is the *smallest*, we assume another super-martingale X dominates Z, and through backward induction from $t = T = t_N$, we show that $X_n \geq V_n$ for all n. □

Remark on American Calls: For an American Call on a non-dividend paying asset, the discounted payoff process $e^{-Rn}(S_n - K)^+$ is a $\mathbb{Q}$-sub-martingale. By Jensen's Inequality for convex functions, it is mathematically proven that early exercise is never optimal ($\hat{\tau} = T$).

1.11.3 Exercise: Pricing an American Put Option

We want to price an American Put and a European Put on a 2-step binomial tree. **Parameters:** $S_0 = 100, K = 100, u = 1.1, d = 0.9, R = 0.05, T = 1$ (so $\Delta t = 1/2$).

Step 1: Risk-neutral probability

$$q = \frac{e^{R\Delta t} - d}{u - d} = \frac{e^{0.025} - 0.9}{1.1 - 0.9} = 0.626576$$

Step 2: European Put Price Using the standard binomial formula (expectation of final payoffs):

$$P_0^{EU} = e^{-RT}\mathbb{E}^{\mathbb{Q}}[P_2^{EU}] = e^{-0.05}[0 + 2q(1-q)(1) + 1 \cdot (1-q)^2 \cdot 19] = 2.965388$$

Step 3: American Put Price (Backward Induction) We must check for early exercise at the intermediate nodes (Time $t = 1$).

- **Node A** ($S_1 = 110$)**:**

$$P_A^{US} = \max\{(100 - 110)^+, \text{Continuation Value}\} = \max\{0, 0.364\} = 0.364$$

  (Optimal to WAIT).

- **Node B** ($S_1 = 90$)**:**

$$P_B^{US} = \max\{(100 - 90)^+, e^{-0.025}[q \cdot 1 + (1-q) \cdot 19]\} = \max\{10, 7.53\} = 10$$

  (Optimal to **EXERCISE EARLY**: the intrinsic value 10 is strictly greater than the continuation value 7.53).

Finally, at Time $t = 0$:

$$P_0^{US} = \max\{(100 - 100)^+, e^{-0.025}[qP_A^{US} + (1-q)P_B^{US}]\} = 3.962446$$

Conclusion: The price of the American Put (3.96) is strictly higher than the European Put (2.96), perfectly demonstrating the added value of the early exercise premium.

1.12 More General Models: Multi-Asset and Absence of Arbitrage

In this lecture, we will go through a more general theory for the multi-asset model. We need to formalize the concepts of Absence of Arbitrage and Completeness. Reference: Ch. 3, T. Björk.

1.12.1 The Multi-Asset Model (1-period)

Let us introduce a general model over 1 period (from $t = 0$ to $t = 1$). We have N assets, represented as column vectors:

$$S_0 = \begin{pmatrix} S_0^1 \\ \vdots \\ S_0^N \end{pmatrix}, \quad S_1 = \begin{pmatrix} S_1^1 \\ \vdots \\ S_1^N \end{pmatrix}$$

We consider a finite probability space $\Omega = \{\omega_1, \omega_2, \ldots, \omega_M\}$ with $\mathbb{P}(\{\omega_j\}) > 0 \ \forall j$.

We can visualize all the randomness in a matrix $(N \times M)$:

$$\Sigma = \begin{pmatrix} S_1^1(\omega_1) & S_1^1(\omega_2) & \ldots & S_1^1(\omega_M) \\ S_1^2(\omega_1) & S_1^2(\omega_2) & \ldots & S_1^2(\omega_M) \\ \vdots & \vdots & \ddots & \vdots \\ S_1^N(\omega_1) & S_1^N(\omega_2) & \ldots & S_1^N(\omega_M) \end{pmatrix}$$

The Numéraire: We assume everything is risky, but we want to normalize the prices using a **Numéraire** (e.g., Asset 1). We assume $S_0^1 > 0$ and $S_1^1(\omega_j) > 0 \ \forall j$.

1.12.2 Portfolio Strategies and Arbitrage

Portfolio Strategy

A strategy $h \in \mathbb{R}^N$ is a deterministic vector where h^i represents the number of units of asset i. The strategy is decided at $t = 0$ and kept until $t = 1$. The portfolio value is:

- At $t = 0$: $V_0^h = h \cdot S_0 = \sum_{i=1}^N h^i S_0^i$ (scalar)
- At $t = 1$: $V_1^h = h \cdot S_1$ (random variable)

Notice that $V_1^h(\omega_j) = (h^T \Sigma)_j$ (row vector times matrix).

Arbitrage

An arbitrage is a particular strategy $h \in \mathbb{R}^N$ such that:

- $V_0^h = 0$
- $\mathbb{P}(V_1^h \geq 0) = 1$
- $\mathbb{P}(V_1^h > 0) > 0$

Let us now move to normalized prices $Z^i := S^i / S^1$. The normalized portfolio value is $V_t^{h,Z} = V_t^h / S_t^1$.

1.12.3 Farkas' Lemma and Arbitrage Pricing

To prove the absence of arbitrage, we need an intermediate lemma from linear algebra.

Farkas' Lemma

Let $x_0, x_1, \ldots, x_M \in \mathbb{R}^N$ be column vectors. Then **exactly one** of the following problems has a solution:

1. Find nonnegative $\lambda_1, \ldots, \lambda_M \geq 0$ such that $x_0 = \sum_{j=1}^{M} \lambda_j x_j$
2. Find a row vector $h \in \mathbb{R}^N$ such that:

$$\begin{cases} h \cdot x_0 < 0 \\ h \cdot x_j \geq 0 \quad \forall j = 1, \ldots, M \end{cases}$$

Proof. Let K be the convex cone of all nonnegative combinations of $x_1, \ldots, x_M$. If $x_0 \in K$, Problem 1 has a solution. If $x_0 \notin K$, by the **Hyperplane Separation Theorem**, there exists a separating hyperplane H strictly separating x_0 from K. The normal vector h to this hyperplane is the solution to Problem 2. □

Proposition: The market is ARBITRAGE FREE if and only if there exist nonnegative numbers $q_1, \ldots, q_M \in \mathbb{R}_{>0}$ with $\sum_{j=1}^{M} q_j = 1$ such that:

$$Z_0 = \sum_{j=1}^{M} q_j Z_1(\omega_j) \quad \iff \quad Z_0^i = \sum_{j=1}^{M} q_j Z_1^i(\omega_j) \quad \forall i = 1, \ldots, N$$

1.12.4 The Fundamental Theorems of Asset Pricing (FTAP)

First Fundamental Theorem of Asset Pricing

Given a fixed numéraire, the market is **ARBITRAGE FREE** $\iff$ $\exists$ at least one Equivalent Probability Measure $\mathbb{Q} \sim \mathbb{P}$ such that:

$$Z_0 = \mathbb{E}^{\mathbb{Q}}[Z_1]$$

Remarks:

- $\mathbb{Q}$ is equivalent to $\mathbb{P}$ means they agree on impossible events: $\mathbb{P}(A) = 0 \iff \mathbb{Q}(A) = 0$.
- $\mathbb{Q}$ is called a **Martingale Measure** (discounted prices are martingales under $\mathbb{Q}$) or a **Risk-Neutral Measure**.
- If the numéraire is the deterministic bank account ($S^1 = e^{RT}$), we get the standard formula: $S_0^i = e^{-RT}\mathbb{E}^{\mathbb{Q}}[S_T^i]$.

Market Completeness and the 2° FTAP: Definition (Completeness): A claim X (random variable) is **reachable** if there exists a portfolio strategy h such that $\mathbb{P}(V_1^h = X) = 1$. If every claim is reachable, the market is **COMPLETE**.

Second Fundamental Theorem of Asset Pricing

Assume the market is arbitrage-free. Then the market is **COMPLETE** $\iff$ $\exists!$ (a unique) Equivalent Martingale Measure $\mathbb{Q}$.

Chapter 2

Continuous Time Finance

2.1 From Discrete to Continuous Time: Trees, Martingales, and Itô Calculus

2.1.1 Risk-Neutral Pricing and Information Structures

The price of any contingent claim can be written in terms of the expected value under the objective probability measure $\mathbb{Q}$ (Risk-Neutral probability). The market value at any intermediate time t (Mark-to-Market) is given by the conditional expected value of the discounted payoff:

$$P_t = \mathbb{E}^{\mathbb{Q}}[e^{-R(T-t)}\text{Payoff}|\mathcal{F}_t]$$

Where $\mathcal{F}_t$ represents the current information structure (filtration), meaning the expectation is conditional on the information available at time t.

Remark on Fundamental Theorems of Asset Pricing (FTAP):

- **1st FTAP:** The absence of arbitrage opportunities is equivalent to the existence of an equivalent risk-neutral probability measure $\mathbb{Q}$.
- **2nd FTAP:** The uniqueness of this measure $\mathbb{Q}$ is related to the completeness of the market (the ability to replicate whatever contingent claim). The binomial model is complete, hence there is no residual risk.

2.1.2 Binomial Trees: Lattice vs. Non-Recombining

When moving to a dynamic model, we face a computational trade-off.

- **Non-Recombining Tree:** The number of nodes increases exponentially (2^N), making it computationally impossible for large N. However, it preserves *path-dependency* and allows us to model stylized facts like volatility increasing in a bearish market (downward trajectories having larger spreads than upward ones).
- **Recombining Tree (Lattice):** The number of nodes is linear ($N+1$), perfectly feasible computationally. We lose path-dependency. It is ideal for Vanilla European options, which only depend on the final "photograph" of the underlying.

Type	Lose	Gain
Non-Recombining	computational complexity	path-dependency
Lattice	path-dependency	computational efficiency

Asian Options: Introduced to prevent market manipulation (e.g., in the crude oil market), their payoff depends on the historical arithmetic average of prices. Being strongly path-dependent, they cannot be priced using a simple lattice; they require non-recombining trees or Monte Carlo simulations.

2.1.3 Standard Brownian Motion and Stochastic Integrals

The continuous-time counterpart of the binomial model is the Brownian Motion (W_t).

- **Distribution:** $W_t \sim \mathcal{N}(0, t)$. The variance grows linearly over time.
- **Irregularity:** The trajectories are highly irregular, non-differentiable, and have vertical behavior around 0 (variation explores to infinity).

Because of this irregularity, we cannot use standard Riemann derivatives/integrals. Kiyosi Itô solved this by defining the stochastic integral first for piecewise constant (elementary/horizontal) processes, and then extending it to a broader class of regular, non-anticipative (adapted) processes.

The Classes of Processes ($\mathcal{H}$ vs $\tilde{\mathcal{H}}$):

1. **Class $\mathcal{H}$ (True Martingales):** Processes where the expected value of the integral of the square is bounded: $\mathbb{E}[\int_0^T H_s^2 ds] < \infty$. If $H \in \mathcal{H}$, the stochastic integral $\int H_s dW_s$ is a **true Martingale** (its expected value is exactly zero), and the **Itô Isometry** holds:

$$\mathbb{E}\left[\left(\int_0^T H_s dW_s\right)^2\right] = \mathbb{E}\left[\int_0^T H_s^2 ds\right]$$

2. **Class $\tilde{\mathcal{H}}$ (Local Martingales):** A weaker class requiring only that the integral is finite almost surely: $\mathbb{P}(\int_0^T H_s^2 ds < \infty) = 1$. Integrals of these processes are only *local* martingales, meaning their expected value might diverge or not be zero.

2.1.4 Itô's Formula and Geometric Brownian Motion

Let us solve the linear Stochastic Differential Equation (SDE) for the Geometric Brownian Motion:

$$dX_t = \mu X_t dt + \sigma X_t dW_t$$

The Naive (Wrong) Approach: If we divide by X_t and treat it as standard deterministic calculus ($d(\ln X) = dX/X$), we get:

$$dX_t = \mu X_t dt + \sigma X_t dW_t \implies \frac{dX_t}{X_t} = \mu dt + \sigma dW_t$$

$$d(\ln X) = \mu dt + \sigma dW_t \implies X_t = X_0 e^{\mu t + \sigma W_t}$$

Error: Taking the expected value using the moment-generating function for a Gaussian variable (W_t), we find $\mathbb{E}[X_t] = X_0 \exp(\mu t) \exp(\frac{1}{2}\sigma^2 t)$. The noise introduces a systematic upward bias ($\frac{1}{2}\sigma^2 t$), failing to revert around the deterministic mean.

The Itô (Correct) Approach: In stochastic calculus, second-order terms are not neglectable because the quadratic variation of the Brownian Motion is non-zero:

$$(dW_t)^2 = dt$$

Applying Itô's Lemma to the logarithmic transformation $f(X) = \ln(X)$, we must include the second derivative ($f_{xx} = -1/x^2$):

$$d(\ln X_t) = \frac{1}{X_t} dX_t - \frac{1}{2X_t^2}(dX_t)^2$$

Substituting $(dX_t)^2 = \sigma^2 X_t^2 dt$, the equation beautifully simplifies to:

$$d(\ln X_t) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t$$

Integrating this gives the correct solution for the Geometric Brownian Motion:

$$X_t = X_0 e^{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t}$$

This extra $-\frac{1}{2}\sigma^2 t$ term (the Itô correction) perfectly cancels out the variance bias, ensuring that the expected value $\mathbb{E}[X_t]$ is exactly $X_0 e^{\mu t}$, as required.

2.2 The Feynman-Kac Formula: Connecting PDEs and Stochastic Processes

The Feynman-Kac methodology bridges the gap between deterministic Partial Differential Equations (PDEs) and probabilistic expectations. It allows us to construct the solution of a PDE by taking an expected value of a stochastic process.

2.2.1 Feynman-Kac: Version 1 (The Elementary Case)

Assume we have a function $F(t,x)$ that solves the following linear parabolic PDE:

$$\text{PDE: } \begin{cases} \frac{\partial F}{\partial t} + K(t,x)\frac{\partial F}{\partial x} + \frac{1}{2}H(t,x)^2\frac{\partial^2 F}{\partial x^2} = 0 \\ F(T,x) = \Phi(x) \leftarrow \text{Payoff at maturity} \end{cases}$$

With a terminal boundary condition at maturity T

The Solution: We introduce a probability space with a standard Brownian Motion W_t. We define an Itô process X_t whose dynamics are built directly from the PDE coefficients:

$$\text{SDE: } dX_u = K(u,X_u)du + H(u,X_u)dW_u$$

Now applyng Itô's Lemma and substituting the quadratic variation:

$$dF(u,X_u) = \frac{\partial F}{\partial u}du + \frac{\partial F}{\partial x}dX_u + \frac{1}{2}\frac{\partial^2 F}{\partial x^2}(dX_u)^2 \implies dF = \frac{\partial F}{\partial u}du + \frac{\partial F}{\partial x}(Kdu + HdW_u) + \frac{1}{2}\frac{\partial^2 F}{\partial x^2}H^2du$$

Now, grouping the deterministic terms and using the PDE to cancel them out:

$$dF = \left(\frac{\partial F}{\partial u} + K\frac{\partial F}{\partial x} + \frac{1}{2}H^2\frac{\partial^2 F}{\partial x^2}\right)du + \frac{\partial F}{\partial x}HdW_u$$

And by the PDE, the deterministic part is zero, leaving us with:

$$dF = \frac{\partial F}{\partial x}HdW_u$$

The solution to the PDE is simply the conditional expected value of the terminal condition:

$$\int_t^T dF = \int_t^T \frac{\partial F}{\partial x}HdW_u \implies \mathbb{E}[\underbrace{F(T,X_T)}_{=\Phi(X_T)}|X_t = x] - F(t,x) = \underbrace{\mathbb{E}\left[\int_t^T \frac{\partial F}{\partial x}HdW_u \middle| X_t = x\right]}_{=0}$$

$$\implies F(t,x) = \mathbb{E}[\Phi(X_T)|X_t = x]$$

2.2.2 Feynman-Kac Formula: Version 2 (The Discounting Term)

In finance, we must account for the time value of money (the risk-free rate R). If we add a linear term $-rF$ to the PDE:

$$\begin{cases} \frac{\partial F}{\partial t} + K(t,x)\frac{\partial F}{\partial x} + \frac{1}{2}H(t,x)^2\frac{\partial^2 F}{\partial x^2} - rF = 0 \\ F(T,x) = \Phi(x) \end{cases}$$

The Solution: We introduce the same process $dX_s = Kds + HdW_s$. To solve this, we apply the two-dimensional Itô formula (using the product rule for stochastic calculus) to the auxiliary function $e^{-rt}F(t,X_t)$. The $-rF$ term perfectly cancels out the deterministic drift. The result is the **discounted conditional expected value**:

$$F(t,x) = \mathbb{E}[e^{-r(T-t)}\Phi(X_T)|X_t = x]$$

Financial meaning: This is exactly the risk-neutral pricing formula for a European derivative, where Φ is the payoff and r is the interest rate!

2.2.3 Feynman-Kac Formula: Version 3 (The Source Term)

If the PDE has an additional generic function $G(t, x)$ (a "source term"):

$$\frac{\partial F}{\partial t} + K\frac{\partial F}{\partial x} + \frac{1}{2}H^2\frac{\partial^2 F}{\partial x^2} + G(t, x) = 0$$

The Solution: The deterministic integral of G remains in the equation. We take the expected value, and the solution becomes:

$$F(t, x) = \mathbb{E}[\Phi(X_T)|X_t = x] + \int_t^T \mathbb{E}[G(s, X_s)|X_t = x]ds$$

Financial meaning: This is used for pricing assets that pay continuous dividends or coupons over time ($G(t, x)$ represents the continuous cash flow).

2.2.4 Example: Solving a PDE with Feynman-Kac

Solve the following PDE:

$$\begin{cases} \frac{\partial F}{\partial t} + \frac{1}{2}x^2\frac{\partial^2 F}{\partial x^2} + x = 0 \\ F(T, x) = 2\ln(x) \end{cases}$$

Step 1: Identification Compare the PDE to Version 3 of Feynman-Kac:

- $K = 0$ (No first derivative term)
- $H(t, x) = x$ (Since $\frac{1}{2}H^2 = \frac{1}{2}x^2$)
- $G(t, x) = x$ (The standalone term)

Step 2: Process Dynamics Construct the Itô process: $dX_s = 0ds + X_s dW_s \implies \frac{dX_s}{X_s} = dW_s$. This is a Geometric Brownian Motion with $\mu = 0$ and $\sigma = 1$. Its explicit solution is:

$$X_s = x\exp\left(-\frac{1}{2}(s - t) + (W_s - W_t)\right)$$

Step 3: Compute the Expectation Apply Version 3 formula:

$$F(t, x) = \mathbb{E}[2\ln(X_T)|X_t = x] + \int_t^T \mathbb{E}[X_s|X_t = x]ds$$

Using the properties of the lognormal distribution and the independence of Brownian increments:

- $\mathbb{E}[2\ln(X_T)] = 2\ln(x) - (T - t)$ (since the expected value of the Brownian increment is 0).
- $\mathbb{E}[X_s] = x$ (since X_s is a Martingale when drift is zero).

$$F(t, x) = 2\ln(x) - (T - t) + x(T - t)$$

2.3 The Black-Scholes Market Model (1973)

The Black-Scholes model assumes a frictionless market, no arbitrage, and constant risk-free rate R. The asset's infinitesimal returns follow a Geometric Brownian Motion under the subjective, historical probability $\mathbb{P}$:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t^{\mathbb{P}}$$

The "Scandalous" Insight (μ Independence): A deeply counterintuitive result of the model is that the option price **does not depend on the real-world drift** μ. Whether an investor expects the stock to soar ($\mu > 0$) or crash ($\mu < 0$), the fair value of the option is identical. This is because μ is subjective, and in a complete market, it is perfectly hedged away and replaced by the risk-free rate r under the risk-neutral measure $\mathbb{Q}$.

2.4 The Black-Scholes Model: Portfolio Dynamics and the PDE

2.4.1 The Problem with the "Naive" Limit in Continuous Time

We want to construct the continuous-time dynamics of a portfolio V_t. In a stylized model, the portfolio consists of a risky asset S_t and a riskless asset (e.g., a bank account). Let the strategy be denoted by $h(t)$, representing the number of shares held at time t.

The Discrete Approach: Suppose we have a time step Δt. We analyze the portfolio value at two points in time:

1. **At the end of the period (before rebalancing):** The value is based on the strategy chosen at the previous step $(t - \Delta t)$.
$$V(t^-) = h(t - \Delta t)S(t)$$

2. **At the beginning of the new period (after rebalancing):** We choose a new strategy $h(t)$. We can also include a continuous consumption rate $c(t)\Delta t$.
$$V(t) = h(t)S(t) + c(t)\Delta t$$

To ensure no external money is injected (the Budget Equation), these two values must be equal:

$$h(t - \Delta t)S(t) = h(t)S(t) + c(t)\Delta t$$

Rearranging to isolate the change in strategy $\Delta h(t) = h(t) - h(t - \Delta t)$:

$$\Delta h(t)S(t) + c(t)\Delta t = 0$$

The Trap (Anticipative Integral): If we naively take the limit as $\Delta t \to 0$, we get $dh \cdot S + cdt = 0$. However, this formulation implies an **anticipative** stochastic integral, which violates the fundamental construction of the Itô integral (where integrands must be measurable with respect to the past, not the future increments). We cannot forecast the future to determine our strategy.

2.4.2 The Correct Formulation (Self-Financing Portfolio)

To recover a proper Itô formulation, we must perform an algebraic trick: we add and subtract the term $h(t - \Delta t)S(t - \Delta t)$ from the budget equation.

$$h(t - \Delta t)S(t) = h(t)S(t) + c(t)\Delta t$$

Let $h^- = h(t - \Delta t)$ and $S^- = S(t - \Delta t)$ for simplicity.

$$h^- S = hS + S^-(h - h^-) - S^-(h - h^-) + c\Delta t$$

By rearranging and taking the limit $\Delta t \to 0$, the correct differential equation emerges:

$$dhdS + dhS + cdt = 0$$

Now, apply the Itô product rule to the portfolio value $V = hS$:

$$dV = dh \cdot S + hdS + dhdS$$

Substitute the previously found relation $(dhdS + dhS = -cdt)$ into this equation:

$$dV = hdS - cdt$$

The Self-Financing Condition: In option pricing, we assume zero consumption $(c = 0)$. The equation simplifies to:

$$dV_t = hdS_t$$

This is the **Self-Financing Condition**: The only reason the portfolio value fluctuates (dV) is due to the fluctuations of the underlying asset prices (dS), not because we inject or withdraw money (changes in h).

2.4.3 Deriving the Black-Scholes PDE

We operate under the Black-Scholes market assumptions:

- Any derivative can be bought or sold without frictions (no transaction costs, same lending/borrowing rate r).
- Absence of Arbitrage Opportunities (AAO).
- The derivative price is a Markovian function $F(t, S_t) \in \mathcal{C}^{1,2}$.

The dynamics of the risky asset S_t and the riskless bond B_t are:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t \quad \implies dS_t = S_t(\mu dt + \sigma dW_t)$$

$$\frac{dB_t}{B_t} = r dt \quad \implies dB_t = r B_t dt$$

Step 1: The Replicating Portfolio Construct a self-financing portfolio $V_t = \alpha_t S_t + \beta_t B_t$ that perfectly replicates the option payoff at maturity T: $V_T = F(T, S_T)$. By AAO, the value of the portfolio must equal the option price at any time t: $V_t = F(t, S_t)$. Furthermore, their *dynamics* must also perfectly match:

$$dV_t = dF(t, S_t)$$

Step 2: Equating the Dynamics Using the self-financing condition, the portfolio dynamics are:

$$dV_t = \alpha dS_t + \beta dB_t = \alpha S_t(\mu dt + \sigma dW_t) + \beta(r B_t dt)$$

Using Itô's Lemma, the dynamics of the option price F are:

$$dF = \left(\frac{\partial F}{\partial t} + \mu S \frac{\partial F}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 F}{\partial S^2}\right) dt + \left(\sigma S \frac{\partial F}{\partial S}\right) dW_t$$

Step 3: Matching Coefficients (Delta Hedging) Since the equality must hold, we equate the stochastic (dW) and deterministic (dt) parts separately. Matching the dW terms reveals the Delta Hedging strategy:

$$\alpha \sigma S = \sigma S \frac{\partial F}{\partial S} \implies \alpha_t = \frac{\partial F}{\partial S}$$

The amount of underlying stock to hold (α) is exactly the sensitivity (Delta) of the option price with respect to the stock price.

Step 4: The PDE Matching the dt terms and substituting $\alpha = \frac{\partial F}{\partial S}$:

$$\alpha \mu S dt + \beta r B dt = \frac{\partial F}{\partial t} dt + \mu S \frac{\partial F}{\partial S} dt + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 F}{\partial S^2} dt$$

The terms containing μ cancel out perfectly. We know the portfolio value is $V = \alpha S + \beta B = F$, so $\beta B = F - \alpha S = F - \frac{\partial F}{\partial S} S$. Substituting βB into the equation and dividing by dt yields the Black-Scholes PDE:

$$\frac{\partial F}{\partial t} + r S \frac{\partial F}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 F}{\partial S^2} - rF = 0$$

2.4.4 Risk-Neutral Evaluation (1st Fundamental Theorem)

The solution to the PDE, via the Feynman-Kac formula, shows that the drift μ (subjective sentiment) is completely replaced by the risk-free rate r. This leads to the Martingale approach and the First Fundamental Theorem of Asset Pricing: A model is arbitrage-free if and only if there exists an equivalent **Risk-Neutral Martingale Measure** $\mathbb{Q}$ under which all discounted asset prices are local martingales. Under $\mathbb{Q}$, the asset dynamics become:

$$\frac{dS_t}{S_t} = r dt + \sigma dW_t^{\mathbb{Q}}$$

And the option price is given by the discounted conditional expected value:

$$F(t, S_t) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[Payoff_T]$$

2.5 The Martingale Approach and The Greeks

2.5.1 Imposing Q-Martingality (The Girsanov Theorem)

We want to derive the Black-Scholes pricing formula not through a PDE, but by directly imposing the condition of No Arbitrage. The Fundamental Theorem of Asset Pricing states that No Arbitrage is equivalent to the existence of a Risk-Neutral Measure $\mathbb{Q}$ (equivalent to the historical measure $\mathbb{P}$) under which all discounted assets are martingales.

The Starting Point: Under the real-world measure $\mathbb{P}$, the asset and the riskless bond evolve as:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t^{\mathbb{P}} \qquad B_t = e^{rt}$$

Discounting the Asset: We calculate the differential of the discounted asset $S_t e^{-rt}$ using integration by parts (Itô's product rule). (but if you prefer, this works also with itô's lemma):

$$d(S_t e^{-rt}) = (dS_t)e^{-rt} - re^{-rt}S_t dt$$

Substituting dS_t with the asset defined under $\mathbb{P}$:

$$d(S_t e^{-rt}) = e^{-rt}S_t\left(\mu dt + \sigma dW_t^{\mathbb{P}} - rdt\right) = e^{-rt}S_t\left((\mu - r)dt + \sigma dW_t^{\mathbb{P}}\right)$$

By factoring out σ, we expose the **Market Price of Risk** ($\frac{\mu - r}{\sigma}$):

$$d(S_t e^{-rt}) = e^{-rt}S_t\sigma\left[\frac{\mu - r}{\sigma}dt + dW_t^{\mathbb{P}}\right]$$

Since there is a drift term ($\frac{\mu - r}{\sigma}dt$), the discounted asset is **not** a martingale under $\mathbb{P}$.

Changing the Measure (Girsanov Theorem): To make the discounted asset a martingale, the drift term inside the bracket must disappear. Girsanov's Theorem allows us to change the probability measure from $\mathbb{P}$ to $\mathbb{Q}$ by defining a new Brownian Motion.

Girsanov's Theorem

Given a Brownian Motion $W_t^{\mathbb{P}}$ under the original measure $\mathbb{P}$, we can define a new Brownian Motion under the new measure $\mathbb{Q}$ as follows:

$$dW_t^{\mathbb{Q}} = \frac{\mu - r}{\sigma}dt + dW_t^{\mathbb{P}}$$

Substituting this new Brownian Motion back into the asset dynamics, the physical drift μ is completely replaced by the risk-free rate r:

$$\frac{dS_t}{S_t} = \mu dt + \sigma\left(dW_t^{\mathbb{Q}} - \frac{\mu - r}{\sigma}dt\right) \implies \frac{dS_t}{S_t} = \mu dt + \sigma dW_t^{\mathbb{Q}} - (\mu - r)dt \implies \frac{dS_t}{S_t} = rdt + \sigma dW_t^{\mathbb{Q}}$$

Under $\mathbb{Q}$, the differential $d(S_t e^{-rt})$ has zero drift ($0dt$), making it a true martingale.

Technical Note (Novikov Condition): For the measure change to be valid, the Radon-Nikodym derivative Z_t must be a true martingale. This is guaranteed by the Novikov condition: $\mathbb{E}[e^{\frac{1}{2}\int_0^T H_s^2 ds}] < \infty$. Since our shift $H = \frac{\mu - r}{\sigma}$ is constant, the condition is trivially satisfied.

$$\left[\frac{\mu - r}{\sigma}dt + dW_t^{\mathbb{P}}\right] \implies Z_t = \exp\left[-\frac{1}{2}\int_0^t H_s^2 ds + \underbrace{\int_0^t H_s dW_s^{\mathbb{P}}}_{\text{Itô correction}}\right] = e^{-\frac{1}{2}\frac{(\mu - r)^2}{\sigma^2}t + \frac{\mu - r}{\sigma}W_t^{\mathbb{P}}}$$

2.5.2 Pricing under $\mathbb{Q}$

Because both the discounted asset and any self-financing portfolio V_t are martingales under $\mathbb{Q}$, their current value is the expected value of their terminal payoff:

$$\mathbb{E}_t^{\mathbb{Q}}[M_T] = M_t \implies \mathbb{E}_t^{\mathbb{Q}}[V_T e^{-rT}] = V_t e^{-rt} \implies V_t = e^{-r(T-t)}\mathbb{E}_t^{\mathbb{Q}}[Payoff_T]$$

2.5.3 The Greeks (Sensitivities)

The Greeks measure how the option price changes in response to small changes in the underlying parameters.

Delta (Δ): Sensitivity to the underlying price S.

$$\Delta_{call} = \frac{\partial P_t}{\partial S_t} = \Phi(d_1) > 0$$

- Represents the amount of stock needed to hedge the option (Delta Hedging).
- It is always positive for a call and bounded between 0 and 1.
- The curve is steepest around the strike price (At-The-Money).

Gamma (Γ): Sensitivity of Delta to the underlying price S (Convexity).

$$\Gamma_{call} = \frac{\partial^2 P_t}{\partial S_t^2} = \frac{\partial \Delta}{\partial S} = \frac{e^{-d_1^2/2}}{S\sigma\sqrt{2\pi(T-t)}} > 0$$

- Measures the frequency of rebalancing required for the hedge.
- It is highly concentrated around the strike price, acting almost like a Dirac delta function as maturity approaches.

Vega ($\mathcal{V}$): Sensitivity to volatility σ.

$$\mathcal{V}_{call} = \frac{\partial P_t}{\partial \sigma} = \frac{Se^{-d_1^2/2}\sqrt{T-t}}{\sqrt{2\pi}} > 0$$

- Always positive: higher volatility increases the option's value because it increases the probability of ending up deep in the money.
- Like Gamma, Vega risk is highly concentrated around the strike price.
- *Volatility Shock Effect:* An increase in σ "smoothes" the Delta and Gamma curves, meaning the trader must start hedging earlier even when the option is Out-Of-The-Money.

Theta (Θ): Sensitivity to the passage of time (Time Decay).

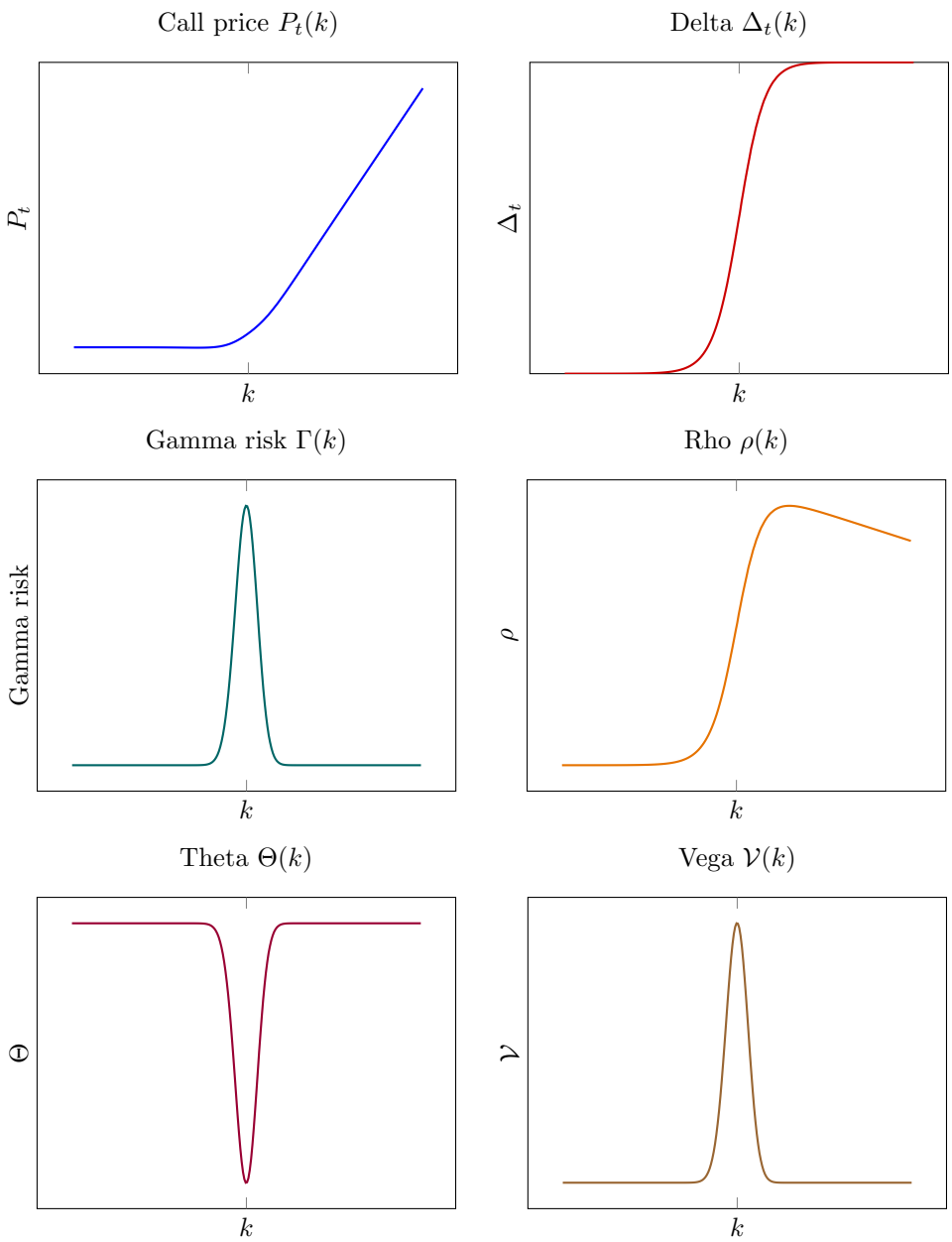
$$\Theta_{call} = \frac{\partial P_t}{\partial t} < 0$$

- Generally negative for options: as time passes with no movement in the underlying, the option loses value (opportunity loss).

Rho (ρ): Sensitivity to the risk-free interest rate r.

$$\rho_{call} = \frac{\partial P_t}{\partial r} = K(T-t)e^{-r(T-t)}\Phi(d_2) > 0$$

Greeks plots as functions of strike k: For the following plots we fix $S_t = 100$, $r = 0\%$, $\sigma = 8\%$, and $\tau = T - t = 0.1$ years. The horizontal axis is always the strike k.



2.6 Put Options, Portfolio Immunization, and Digital Options

2.6.1 Pricing and Greeks of a Put Option

Instead of deriving the Put option price from scratch using Feynman-Kac, we can exploit the **Put-Call Parity** under the risk-neutral measure $\mathbb{Q}$. The payoff relationship is:

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K$$

Taking the discounted expected value under $\mathbb{Q}$:

$$Call_t - Put_t = e^{-r(T-t)}\mathbb{E}_t^{\mathbb{Q}}[S_T] - Ke^{-r(T-t)}$$

Since the discounted asset $S_t e^{-rt}$ is a martingale under $\mathbb{Q}$, we know that $\mathbb{E}_t^{\mathbb{Q}}[S_T e^{-rT}] = S_t e^{-rt}$, which implies $\mathbb{E}_t^{\mathbb{Q}}[S_T] = S_t e^{r(T-t)}$. Substituting this into the equation:

$$Call_t - Put_t = S_t - Ke^{-r(T-t)}$$

Isolating the Put price and substituting the Black-Scholes Call formula $(S_t\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2))$ yields theBlack-Scholes Put Formula:

Black-Scholes Put Formula

$$Put_t = Ke^{-r(T-t)}(1 - \Phi(d_2)) - S_t(1 - \Phi(d_1))$$

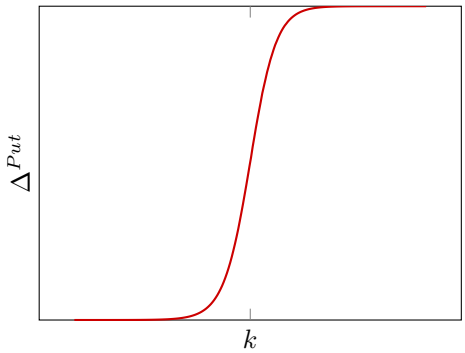
Greeks of the Put Option:

- **Delta:** Taking the derivative of the Put-Call parity with respect to S_t gives $\Delta^{Call} - \Delta^{Put} = 1$. Therefore:

  $$\Delta^{Put} = \Delta^{Call} - 1 = \Phi(d_1) - 1$$

  The Put Delta is always negative (between -1 and 0), reflecting the need to short the underlying to hedge a short Put position.

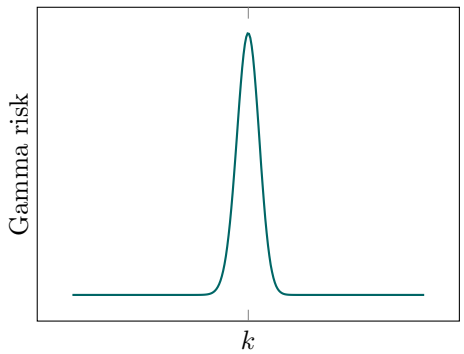
Put Delta $\Delta^{Put}(k)$



- **Gamma (Γ^{Put}):** Since Δ^{Put} differs from Δ^{Call} only by a constant (-1), their derivatives are identical.

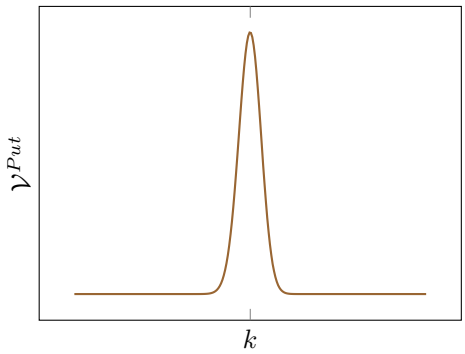
  $$\Gamma^{Put} = \Gamma^{Call} > 0$$

Put Gamma $\Gamma^{Put}(k)$



- **Vega ($\mathcal{V}^{Put}$):** Since the forward components (S_t and $Ke^{-r(T-t)}$) have zero Vega, $\mathcal{V}^{Put} = \mathcal{V}^{Call} > 0$.

Put Vega $\mathcal{V}^{Put}(k)$



2.6.2 Portfolio Immunization (Delta, Gamma, Vega Neutrality)

Traders seek to "immunize" their global portfolio F against overnight market shocks by achieving a neutral position (Greeks equal to zero).

1. Delta Neutrality ($\Delta^{Total} = 0$): To neutralize a non-zero Δ^F, we must add a financial instrument sensitive to the underlying S. The riskless asset is useless ($\Delta = 0$). Buying the underlying itself ($\Delta = 1$) is effective but extremely capital-intensive. The optimal, cost-effective instrument is the **Forward contract** ($\Delta^{Forward} = 1$). We construct a new portfolio $F^{Tot} = F + \alpha \cdot Forward$.

$$\Delta^{F^{Tot}} = \Delta^F + \alpha \cdot 1 = 0 \implies \alpha = -\Delta^F$$

2. Gamma Neutrality ($\Gamma^{Total} = 0$): If the portfolio is Delta-neutral but $\Gamma^F \neq 0$, the Delta will change rapidly, requiring frequent rebalancing. We cannot use Forwards to hedge Gamma because $\Gamma^{Forward} = 0$ (it is a linear contract). We must introduce a non-linear instrument, such as a **Call option** ($\Gamma^{Call} > 0$).

$$F^{Tot} = F_{old} + \beta \cdot Call(S, K)$$

$$\Gamma^{F^{Tot}} = \Gamma^{F_{old}} + \beta \cdot \Gamma^{Call} = 0 \implies \beta = -\frac{\Gamma^{F_{old}}}{\Gamma^{Call}}$$

Warning: Adding the Call alters the previously neutralized Delta. Therefore, Delta and Gamma neutrality must be solved jointly as a system of equations.

3. Vega Neutrality ($\mathcal{V}^{Total} = 0$): To also immunize against volatility shocks, we must add another independent instrument sensitive to Vega. We add another Call option with a significantly different strike K_2 to avoid collinearity:

$$F^{Total} = F + \alpha \cdot Forward + \beta_1 \cdot Call(S, K_1) + \beta_2 \cdot Call(S, K_2)$$

We solve a 3×3 system to find α, β_1, β_2 to simultaneously zero out Delta, Gamma, and Vega.

2.6.3 The Black-Scholes PDE and the Sign of Gamma

From the Black-Scholes PDE for a Delta-neutral portfolio ($\Delta = 0$), we get the relation:

$$\Theta + \mu S \Delta + \frac{1}{2}\sigma^2 S^2 \Gamma - rF = 0 \implies \Theta + \frac{1}{2}\sigma^2 S^2 \Gamma = rF$$

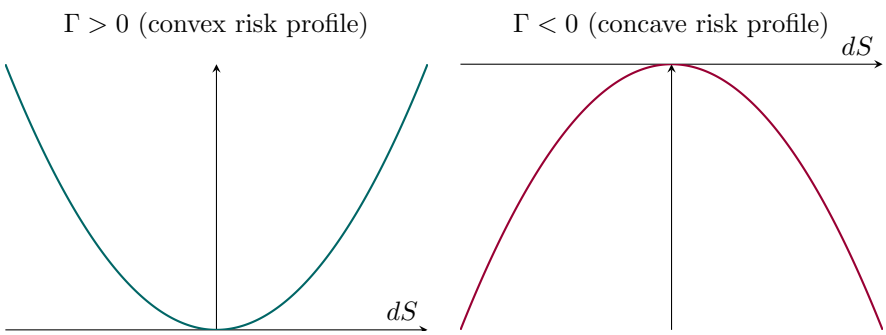
This implies an inverse relationship: if $\Gamma > 0$, then typically $\Theta < 0$.

Equation of a Δ-Neutral Portfolio

From a P&L (Profit and Loss) perspective, the dynamic portfolio variation is roughly

$$dP\&L \approx \Theta dt + \frac{1}{2}\Gamma (dS)^2$$

This type of approach is useful in practice to neutralize the portfolio against the fluttuations of the underlying, but it is not exact and does not capture the full dynamics of the option price (e.g., it ignores Vega risk and higher-order terms).



- $\Gamma > 0$ **(e.g., long Calls/Puts):** The quadratic term is positive. The portfolio is safe against large market crashes or booms (bounded losses, unlimited upside).
- $\Gamma < 0$**:** The quadratic term is negative. The portfolio faces potential **unbounded losses** for huge variations of the underlying. This is the "nightmare" scenario for a trader.

2.6.4 Digital Options (or Binary)

A Digital Call option pays a fixed amount (e.g., 1) if $S_T \geq K$, and 0 otherwise. Its payoff is the indicator function:

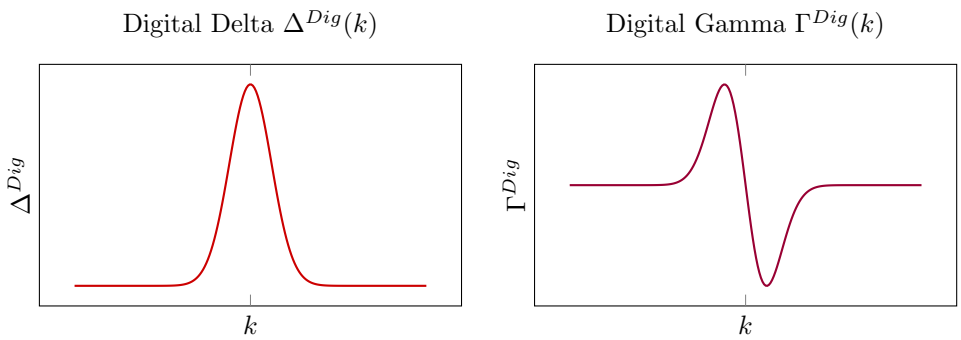
$$Payoff_{Dig} = \mathbb{I}_{S_T \geq K}$$

Pricing: The price is the discounted risk-neutral expected value of the indicator function, which is simply the risk-neutral probability that the option finishes in the money. This probability is exactly the $\Phi(d_2)$ term from the standard Black-Scholes formula:

$$Dig_t = e^{-r(T-t)}\Phi(d_2)$$

The Hedging Nightmare ($\Gamma < 0$): While pricing is trivial, hedging a Digital option is practically impossible.

- **Delta:** Δ^{Dig} behaves like the Gamma of a Vanilla call (a sharp peak around the strike). It requires massive and instantaneous buying/selling of the underlying just before maturity if the price is near the strike, which is unrealistic.
- **Gamma:** Γ^{Dig} is the derivative of the peak, resulting in a shape that plunges into negative territory. $\Gamma^{Dig} < 0$ implies that holding short/long digital components exposes the trader to severe, unhedgeable gap risks and potential unbounded losses if the underlying price jumps across the strike at maturity.



Solution to $\Gamma < 0$: To mitigate this, traders approximate the discontinuous payoff using a "Call Spread" (a portfolio of a long Call at K_1 and a short Call at K_2), sacrificing exact payoff replication for a manageable, continuous Greeks profile.

Practical Structuring Example: Suppose a client asks for a Digital option with strike $K = 100$ and a payout of 1,000 €. Instead of issuing the pure digital derivative, the investment bank structures a Call Spread with $K_1 = 99.5$ and $K_2 = 100.5$. The spread width is 1. To replicate a 1,000 € payout, the bank buys 1,000 units of the 99.5 Call and sells 1,000 units of the 100.5 Call.

If $S_T < 99.5$: both options expire OTM, so payout = 0. If $S_T > 100.5$: both options expire ITM, and the portfolio value is exactly $(S_T - 99.5) - (S_T - 100.5) = 1$ per unit. Multiplying by 1,000 contracts, the payout is locked at 1,000 €. If $99.5 \leq S_T \leq 100.5$: in this narrow window (the spread slope), the payoff is not a pure jump (0 or 1), but increases linearly.

Immunization Result: Through this overlay, the institution deliberately sacrifices exact replication, effectively smoothing the vertical jump in the payoff. This engineering trade-off is crucial because it transforms pathological Greeks into continuous and manageable curves, places an upper cap on Delta and Gamma fluctuations, and makes hedging mathematically and financially sustainable.

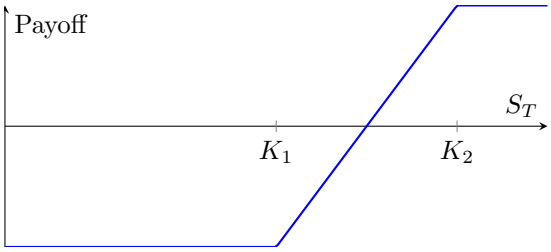
2.7 Option Strategies, Dividends, and Implied Volatility

2.7.1 Static Option Strategies

Traders construct static portfolios combining standard European options (Vanilla) to bet on specific market views while capping potential losses, that useful in cases where the trader has a directional view but wants to limit risk, or when they want to express a view on volatility without taking a directional bet.

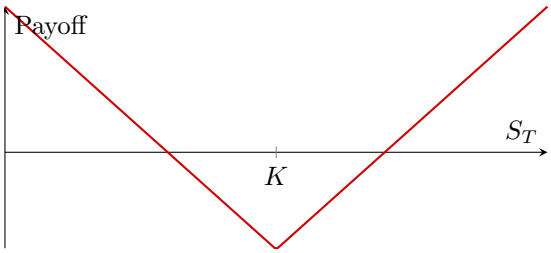
- **Bull Spread:** A bet on a moderate market increase. Constructed by going Long a Call at strike K_1 and Short a Call at strike K_2 (with $K_1 < K_2$). This caps both the maximum loss (the net premium paid) and the maximum profit (above K_2).

Bull Spread (schematic)



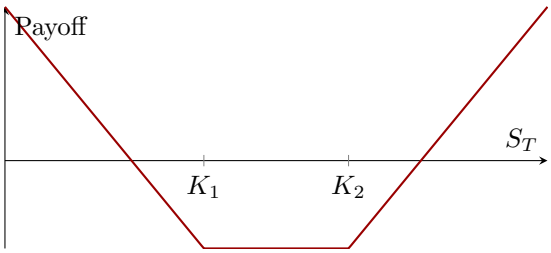
- **Straddle:** A bet on high volatility (a large movement in either direction). Constructed by going Long a Call and Long a Put at the exact same strike K. The trader profits if the underlying moves significantly away from K, offsetting the cost of buying both options.

Straddle (schematic)



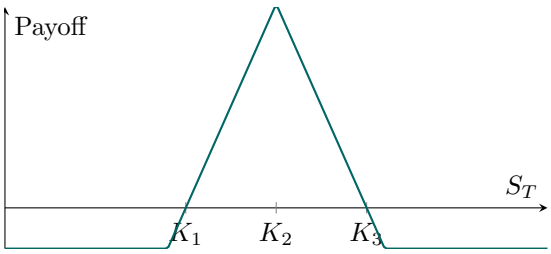
A common two-strike variant is the **Strangle** (sometimes called an asymmetric straddle): Long a Put at K_1 and Long a Call at K_2, with $K_1 < K_2$.

Strangle (schematic)



- **Butterfly Spread:** A bet on low volatility (market staying near a specific level). Constructed by going Long 1 Call at K_1, Short 2 Calls at K_2, and Long 1 Call at K_3 (where K_2 is in the middle). It yields maximum profit if $S_T = K_2$ and has strictly bounded losses.

Butterfly Spread (schematic)



2.7.2 Decomposing Complex Payoffs

A generic, complex, or discontinuous payoff can be priced and hedged by decomposing it into a linear combination of basic financial building blocks.

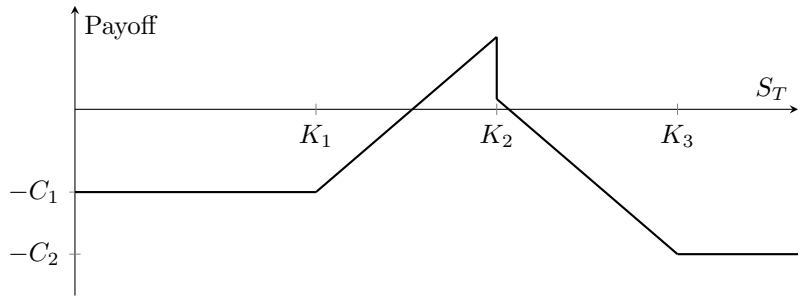
Example Decomposition: Suppose we have a payoff that starts at $-C_1$, becomes a straight line with slope $+1$ at K_1, has a downward jump at K_2, changes to slope -1, and flattens at K_3. We decompose it as follows:

$$Payoff_T = \begin{cases} -C_1 & \text{if } S_T \in (0, K_1) \\ (S_T - K_1) - C_1 & \text{if } S_T \in [K_1, K_2) \\ (K_3 - S_T) - C_2 & \text{if } S_T \in [K_2, K_3) \\ -C_2 & \text{if } S_T \in [K_3, +\infty) \end{cases}$$

As requested in the problem we need to find: the schematic of the payoff, the decomposition into building blocks, the total pricing formula, and the total Delta of the global position.

1. Schematic of the Payoff

Target payoff for decomposition (schematic)



2. Payoff Decomposition To price this contract, we must decompose the piecewise function into a linear combination of standard financial building blocks (Zero-Coupon Bonds, European Calls, and Digital Options).

We start from the left ($S_T = 0$) and move rightward:

1. **Base Level:** We start with a constant offset $-C_1$.
2. **Slope $+1$ at K_1:** We add a Long Call at strike K_1, adding $(S_T - K_1)^+$.
3. **Discontinuous Jump at K_2:** We evaluate the right limit minus the left limit at K_2:

   $$Jump = \lim_{S \to K_2^+} (K_3 - S - C_2) - \lim_{S \to K_2^-} (S - K_1 - C_1) = (K_3 - K_2 - C_2) - (K_2 - K_1 - C_1)$$

   We manage this gap using a Digital Call option at K_2: $Jump \cdot \mathbb{I}_{S_T \geq K_2}$.
4. **Slope Reversal at K_2:** The slope changes from $+1$ to -1. We need -2 units of a Call at K_2: $-2(S_T - K_2)^+$.
5. **Flattening the Slope at K_3:** The slope changes from -1 to 0. We add $+1$ unit of a Call at K_3: $+(S_T - K_3)^+$.

The total terminal payoff can be expressed as:

$$Payoff_T = -C_1 + (S_T - K_1)^+ + [Jump]\mathbb{I}_{S_T \geq K_2} - 2(S_T - K_2)^+ + (S_T - K_3)^+$$

3. Total Pricing Formula Using the risk-neutral valuation framework, the price at time t is the discounted expected value of the payoff under the risk-neutral measure $\mathbb{Q}$:

$$Price_t = e^{-r(T-t)} \mathbb{E}_t^{\mathbb{Q}}[Payoff_T]$$

By linearity of the expectation operator, we substitute the Black-Scholes formulas for each component:

$$\begin{aligned} Price_t = &- C_1 e^{-r(T-t)} + \left(S_t \Phi(d_1^{K_1}) - K_1 e^{-r(T-t)} \Phi(d_2^{K_1}) \right) + [Jump] e^{-r(T-t)} \Phi(d_2^{K_2}) \\ &- 2 \left(S_t \Phi(d_1^{K_2}) - K_2 e^{-r(T-t)} \Phi(d_2^{K_2}) \right) + \left(S_t \Phi(d_1^{K_3}) - K_3 e^{-r(T-t)} \Phi(d_2^{K_3}) \right) \end{aligned}$$

Where d_1 and d_2 for each generic strike K_i are defined as:

$$d_{1,2}^{K_i} = \frac{\ln\left(\frac{S_t}{K_i}\right) + \left(r \pm \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}$$

4. Total Delta (Δ_t) The Delta of the global position is the linear combination of the Deltas of the individual building blocks. The constant term (Zero-Coupon Bond) has zero Delta.

$$\Delta_t = 0 + \Phi(d_1^{K_1}) + [Jump]\Delta^{Dig(K_2)} - 2\Phi(d_1^{K_2}) + \Phi(d_1^{K_3})$$

Note: The term $\Delta^{Dig(K_2)}$ represents the Delta of the Digital Option, which is mathematically related to the Gamma of a standard Vanilla Call and is the main source of hedging instability (unbounded risk) around the strike K_2.

2.7.3 Black-Scholes with Continuous Proportional Dividends

When the underlying asset pays a continuous dividend yield q (proportional to the asset price, $dD_t = qS_t dt$), holding the asset generates an extra cash flow.

The New Self-Financing Condition: The budget equation must account for the dividend received between $t - \Delta t$ and t. Taking the continuous limit, the portfolio dynamics become:

Self-Financing Portfolio with Dividends

$$dV_t = h(t)dS_t + h(t)dD_t = h(t)(dS_t + qS_t dt)$$

The Normalized Gain Process: Under the risk-neutral measure $\mathbb{Q}$, the simple discounted asset $S_t e^{-rt}$ is no longer a martingale. Instead, the **Normalized Gain Process**, which includes the reinvested discounted dividends, is a $\mathbb{Q}$-martingale:

$$M_t = S_t e^{-rt} + \int_0^t e^{-rs} dD_s$$

The Modified PDE and Dynamics: Matching the portfolio dynamics with Itô's Lemma on the pricing function $F(t, S_t)$ yields a modified Black-Scholes PDE where the drift r is replaced by $r - q$:

$$\frac{\partial F}{\partial t} + (r-q)S\frac{\partial F}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 F}{\partial S^2} - rF = 0$$

Under $\mathbb{Q}$, the asset dynamics are:

$$\frac{dS_t}{S_t} = (r-q)dt + \sigma dW_t^{\mathbb{Q}}$$

The Pricing Shortcut: Because the $\mathbb{Q}$-dynamics merely shift the drift, the solution to the PDE is identical to the standard Black-Scholes formula, provided we replace the current stock price S_t with $S_t e^{-q(T-t)}$.

$$Call = S_t e^{-q(T-t)}\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2)$$

Where $d_{1,2}$ are computed using the modified underlying $S_t e^{-q(T-t)}$.

2.7.4 Implied Volatility and The Smile Effect

In the market, the option price is observable, but the instantaneous volatility σ is not.

Implied Volatility

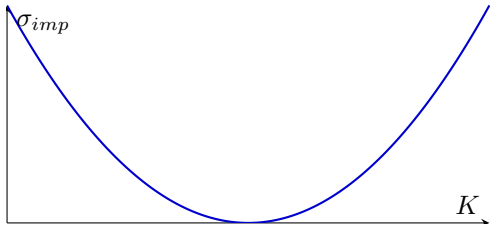
The **Implied Volatility** σ_{imp} is the unique value of σ that, when plugged into the Black-Scholes formula, yields the observed market price of the option.

By setting the Black-Scholes formula equal to the empirical market price, traders invert the formula to find the implied volatility. If the Black-Scholes model were perfectly true (log-normal distribution, constant σ), plotting σ_{imp} against different strikes K would result in a perfectly flat horizontal line.

The Smile and Skew (Empirical Reality): Empirically, the plot is not flat, proving the Black-Scholes model is flawed:

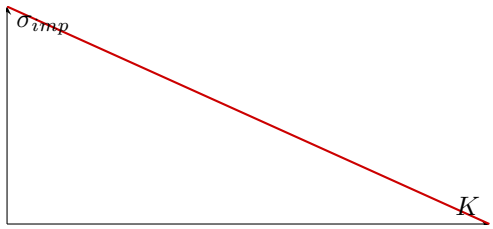
- **Smile Effect:** In Forex markets, σ_{imp} forms a U-shape ("smile"), meaning Deep-Out-Of-The-Money and Deep-In-The-Money options are priced higher than At-The-Money options (reflecting fatter tails in the actual distribution).

Smile effect (schematic)



- **Skew/Smirk Effect:** In Equity markets, the curve is asymmetric, sloping downward. Out-Of-The-Money puts are priced with a much higher implied volatility than Out-Of-The-Money calls, reflecting the market's fear of sudden crashes.

Skew/smirk effect (schematic)



Term Structure: For short maturities (e.g., 1 month), the smile is very pronounced. For long maturities, the Central Limit Theorem takes over, making the distribution more log-normal, and the smile flattens out.

2.8 Multidimensional Market Model and Correlation

1. Asset Dynamics and Scalar Volatility In a realistic market, we model n risky assets driven by d independent Brownian motions $W_t = (W_t^1, \ldots, W_t^d)$. The dynamics of the i-th asset are:

$$\frac{dS_t^i}{S_t^i} = \mu_t^i dt + \sum_{j=1}^{d} \sigma_t^{ij} dW_t^j$$

To simplify pricing for a single asset, we can collapse the linear combination of Brownian motions into a single, scalar Brownian motion $\hat{W}_t^i$ by defining a **scalar volatility** Σ_t^i:

$$\Sigma_t^i = \sqrt{\sum_{j=1}^{d} (\sigma_t^{ij})^2} \implies \frac{dS_t^i}{S_t^i} = \mu_t^i dt + \Sigma_t^i d\hat{W}_t^i$$

2. Correlation and The Itô Multiplication Table The historical correlation is estimated using time series data:

$$\hat{\rho}(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

However, in stochastic calculus, the **instantaneous correlation** is derived from the covariation of the assets. We compute this using the extended Itô multiplication table for independent Brownian motions:

Using this table, the covariation between two assets is:

$$d\langle S^1, S^2 \rangle_t = S_t^1 S_t^2 (\sigma^{11}\sigma^{21} + \sigma^{12}\sigma^{22}) dt$$

$\times$	dt	dW_t^1	dW_t^2
dt	0	0	0
dW_t^1	0	dt	0
dW_t^2	0	0	dt

3. A practical approach: The Triangular Formulation (Cholesky Decomposition) Instead of defining an arbitrary $n \times d$ matrix, quants build the market step-by-step using a triangular structure to properly assign correlations without altering previously modeled assets:

$$\begin{aligned}
\text{Asset 1: } \frac{dS_t^1}{S_t^1} &= \mu^1 dt + \sigma^1 dW_t^1 \\
\text{Asset 2: } \frac{dS_t^2}{S_t^2} &= \mu^2 dt + \sigma^2 \left(\rho_{1,2} dW_t^1 + \sqrt{1-\rho_{1,2}^2} dW_t^2\right) \\
\text{Asset 3: } \frac{dS_t^3}{S_t^3} &= \mu^3 dt + \sigma^3 \left(\rho_{1,3} dW_t^1 + \hat{\rho} dW_t^2 + \sqrt{1-\rho_{1,3}^2-\hat{\rho}^2} dW_t^3\right)
\end{aligned}$$

Note: For Asset 3, $\hat{\rho}$ is not directly the correlation $\rho_{2,3}$. The true correlation $\rho_{2,3}$ must be found by computing the covariation $d\langle S^2, S^3\rangle_t$ and solving for $\hat{\rho}$.

4. No Arbitrage and Completeness Let Σ be the $n \times d$ volatility matrix. To change the drift from μ to the risk-free rate r under the Equivalent Martingale Measure $\mathbb{Q}$, we must find a Market Price of Risk vector $\lambda_t \in \mathbb{R}^d$ such that:

$$\Sigma \lambda_t = r\mathbf{1} - \mu$$

- **No Arbitrage (Existence of $\mathbb{Q}$):** The system must have at least one solution. The vector $(r-\mu)$ must be in the image of Σ, requiring $rank(\Sigma) = n$. This implies $\mathbf{d} \geq \mathbf{n}$ (we need at least as many risk sources as assets).
- **Completeness (Uniqueness of $\mathbb{Q}$):** The solution must be unique to perfectly replicate any payoff. The kernel of Σ must be trivial ($Ker(\Sigma) = \{0\}$), which implies $\mathbf{d} \leq \mathbf{n}$.
- **Conclusion:** A market is both arbitrage-free and complete if and only if $\mathbf{d} = \mathbf{n}$ (and Σ is invertible). Furthermore, the market must have **no jumps** (the filtration is generated solely by continuous Brownian motions).

2.8.1 Pricing a Multi-Asset Payoff: The Exchange Option

The Exchange Option (Margrabe's Option) gives the right to exchange asset S^2 for asset S^1 at maturity T. The payoff is:

$$Payoff_T = (S_T^1 - S_T^2)^+$$

The PDE Approach and Dimensionality Reduction: Let $F(t, x_1, x_2)$ be the 2D pricing function. The associated 2D Black-Scholes PDE under $\mathbb{Q}$ is:

$$\frac{\partial F}{\partial t} + rx_1 \frac{\partial F}{\partial x_1} + rx_2 \frac{\partial F}{\partial x_2} + \frac{1}{2} x_1^2 \sigma_1^2 \frac{\partial^2 F}{\partial x_1^2} + \frac{1}{2} x_2^2 \sigma_2^2 \frac{\partial^2 F}{\partial x_2^2} + x_1 x_2 \rho \sigma_1 \sigma_2 \frac{\partial^2 F}{\partial x_1 \partial x_2} - rF = 0$$

We exploit the linear homogeneity of the payoff: $(x_1 - x_2)^+ = x_2(x_1/x_2 - 1)^+$. We define a new state variable $z = x_1/x_2$ and set $F(t, x_1, x_2) = x_2 G(t, z)$. Applying the chain rule, we compute the partial derivatives:

$$\begin{aligned}
\frac{\partial F}{\partial x_1} &= \frac{\partial G}{\partial z} \\
\frac{\partial F}{\partial x_2} &= G - z\frac{\partial G}{\partial z} \\
\frac{\partial^2 F}{\partial x_1^2} &= \frac{1}{x_2}\frac{\partial^2 G}{\partial z^2} \\
\frac{\partial^2 F}{\partial x_2^2} &= \frac{z^2}{x_2}\frac{\partial^2 G}{\partial z^2} \\
\frac{\partial^2 F}{\partial x_1 \partial x_2} &= -\frac{z}{x_2}\frac{\partial^2 G}{\partial z^2}
\end{aligned}$$

Substituting these into the 2D PDE and dividing by x_2, all terms involving r and the first derivative with respect to z perfectly cancel out. The equation reduces to a 1D Black-Scholes PDE with zero drift ($r = 0$) and a combined volatility σ_z:

$$\implies \frac{\partial G}{\partial t} + \frac{1}{2} z^2 \sigma_z^2 \frac{\partial^2 G}{\partial z^2} = 0 \quad \text{where} \quad \sigma_z = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$$

Solving for $G(t, z)$ using the standard formula (with $K = 1, r = 0$) and multiplying back by x_2, we obtain the exact:

Margrabe's Formula for the Exchange Option

The price of the exchange option at time t is given by:

$$Price_t = S_t^1 \Phi(d_1) - S_t^2 \Phi(d_2) \quad \text{where} \quad d_{1,2} = \frac{\ln\left(\frac{S_t^1}{S_t^2}\right) \pm \frac{1}{2}\sigma_z^2 (T - t)}{\sigma_z \sqrt{T - t}}$$

2.8.2 Intuition Behind Margrabe's Formula

Comparing the exact formula with a "naive" application of Black-Scholes reveals two key insights:

- **The Disappearance of the Interest Rate (r):** In a standard option, the strike K is a deterministic cash amount that must be discounted by $e^{-r(T-t)}$. Here, the "strike" is S_t^2, which is a traded asset subject to market risk. Its current market price already incorporates all necessary discounting, making r disappear from the formula.
- **The Crucial Role of Correlation (ρ):** The volatility of the ratio $z = S^1/S^2$ heavily depends on how the two assets move together:
  - **High Positive Correlation ($\rho \to 1$):** The assets move together, the spread remains stable, and σ_z drops. The option is cheaper because a massive divergence is unlikely.
  - **High Negative Correlation ($\rho \to -1$):** The assets move in opposite directions, meaning the spread can explode. The combined volatility σ_z approaches $\sigma_1 + \sigma_2$, making the option highly valuable.

2.9 The Change of Numéraire Technique

The standard risk-neutral pricing framework relies on using the riskless bank account $B_t = e^{\int_0^t r_s ds}$ as the numéraire. Under the risk-neutral measure $\mathbb{Q}$, any tradable asset discounted by B_t is a martingale.

Numéraire

A **numéraire** is a strictly positive traded asset N_t used as a benchmark to express the value of other assets. The choice of numéraire determines the equivalent martingale measure under which discounted asset prices are martingales.

When we change the numéraire from B_t to N_t, we move from the probability measure $\mathbb{Q}$ to a new equivalent martingale measure $\mathbb{Q}^N$. Under $\mathbb{Q}^N$, the price of any asset X_t relative to N_t is a martingale:

$$\frac{X_t}{N_t} = \mathbb{E}_t^{\mathbb{Q}^N}\left[\frac{X_T}{N_T}\right] \implies X_t = N_t \mathbb{E}_t^{\mathbb{Q}^N}\left[\frac{X_T}{N_T}\right]$$

Note: This equation reveals the power of the technique. If an analyst needs to price a derivative whose terminal payoff (X_T) is complex, and cleverly chooses a numéraire N_t such that the ratio $\frac{X_T}{N_T}$ simplifies dramatically, the expected-value calculation becomes elementary. Moreover, the stochastic discount factor (which creates major analytical difficulties) disappears inside the expectation and is replaced by the observed spot value of the new numéraire (N_t).

The Radon-Nikodym derivative: thanks to the Radon-Nikodym derivative we can explicitly compute the change of measure from $\mathbb{Q}$ to $\mathbb{Q}^N$:

$$\left.\frac{d\mathbb{Q}^N}{d\mathbb{Q}}\right|_t = \frac{N_t/N_0}{B_t/B_0}$$

$$\begin{tikzcd}
 & \mathbb{P} \text{ (Real World)} \arrow[dl, "\text{Girsanov (Risk Premium)}"'] \arrow[dr, dashed] & \\
\mathbb{Q} \text{ (Numéraire: } B_t) \arrow[rr, "\text{Change of Numéraire}"] & & \mathbb{Q}^N \text{ (Numéraire: } N_t)
\end{tikzcd}$$

Figure 2.1: Transitions between Probability Measures.

2.9.1 Application 1: The Exchange Option

Consider an exchange option with payoff at maturity T:

$$\text{Payoff}_T = (S_1(T) - S_2(T))^+$$

Instead of solving a 2-dimensional PDE, we change the numéraire to $S_2(t)$. We define a new probability measure $\mathbb{Q}^{S_2}$. The pricing formula becomes:

$$V_t = S_2(t)\mathbb{E}_t^{\mathbb{Q}^{S_2}}\left[\frac{(S_1(T) - S_2(T))^+}{S_2(T)}\right] = S_2(t)\mathbb{E}_t^{\mathbb{Q}^{S_2}}\left[(Z_T - 1)^+\right]$$

where $Z_t = \frac{S_1(t)}{S_2(t)}$. This matches the payoff of a standard call option on Z_t with strike $K = 1$ and zero interest rate.

Dynamics of Z_t To apply the Black-Scholes formula, we need the volatility of Z_t under $\mathbb{Q}^{S_2}$. Applying Itô's Lemma to the ratio $Z_t = S_1(t) \cdot (S_2(t))^{-1}$:

$$dZ_t = Z_t(\dots)dt + Z_t\left(\sigma_{11} - \sigma_{21}\right)dW_1^{\mathbb{Q}^{S_2}} + Z_t\left(\sigma_{12} - \sigma_{22}\right)dW_2^{\mathbb{Q}^{S_2}}$$

Under $\mathbb{Q}^{S_2}$, Z_t is a martingale, so the drift is exactly zero. The scalar volatility σ_Z is given by the quadratic variation:

$$\sigma_Z = \sqrt{(\sigma_{11} - \sigma_{21})^2 + (\sigma_{12} - \sigma_{22})^2}$$

Using the Black-Scholes formula for a zero-rate environment ($r = 0$, $K = 1$):

$$V_t = S_2(t)\left[Z_t\Phi(d_1) - 1 \cdot \Phi(d_2)\right] = S_1(t)\Phi(d_1) - S_2(t)\Phi(d_2) \quad \text{where} \quad d_{1,2} = \frac{\ln(S_1(t)/S_2(t)) \pm \frac{1}{2}\sigma_Z^2(T-t)}{\sigma_Z\sqrt{T-t}}$$

2.9.2 Application 2: Stochastic Interest Rates

When the short rate r_t is stochastic, the standard risk-neutral pricing formula is problematic because the discount factor and the asset payoff are correlated:

$$V_t = \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_s ds}\text{Payoff}_T\right] \neq \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_s ds}\right]\mathbb{E}_t^{\mathbb{Q}}\left[\text{Payoff}_T\right]$$

1. The T-Forward Measure ($\mathbb{Q}^T$): To disentangle the discounting, we choose the Zero-Coupon Bond $P(t,T)$ as the numéraire. $P(t,T)$ is the price at t of a contract paying 1 at maturity T. Switching to the **T-Forward Risk-Neutral Measure** $\mathbb{Q}^T$, the pricing formula becomes:

$$V_t = P(t,T)\mathbb{E}_t^{\mathbb{Q}^T}\left[\frac{\text{Payoff}_T}{P(T,T)}\right] = P(t,T)\mathbb{E}_t^{\mathbb{Q}^T}\left[\text{Payoff}_T\right]$$

Since $P(T,T) = 1$, the stochastic discount factor vanishes inside the expectation, effectively reducing the problem to a zero-interest-rate valuation under a new measure.

2. Call Option with Stochastic Rates Consider a standard European Call Option with payoff $(S_T - K)^+ = S_T \mathbf{1}_{\{S_T \geq K\}} - K\mathbf{1}_{\{S_T \geq K\}}$. We split the pricing into two parts using two different numéraires:

$$C_t = \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_s ds} S_T \mathbf{1}_{\{S_T \geq K\}}\right] - \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_s ds} K\mathbf{1}_{\{S_T \geq K\}}\right]$$

1. **The Asset Measure ($\mathbb{Q}^S$)** For the first term, we use the underlying asset S_t as the numéraire. The associated measure is $\mathbb{Q}^S$. The terminal value of the numéraire is S_T, so it cancels out:

$$\text{Term}_1 = S_t \mathbb{E}_t^{\mathbb{Q}^S}\left[\mathbf{1}_{\{S_T \geq K\}}\right] = S_t \mathbb{Q}_t^S(S_T \geq K)$$

2. **The Forward Measure ($\mathbb{Q}^T$)** For the second term, we use the Zero-Coupon Bond $P(t,T)$ as the numéraire, moving to the T-forward measure $\mathbb{Q}^T$:

$$\text{Term}_2 = KP(t,T)\mathbb{E}_t^{\mathbb{Q}^T}\left[\mathbf{1}_{\{S_T \geq K\}}\right] = KP(t,T)\mathbb{Q}_t^T(S_T \geq K)$$

Final Pricing Formula Combining both parts yields a Black-Scholes-like formula generalized for stochastic interest rates:

$$C_t = S_t \mathbb{Q}_t^S(S_T \geq K) - KP(t,T)\mathbb{Q}_t^T(S_T \geq K)$$

Here, the complex expectations have been successfully reduced to computing the probabilities of the option ending in-the-money under two distinctly chosen probability measures.

2.10 Itô's Quotient Rule and the Forward Price

When dealing with stochastic interest rates, we need to switch to the T-forward measure $\mathbb{Q}^T$. This requires computing the dynamics of the forward price $Z_t = \frac{S_t}{P(t,T)}$.

Consider two general Itô processes:

$$\frac{dX_1}{X_1} = (\dots)dt + \sigma_1 dW_t$$
$$\frac{dX_2}{X_2} = (\dots)dt + \sigma_2 dW_t$$

To find the dynamics of the ratio, we apply Itô's product rule to $X_1 \cdot \frac{1}{X_2}$:

$$d\left(\frac{X_1}{X_2}\right) = \frac{1}{X_2}dX_1 + X_1 d\left(\frac{1}{X_2}\right) + d\left\langle X_1, \frac{1}{X_2}\right\rangle$$

Focusing only on the diffusion (Brownian) terms, the differential of the inverse is $d(1/X_2) = -\frac{1}{X_2^2}dX_2 + \dots$, leading to:

$$d\left(\frac{X_1}{X_2}\right) = (\dots)dt + \frac{X_1}{X_2}(\sigma_1 - \sigma_2)dW_t$$

Thus, the volatility of the ratio involves the difference of the individual volatilities. Assuming the volatility of $Z_t = S_t/P(t,T)$, denoted as $\sigma_Z = \sigma_S - \sigma_P$ where σ_P is the volatility of the zero-coupon bond, is deterministic, Z_t is a martingale under $\mathbb{Q}^T$:

$$\frac{dZ_t}{Z_t} = \sigma_Z dW_t^{\mathbb{Q}^T}$$

Thanks to the deterministic volatility, Z_t is lognormal under $\mathbb{Q}^T$, allowing us to apply the Black-Scholes formula in this new measure.

2.11 Black-Scholes Call Price with Stochastic Rates

The price of a European call option is split into two expected values under different measures:

$$Call_t = S_t \mathbb{Q}_t^S(S_T \geq K) - P(t,T)K\mathbb{Q}_t^T(S_T \geq K)$$

2.11.1 Computing the Forward Measure Part ($\mathbb{Q}^T$)

Since $P(T,T) = 1$, we have:

$$\mathbb{Q}_t^T(S_T \geq K) = \mathbb{Q}_t^T\left(\frac{S_T}{P(T,T)} \geq K\right) = \mathbb{Q}_t^T(Z_T \geq K)$$

Solving the driftless geometric Brownian motion for Z_T:

$$Z_T = Z_t \exp\left(-\frac{1}{2}\int_t^T ||\sigma_Z(s)||^2 ds + \int_t^T \sigma_Z(s) dW_s^{\mathbb{Q}^T}\right)$$

Because σ_Z is deterministic, the stochastic integral is normally distributed, making Z_T lognormal. This yields $\Phi(d_2)$, where:

$$d_2 = \frac{\ln\left(\frac{S_t}{KP(t,T)}\right) - \frac{1}{2}\int_t^T ||\sigma_Z(s)||^2 ds}{\sqrt{\int_t^T ||\sigma_Z(s)||^2 ds}}$$

2.11.2 Computing the Stock Measure Part ($\mathbb{Q}^S$)

For the first term, the underlying asset S_t is the numéraire. Under $\mathbb{Q}^S$, the inverse forward price $\frac{1}{Z_t}$ is a martingale:

$$\frac{d(1/Z_t)}{1/Z_t} = 0 \cdot dt - \sigma_Z dW_t^{\mathbb{Q}^S}$$

We rewrite the probability as $\mathbb{Q}_t^S\left(\frac{1}{Z_T} \leq \frac{1}{K}\right)$. Solving for $1/Z_T$:

$$\frac{1}{Z_T} = \frac{1}{Z_t}\exp\left(-\frac{1}{2}\int_t^T ||\sigma_Z||^2 ds - \int_t^T \sigma_Z dW_s^{\mathbb{Q}^S}\right)$$

Evaluating this yields $\Phi(d_1)$, where $d_1 = d_2 + \sqrt{\int_t^T ||\sigma_Z(s)||^2 ds}$. This gives the **Generalized Black-Scholes formula**:

$$Call_t = S_t\Phi(d_1) - P(t,T)K\Phi(d_2)$$

2.12 The Vasiček Interest Rate Model (1977)

Unlike stock prices, which can theoretically grow without bound, interest rates are monetary policy instruments influenced by central banks. They therefore tend to fluctuate around a long-run equilibrium level.

The Vasiček model describes the short rate r_t via a mean-reverting Gaussian SDE:

$$dr_t = (a - br_t)dt + \sigma_r dW_t^{\mathbb{Q}}$$

Solving the SDE via Integrating Factor We introduce the auxiliary variable $y_t = r_t e^{bt}$. Applying Itô's formula:

$$\begin{aligned} dy_t &= (dr_t)e^{bt} + r_t b e^{bt} dt = (a - br_t)e^{bt}dt + \sigma_r e^{bt} dW_t^{\mathbb{Q}} + br_t e^{bt} dt \\ &= ae^{bt}dt + \sigma_r e^{bt} dW_t^{\mathbb{Q}} \end{aligned}$$

Integrating both sides from 0 to t:

$$r_t e^{bt} - r_0 = \frac{a}{b}(e^{bt} - 1) + \sigma_r \int_0^t e^{bs} dW_s^{\mathbb{Q}}$$

Dividing by e^{bt} yields the explicit solution:

$$r_t = r_0 e^{-bt} + \frac{a}{b}(1 - e^{-bt}) + \sigma_r \int_0^t e^{-b(t-s)} dW_s^{\mathbb{Q}}$$

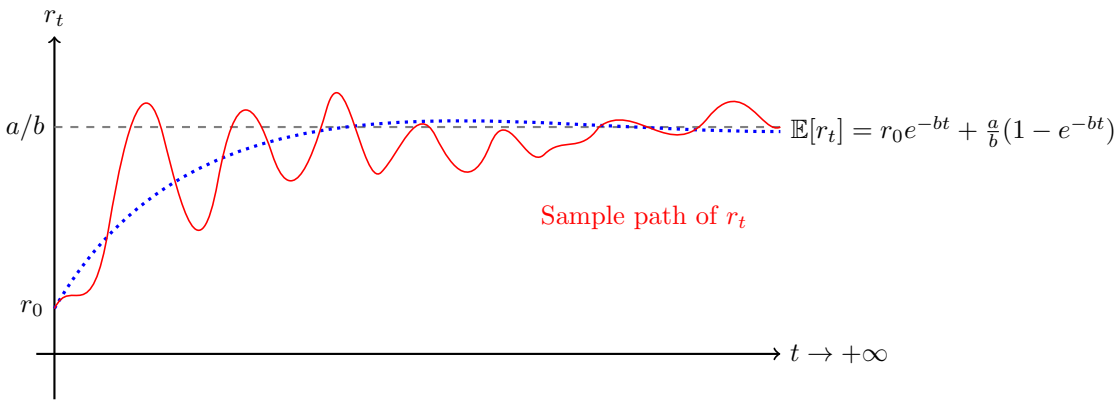


Figure 2.2: Mean reversion in the Vasiček model. The short rate fluctuates but is pulled toward the long-run mean a/b.

2.12.1 Bond Pricing and Volatility

Because r_t is normally distributed, the integral $\int_t^T r_s ds$ is also Gaussian. The zero-coupon bond price $P(t,T) = \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_s ds}\right]$ admits an affine term structure:

$$P(t,T) = \exp\{A(t,T) - B(t,T)r_t\}$$

where $B(t,T) = \frac{1}{b}(1 - e^{-b(T-t)})$ and $A(t,T)$ is a deterministic function of time.

To prove that the volatility of the bond is deterministic, we apply Itô's formula to $P(t,T)$:

$$dP = \frac{\partial P}{\partial t}dt + \frac{\partial P}{\partial r}dr_t + \frac{1}{2}\frac{\partial^2 P}{\partial r^2}(dr_t)^2$$

Computing the partial derivatives based on the affine form:

$$\begin{aligned}\frac{\partial P}{\partial r} &= -B(t,T)P(t,T) = -\frac{1}{b}(1 - e^{-b(T-t)})P(t,T)\\ \frac{\partial^2 P}{\partial r^2} &= B(t,T)^2 P(t,T)\end{aligned}$$

Substituting $dr_t = (a - br_t)dt + \sigma_r dW_t^{\mathbb{Q}}$ into the expansion and collecting the diffusion term:

$$dP = (\dots)dt + P(t,T)\left[-B(t,T)\right]\sigma_r dW_t^{\mathbb{Q}}$$

Dividing by P, we identify the bond's volatility $\sigma_P(t,T)$:

$$\frac{dP}{P} = (\dots)dt + \underbrace{\frac{\sigma_r}{b}\left(e^{-b(T-t)} - 1\right)}_{\sigma_P(t,T)} dW_t^{\mathbb{Q}}$$

Since $\sigma_P(t,T)$ only depends on time parameters and not on the random variable r_t, it is perfectly deterministic. Therefore, the generalized Black-Scholes formula is strictly valid under the Vasiček model.

2.13 Black Formula (1976): Options on Futures

A European call option on a futures contract has maturity T and strike K. The underlying futures contract $F(t,T_1)$ matures at $T_1 \geq T$. The payoff at option maturity T is:

$$Payoff_T = (F(T,T_1) - K)^+ = (S_T e^{r(T_1-T)} - K)^+ = e^{r(T_1-T)}\left(S_T - Ke^{-r(T_1-T)}\right)^+$$

This is equivalent to $e^{r(T_1-T)}$ standard calls on the asset S_T with a modified strike $\tilde{K} = Ke^{-r(T_1-T)}$. Using the generalized Black-Scholes formula on this adjusted payoff and simplifying, we obtain:

Black 76 Formula for Futures Options

The price of a European call option on a futures contract is given by:

$$Price_t = e^{-r(T-t)} \left[F(t, T_1)\Phi(d_1) - K\Phi(d_2) \right]$$

where d_1, d_2 are evaluated using the futures price $F(t, T_1)$ as the underlying asset instead of the spot price.

2.14 (FOREX) Exchange Rates and Currency Conversion

Consider two economies: a domestic economy (e.g., US, currency \$) and a foreign economy (e.g., Eurozone, currency €). We define the exchange rate X_t as the price of 1 unit of foreign currency expressed in domestic currency:

$$X_t = \text{Price of 1 € in \$}$$

If you hold 1 € and convert it to dollars, you receive X_t dollars. Note that if X_t is the EUR/USD exchange rate, $1/X_t$ is the USD/EUR exchange rate. To avoid arbitrage in a multi-currency environment, triangular relationships must hold (e.g., $X \cdot Y = Z$).

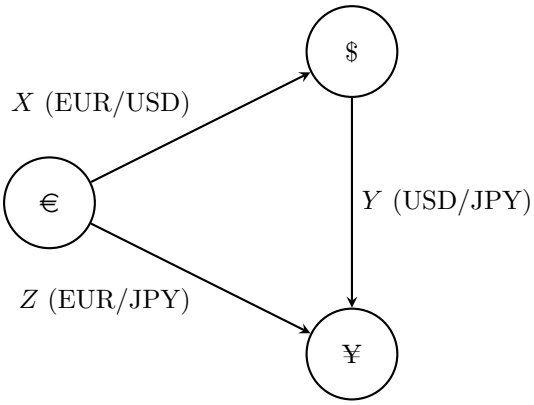


Figure 2.3: Triangular currency conversion. By no-arbitrage: $X \cdot Y = Z$.

Model Hypotheses under the Physical Measure $\mathbb{P}$ We assume the exchange rate and asset prices follow geometric Brownian motions driven by a multi-dimensional Brownian motion $W^{\mathbb{P}}$. The volatility vectors $\sigma_x, \sigma_d, \sigma_f$ are row vectors.

$$\begin{aligned}
\text{Exchange Rate:} \quad & \frac{dX_t}{X_t} = \mu_x dt + \sigma_x dW_t^{\mathbb{P}} \\
\text{Domestic Asset:} \quad & \frac{dS_d}{S_d} = \mu_d dt + \sigma_d dW_t^{\mathbb{P}} \qquad dB_d = B_d r_d dt \\
\text{Foreign Asset:} \quad & \frac{dS_f}{S_f} = \mu_f dt + \sigma_f dW_t^{\mathbb{P}} \qquad dB_f = B_f r_f dt
\end{aligned}$$

The domestic and foreign risk-free bank accounts evolve deterministically.

Crucial Assumption: Buying foreign currency and converting it yields a position that evolves at the foreign risk-free rate r_f.

Change of Measure: Domestic Risk-Neutral Pricing ($\mathbb{Q}^d$) To price options in the domestic economy, we must find a domestic risk-neutral measure $\mathbb{Q}^d$ under which **all discounted domestic assets are martingales**.

We must convert foreign assets into domestic currency to treat them as domestic tradable assets. Let $\tilde{B}_f = X_t B_f$ be the foreign bank account expressed in domestic currency. Applying Itô's product rule (skipping the cross-variation since B_f is deterministic):

$$\begin{aligned}
d\tilde{B}_f &= (dX_t)B_f + X_t(dB_f) = X_t(\mu_x dt + \sigma_x dW_t^{\mathbb{P}})B_f + X_t B_f r_f dt \\
&= \tilde{B}_f[(\mu_x + r_f)dt + \sigma_x dW_t^{\mathbb{P}}]
\end{aligned}$$

To make $\tilde{B}_f$ a martingale when discounted by B_d, its drift under $\mathbb{Q}^d$ must equal r_d. Applying Girsanov's theorem to shift the drift:

$$\frac{d\hat{B}_f}{\hat{B}_f} = r_d dt + \sigma_x dW_t^{\mathbb{Q}^d}$$

Consequently, the dynamics of the exchange rate under the domestic risk-neutral measure $\mathbb{Q}^d$ become:

$$\frac{dX_t}{X_t} = (r_d - r_f)dt + \sigma_x dW_t^{\mathbb{Q}^d}$$

Financial intuition: X_t behaves exactly like a continuous dividend-paying stock, where the foreign interest rate r_f plays the role of the continuous dividend yield q.

Similarly, the foreign stock converted to domestic currency, $\hat{S}_f = X_t S_f$, has the following dynamics under $\mathbb{Q}^d$ (summing the diffusions via Itô's product rule):

$$\frac{d\hat{S}_f}{\hat{S}_f} = r_d dt + (\sigma_f + \sigma_x) dW_t^{\mathbb{Q}^d}$$

We prove that if an investor buys the foreign stock there's a combined volatility from both the foreign stock and the exchange rate, and the expected return is the domestic risk-free rate r_d.

2.14.1 Pricing Options on Foreign Assets

A call option on the exchange rate has payoff $(X_T - K)^+$ in domestic currency. Because X_t behaves like a stock paying dividend r_f, we apply the Black-Scholes formula with dividend yield:

Black-Scholes Formula for Exchange Rate Options

The price of a European call option on the exchange rate is given by:

$$\pi_t = X_t e^{-r_f(T-t)}\Phi(d_1) - Ke^{-r_d(T-t)}\Phi(d_2) \quad \text{where} \quad d_{1,2} = \frac{\ln(X_t/K) + (r_d - r_f \pm \frac{1}{2}||\sigma_x||^2)(T-t)}{||\sigma_x||\sqrt{T-t}}$$

General Pricing PDE For a payoff dependent on $(X_T, S_d(T), \hat{S}_f(T))$, the price $\pi_t = F(t, x, s_d, \hat{s}_f)$ satisfies a 3-dimensional PDE. Due to correlated Brownian motions, the volatility of $\hat{S}_f$ squared is $||\sigma_f + \sigma_x||^2 = ||\sigma_f||^2 + ||\sigma_x||^2 + 2\sigma_f\sigma_x^\top$. The PDE includes cross-derivatives corresponding to the covariance between the assets:

$$\begin{aligned} 0 = &\frac{\partial F}{\partial t} + x(r_d - r_f)\frac{\partial F}{\partial x} + s_d r_d \frac{\partial F}{\partial s_d} + \hat{s}_f r_d \frac{\partial F}{\partial \hat{s}_f} - r_d F \\ &+ \frac{1}{2}\left\{ x^2||\sigma_x||^2 \frac{\partial^2 F}{\partial x^2} + s_d^2||\sigma_d||^2 \frac{\partial^2 F}{\partial s_d^2} + \hat{s}_f^2(||\sigma_f||^2 + ||\sigma_x||^2 + 2\sigma_f\sigma_x^\top)\frac{\partial^2 F}{\partial \hat{s}_f^2} \right\} + \text{cross-derivative terms}\ldots \end{aligned}$$

Example 1: Call on S_f Paid in Foreign Currency (The "Fake Problem") Payoff in foreign currency: $Payoff_T^f = (S_f(T) - K)^+$. Converted to domestic currency: $Payoff_T^d = X_T(S_f(T) - K)^+$. Instead of treating this as a complex multidimensional problem, we evaluate the call in the foreign economy using the foreign risk-neutral measure $\mathbb{Q}^f$ and then convert it using the spot exchange rate X_t:

$$\pi_t^d = X_t \cdot \pi_t^f = X_t \left[S_f(t)\Phi(d_1) - Ke^{-r_f(T-t)}\Phi(d_2) \right]$$

Example 2: Quanto Option (Strike in Domestic Currency) A true Quanto option compares a converted foreign asset to a domestic strike. Payoff in domestic currency:

$$PAYOFF_T^d = (X_T S_f(T) - K)^+ = (\hat{S}_f(T) - K)^+$$

This is a standard Black-Scholes call on the domestic asset $\hat{S}_f$. Using its volatility $||\sigma_f + \sigma_x||$:

$$\pi_t^d = X_t S_f(t)\Phi(d_1) - Ke^{-r_d(T-t)}\Phi(d_2) \quad \text{where} \quad d_{1,2} = \frac{\ln(X_t S_f(t)/K) + (r_d \pm \frac{1}{2}||\sigma_f + \sigma_x||^2)(T-t)}{||\sigma_f + \sigma_x||\sqrt{T-t}}$$

2.14.2 Market Note: Volatility Smile in Forex

Unlike the equity market—where the volatility smile is typically asymmetric (skewed) due to the negative leverage effect (prices drop $\implies$ volatility spikes)—the Forex volatility smile is characteristically **symmetric**. This symmetry arises because an exchange rate is a ratio between two currencies (X vs $1/X$); there is no inherent preference between the numerator and denominator. The primary challenge in advanced FX modeling is not just capturing the smile, but modeling its dynamics over time (e.g., using the Strike Stickiness Ratio, SSR).

2.15 Mathematical Foundations of Barrier Options

2.15.1 Hitting Times and Stopping Times

Barrier options are path-dependent derivatives whose payoff depends on whether the underlying asset reaches a certain level (barrier) during the contract's life. To price them, we need to mathematically capture the "time" a process reaches a level.

Hitting Time

Let $(B_t)_{t\geq 0}$ be a standard Brownian motion. For a level $a > 0$, the *hitting time* of a is defined as:

$$T_a = \inf\{t \geq 0 : B_t = a\}$$

Stopping Time

T_a is a **stopping time**, meaning that the event $\{T_a \leq t\}$ is measurable with respect to the current filtration $\mathcal{F}_t$.

In financial terms, observing the trajectory up to time t is perfectly sufficient to answer the question: "Has the asset touched the barrier yet?"

Conversely, the *last* time the Brownian motion visits a level a, i.e., $\hat{T}_a = \sup\{t : B_t = a\}$, is **not** a stopping time, because knowing if a visit is the "last" one requires anticipating the entire future trajectory.

By the properties of Brownian motion ($\limsup_{t\to\infty} B_t = +\infty$ almost surely), we know that $\mathbb{P}(T_a < \infty) = 1$. The barrier will eventually be hit, though the expected time to hit it might be infinite.

2.15.2 The Reflection Principle

To compute probabilities regarding the maximum of a Brownian motion, we define the **running maximum** process:

$$M(t) := \sup_{0\leq s\leq t} B_s$$

Reflection Principle

Let $(B_t)_{t\geq 0}$ be a standard Brownian motion and $a > 0$. The probability that the running maximum exceeds a is exactly twice the probability that the Brownian motion itself exceeds a at time t:

$$\mathbb{P}(M(t) \geq a) = 2\mathbb{P}(B_t \geq a)$$

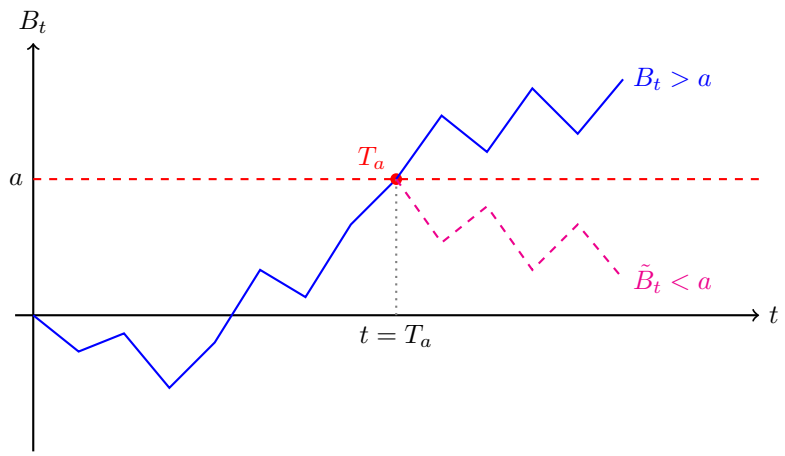


Figure 2.4: The Reflection Principle: For every path that hits a and finishes at $B_t > a$, there is an equally probable reflected path that finishes at $2a - B_t < a$.

Proof. We split the probability $\mathbb{P}(B_t \geq a)$ into two disjoint events based on the maximum:

$$\mathbb{P}(B_t \geq a) = \mathbb{P}(B_t \geq a, M(t) \geq a) + \mathbb{P}(B_t \geq a, M(t) < a)$$

The second term is trivially empty (probability 0) because if the current value $B_t \geq a$, the maximum up to t must necessarily be at least a. Thus:

$$\mathbb{P}(B_t \geq a) = \mathbb{P}(B_t \geq a \mid M(t) \geq a)\mathbb{P}(M(t) \geq a)$$

Conditioning on $M(t) \geq a$ is equivalent to saying that the hitting time $T_a \leq t$. By the **Strong Markov Property**, the process restarted at the random time T_a, defined as $\tilde{B}_{t-T_a} = B_t - B_{T_a} = B_t - a$, is a new standard Brownian motion independent of the past. By the symmetry of Brownian motion:

$$\mathbb{P}(B_t \geq a \mid M(t) \geq a) = \mathbb{P}(B_t - a \geq 0 \mid T_a \leq t) = \mathbb{P}(\tilde{B}_{t-T_a} \geq 0) = \frac{1}{2}$$

Substituting this back yields the result: $\mathbb{P}(B_t \geq a) = \frac{1}{2}\mathbb{P}(M(t) \geq a)$. □

Joint Distribution and Density of (M_t, B_t)

To price barrier options, we need the joint behavior of the running maximum and the terminal asset value.

Proposition. For $a > 0$ and $y \geq 0$:

$$\mathbb{P}(M(t) \geq a, B_t \leq a - y) = \mathbb{P}(B_t \geq a + y)$$

Proof. Following the reflection logic, if $M(t) \geq a$, the path has hit a. The event that it subsequently ends up at a distance y below a (i.e., $B_t \leq a - y$) has the exact same probability as the reflected path ending up at a distance y above a (i.e., $B_t \geq a + y$). □

Joint CDF

By differentiating the joint cumulative distribution function twice (first with respect to $m = a$, then with respect to $b = a - y$), we obtain the **Joint Density Function** of (M_t, B_t) for $m > 0$ and $b < m$:

$$f_{M_t,B_t}(m, b) = \frac{2(2m - b)}{\sqrt{2\pi t^3}} \exp\left(-\frac{(2m - b)^2}{2t}\right)$$

2.15.3 Moment Generating Function of the Hitting Time (via Localization)

We want to find the MGF (or Laplace transform) of T_a, $\mathbb{E}[e^{-\lambda T_a}]$. We use the exponential martingale:

$$Z_t = \exp\left(\alpha B_t - \frac{1}{2}\alpha^2 t\right)$$

Since Z_t is a martingale, we intuitively want to apply the **Optional Sampling Theorem** and say $\mathbb{E}[Z_{T_a}] = Z_0 = 1$. However, OST only holds strictly for *bounded* stopping times. Since T_a can be infinite, we must apply a rigorous **localization** technique.

We evaluate the martingale at the bounded stopping time $T_a \wedge t = \min(T_a, t)$:

$$1 = \mathbb{E}\left[\exp\left(\alpha B_{T_a \wedge t} - \frac{1}{2}\alpha^2 (T_a \wedge t)\right)\right]$$

Since $B_{T_a \wedge t} \leq a$ almost surely (it stops growing once it hits a), the term inside the expectation is strictly bounded by $e^{\alpha a} \cdot 1$. Because of this uniform bound, we can apply the **Dominated Convergence Theorem** and take the limit as $t \to \infty$ inside the expectation.

As $t \to \infty$, $T_a \wedge t \to T_a$. Also, on the event $\{T_a < \infty\}$, we know strictly that $B_{T_a} = a$. Thus:

$$1 = \mathbb{E}\left[\exp\left(\alpha a - \frac{1}{2}\alpha^2 T_a\right) \mathbf{1}_{\{T_a < \infty\}}\right] = e^{\alpha a}\mathbb{E}\left[\exp\left(-\frac{1}{2}\alpha^2 T_a\right)\right]$$

Setting $\lambda = \frac{1}{2}\alpha^2 \implies \alpha = \sqrt{2\lambda}$, we isolate the MGF for $a > 0$:

$$\mathbb{E}[e^{-\lambda T_a}] = e^{-a\sqrt{2\lambda}}$$

By symmetry, the general formula for any $a \in \mathbb{R}$ is $\mathbb{E}[e^{-\lambda T_a}] = e^{-|a|\sqrt{2\lambda}}$. By taking the derivative with respect to λ and evaluating at 0, we can easily verify that the expected hitting time $\mathbb{E}[T_a] = \infty$.

2.15.4 Brownian Motion with Drift and Barrier Options

When modeling stocks, we deal with geometric Brownian motion, which implies a drifted arithmetic Brownian motion in the exponent:

$$X_t = d + \rho t + \sigma W_t = d + \sigma\left(W_t + \frac{\rho}{\sigma}t\right) = d + \sigma \hat{W}_t$$

By **Girsanov's Theorem**, we can define a new probability measure $\hat{\mathbb{P}}$ where $\hat{W}_t = W_t + \theta t$ (with $\theta = \rho/\sigma$) acts as a standard, driftless Brownian motion. The hitting time T_β for the drifted process can then be evaluated by shifting the measure, resulting in the hitting time distribution:

$$\mathbb{P}(T_\beta \leq t) = \int_0^t \frac{\beta - d}{\sigma} \frac{1}{\sqrt{2\pi y^3}} \exp\left(-\frac{((\beta - d)/\sigma - \theta y)^2}{2y}\right) dy$$

Classic Approach to Barrier Options

Barrier Option

A **barrier option** is a type of path-dependent option whose payoff depends on whether the underlying asset's price reaches a certain barrier level during the option's life. There are two main types:

- **Knock-In**: The option only becomes active if the barrier is breached.
- **Knock-Out**: The option becomes worthless if the barrier is breached.

For example, an:

Up-and-In Barrier Call Option

An **up-and-in** barrier call option only pays off if the asset S_t crosses a predefined upper barrier L before maturity T. Its payoff is:

$$\text{Payoff}_{UIC} = (S_T - k)^+ \cdot \mathbf{1}_{\{M_T^S > L\}}$$

Where $M_T^S = \max_{0 \leq s \leq T} S_s$.

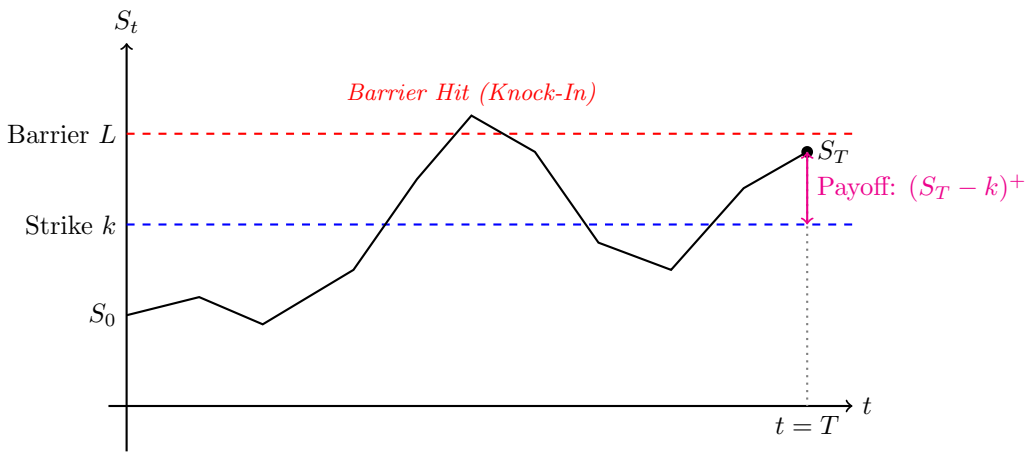


Figure 2.5: Up-and-In Call Option: The option becomes active (knocks in) because the path touches the barrier L. The final payoff at T is calculated based on the strike k.

Pricing this requires computing the joint risk-neutral probability distribution of the terminal stock value and its running maximum, converting the SDE to a drifted Brownian motion, and applying the integrated Girsanov and Reflection logic established in the previous sections.

Up-and-Out Call Option

Up-and-Out Call (UOC) with strike K and upper barrier L (where $S_0 < K < L$) pays the standard call payoff only if the maximum value attained by the asset remains strictly below L:

$$\text{Payoff}_{UOC} = (S_T - K)^+ \cdot \mathbf{1}_{\{\sup_{0 \le t \le T} S_t \le L\}}$$

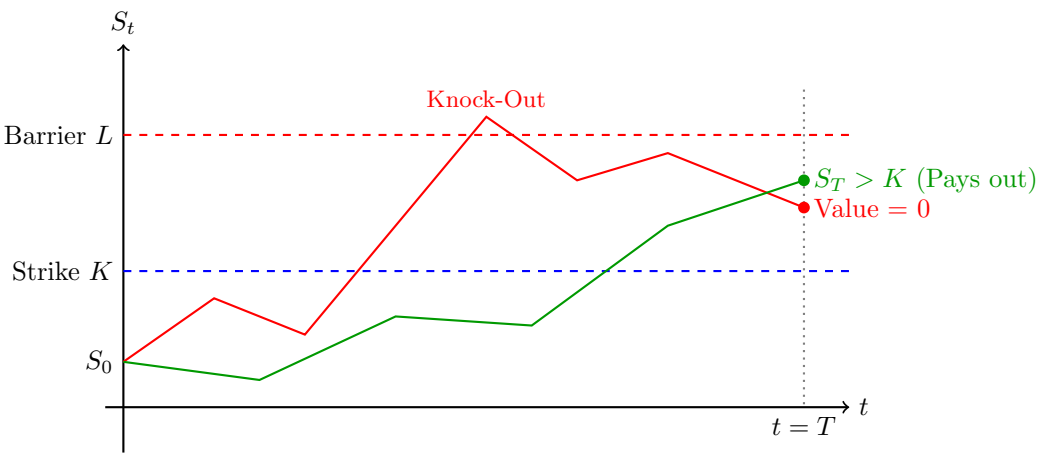


Figure 2.6: Up-and-Out Call: The red path hits the barrier L and becomes worthless. The green path stays below L and ends in-the-money.

The classical approach to pricing this contract involves calculating the joint density of the Brownian motion and its running maximum, leading to complex double integrations. Before showing a more elegant, unified approach (Björk's approach), we must introduce an important probabilistic concept.

2.15.5 Uniform Integrability (UI) and Stopping Times

A sequence of random variables (or a martingale) is **Uniformly Integrable (UI)** if the contribution of the extreme tails to the expected value vanishes uniformly. Crucially, if a martingale $(Y_t)_{t \ge 0}$ is UI, we

can interchange the limit and the expectation:

$$\mathbb{E}\left[\lim_{t\to\infty} Y_t\right] = \lim_{t\to\infty} \mathbb{E}[Y_t]$$

Bounded martingales are always UI. We explore two classic interview questions that exploit this property.

Example 1: Geometric Brownian Motion Hitting a Level Let $Y_t = \exp\left(W_t - \frac{1}{2}t\right)$. We know Y_t is a true martingale, so $\mathbb{E}[Y_t] = 1$ for all t. By the law of the iterated logarithm, $\lim_{t\to\infty} Y_t = 0$ almost surely. Since $\mathbb{E}[Y_\infty] = 0 \neq \lim \mathbb{E}[Y_t] = 1$, the process Y_t is **not** uniformly integrable.

Let $\tau_n = \inf\{t \geq 0 : Y_t = n\}$. What is the probability that the process ever reaches n? We artificially bound the martingale by stopping it at τ_n. The stopped process $Y_{t\wedge\tau_n}$ takes values in $[0, n]$, so it is bounded and therefore **uniformly integrable**. By the Optional Stopping Theorem:

$$1 = \mathbb{E}[Y_0] = \mathbb{E}[Y_{\tau_n}]$$

The random variable Y_{τ_n} takes the value n if $\tau_n < \infty$, and 0 if $\tau_n = \infty$ (since the process decays to 0). Thus:

$$1 = n \cdot \mathbb{P}(\tau_n < \infty) + 0 \cdot \mathbb{P}(\tau_n = \infty) \implies \mathbb{P}(\tau_n < \infty) = \frac{1}{n}$$

Example 2: Standard BM Hitting a Double Barrier Let B_t be a standard Brownian motion. Define stopping times for lower barrier $-a$ and upper barrier b (with $a, b > 0$):

$$T_{-a} = \inf\{t : B_t = -a\}, \quad T_b = \inf\{t : B_t = b\}, \quad T = \min(T_{-a}, T_b)$$

The stopped process $B_{t\wedge T}$ is bounded in $[-a, b]$, hence it is a UI martingale. Applying the Optional Stopping Theorem:

$$0 = \mathbb{E}[B_0] = \mathbb{E}[B_T] = -a \cdot \mathbb{P}(B_T = -a) + b \cdot \mathbb{P}(B_T = b)$$

Since $\mathbb{P}(B_T = -a) + \mathbb{P}(B_T = b) = 1$, we solve the system to find:

$$\mathbb{P}(B_T = b) = \frac{a}{a+b}, \quad \mathbb{P}(B_T = -a) = \frac{b}{a+b}$$

To find the expected hitting time $\mathbb{E}[T]$, we use the martingale $M_t = B_t^2 - t$. The stopped process is UI, yielding:

$$\mathbb{E}[B_T^2 - T] = 0 \implies \mathbb{E}[T] = \mathbb{E}[B_T^2] = a^2\left(\frac{b}{a+b}\right) + b^2\left(\frac{a}{a+b}\right) = ab$$

2.16 Björk's Unified Approach for Barrier Options

Instead of dealing directly with running maximums, Björk's approach uses the concept of the **Absorbed Process**. If τ_L is the hitting time of the barrier L, the absorbed process is $S_{t\wedge\tau_L}$. It behaves like S_t until it hits L, at which point it is permanently "frozen" at L.

Consider a generic European payoff $\Phi(S_T)$. We define the modified payoff $\Phi_L(x)$:

$$\Phi_L(x) = \Phi(x)\mathbf{1}_{\{x>L\}} \quad (\text{or } \mathbf{1}_{\{x<L\}} \text{ depending on the contract})$$

Down-and-Out Option

A **Down-and-Out** option with barrier L pays the payoff $\Phi(S_T)$ only if the underlying asset price S_t has never fallen below L during the life of the option. If S_t hits or goes below L at any time before maturity, the option becomes worthless.

$$\text{Payoff}_{DO} = \Phi(S_T)\mathbf{1}_{\{\inf_{0\leq t\leq T} S_t > L\}}$$

(Theorem) Pricing a Down-and-Out (DO) Contract The price of a Down-and-Out option can be synthetically replicated by a portfolio of standard European options:

$$Price_{DO}(t, S, \Phi) = Price_{Eur}(t, S, \Phi_L) - \left(\frac{L}{S}\right)^{\frac{2\tilde{r}}{\sigma^2}} Price_{Eur}\left(t, \frac{L^2}{S}, \Phi_L\right)$$

where $\tilde{r} = r - \frac{1}{2}\sigma^2$, and $Price_{Eur}$ represents the standard Black-Scholes pricing function for a European payoff.

Proof. Under the risk-neutral measure $\mathbb{Q}$, the underlying asset follows $S_T = S_0 \exp(X_T)$, where X_T is an arithmetic Brownian motion: $dX_t = \tilde{r}dt + \sigma dW_t^{\mathbb{Q}}$, starting at $\alpha = \ln S_0$. The barrier is $\beta = \ln L$.

The density of the absorbed process X^β evaluated at time T (derived via the reflection principle) is:

$$f_{X^\beta}(x; T, \alpha) = \varphi(x; \alpha + \tilde{r}T, \sigma\sqrt{T}) - \exp\left(-2\tilde{r}\frac{\alpha - \beta}{\sigma^2}\right)\varphi(x; 2\beta - \alpha + \tilde{r}T, \sigma\sqrt{T})$$

where $\varphi(x; \mu, \nu)$ is the Gaussian density.

The price of the option is the discounted expectation over the unabsorbed paths ($x > \beta$):

$$\begin{aligned} Price_{DO} &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\Phi(S_T)\mathbf{1}_{\{\inf S_t > L\}}\right] = e^{-rT}\int_{\ln L}^{\infty}\Phi_L(e^x)f_{X^\beta}(x; T, \ln S_0)dx \\ &= e^{-rT}\int_{\ln L}^{\infty}\Phi_L(e^x)\varphi(x; \ln S_0 + \tilde{r}T, \dots)dx \\ &\quad - e^{-rT}\exp\left(-2\tilde{r}\frac{\ln(S_0/L)}{\sigma^2}\right)\int_{\ln L}^{\infty}\Phi_L(e^x)\varphi\left(x; \ln\left(\frac{L^2}{S_0}\right) + \tilde{r}T, \dots\right)dx \end{aligned}$$

Notice that the exponential factor in front of the second integral simplifies perfectly:

$$\exp\left(-2\tilde{r}\frac{\ln(S_0/L)}{\sigma^2}\right) = \exp\left(\ln\left(\frac{L}{S_0}\right)^{\frac{2\tilde{r}}{\sigma^2}}\right) = \left(\frac{L}{S_0}\right)^{\frac{2\tilde{r}}{\sigma^2}}$$

The first integral is simply the expectation of the payoff Φ_L assuming the asset starts at S_0. The second integral is exactly the expectation of the same payoff, but assuming a "fictitious" asset starting at the reflected origin $e^{2\beta-\alpha} = L^2/S_0$. This concludes the proof, linking path-dependent pricing directly to European pricing formulas. □

Linearity of Barrier Contracts A very powerful property of the barrier option pricing framework (Björk's approach) is the linearity of the modified payoffs.

Proposition (Linearity). If the terminal payoff of a barrier contract is a linear combination of two generic payoffs, $\alpha\phi + \beta\psi$, the price of the barrier contract is the linear combination of the prices of the individual barrier contracts:

$$Price_{DO}(t, s, \alpha\phi + \beta\psi) = \alpha Price_{DO}(t, s, \phi) + \beta Price_{DO}(t, s, \psi)$$

Proof. This follows directly from the linearity of the modification operator: $(\alpha\phi + \beta\psi)_L(x) = \alpha\phi_L(x) + \beta\psi_L(x)$.

2.16.1 Building Blocks: Vanilla Contracts

To exploit linearity, we define the basic building blocks and their standard European prices:

1. **Stock (ST):** $\phi(x) = x \implies Price(t, s, ST) = s$
2. **Zero-Coupon Bond (BO):** $\phi(x) = 1 \implies Price(t, s, BO) = e^{-r(T-t)}$
3. **Digital/Heaviside (H_L):** $\phi(x) = \mathbf{1}_{\{x>L\}} \implies Price(t, s, H_L) = e^{-r(T-t)}\Phi(d_2(s, L))$
4. **Vanilla Call (C_K):** $\phi(x) = (x - K)^+ \implies Price(t, s, C_K) = s\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2)$

where $d_2(s, L) = \frac{\ln(s/L) + \tilde{r}(T-t)}{\sigma\sqrt{T-t}}$. Note: H_L represents the risk-neutral probability of finishing above L, heavily used in digital options.

2.16.2 Up-and-Out (UO) Contracts

Up-and-Out contract pays $\phi(S_T)$ provided the underlying never crosses an upper barrier L during $[0, T]$.

$$Z^{UO} = \phi(S_T)\mathbf{1}_{\{\max_{u\in[0,T]} S_u < L\}}$$

The pricing formula mirrors the DO formula, using the upper modification $\phi^L(x) = \phi(x)\mathbf{1}_{\{x<L\}}$:

$$Price_{UO}(t, s, \phi) = \left[Price_{Eur}(t, s, \phi^L) - \left(\frac{L}{s}\right)^{\frac{2\tilde{r}}{\sigma^2}} Price_{Eur}\left(t, \frac{L^2}{s}, \phi^L\right)\right]\mathbf{1}_{\{s<L\}}$$

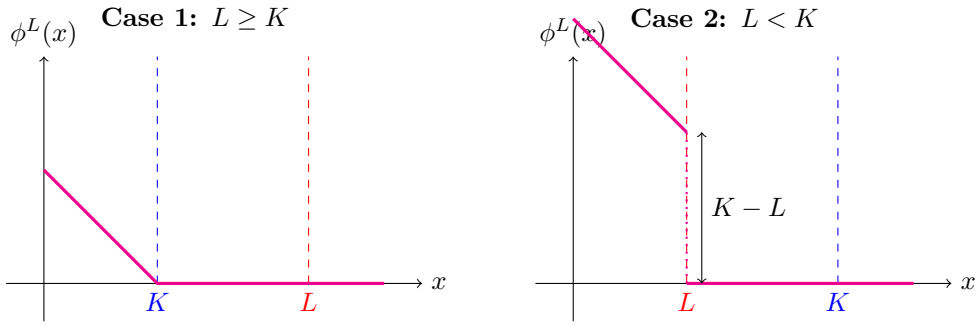
where $\tilde{r} = r - \frac{1}{2}\sigma^2$.

Up-and-Out Put (UOP) and Geometric Decomposition

An Up-and-Out Put pays $(K - S_T)^+$ provided the asset never crosses an upper barrier L. The modified terminal payoff is:

$$\phi^L(x) = (K - x)^+ \mathbf{1}_{\{x<L\}}$$

As with the call, the geometric decomposition and the resulting pricing formula depend heavily on the position of the barrier L relative to the strike K.



Modified payoffs for the Up-and-Out Put. If $L \geq K$, the barrier condition is redundant for the payoff. If $L < K$, the payoff introduces a discontinuity, requiring a combination of a Put and Digital options to replicate.

Case 1: $L \geq K$. If the option finishes in the money ($S_T < K$), it is automatically true that $S_T < L$. The condition $\mathbf{1}_{\{x<L\}}$ is redundant. Thus, $\phi^L(x) = (K - x)^+$, and the price is simply the UO pricing formula applied to a vanilla put:

$$UOP(t, s, K) = \left[Put(t, s, K) - \left(\frac{L}{s}\right)^{\frac{2\tilde{r}}{\sigma^2}} Put\left(t, \frac{L^2}{s}, K\right) \right] \mathbf{1}_{\{s<L\}}$$

Case 2: $L < K$. If $L < K$, the payoff is abruptly cut off at L. The modified payoff is $K - x$ for $x < L$, and 0 otherwise. We decompose this algebraically into a put with strike L and a constant jump component:

$$(K - x)^+ \mathbf{1}_{\{x<L\}} = (L - x)^+ + (K - L)\mathbf{1}_{\{x<L\}}$$

Notice that the indicator can be rewritten as $\mathbf{1}_{\{x<L\}} = 1 - \mathbf{1}_{\{x\geq L\}}$. In terms of our financial building blocks, 1 corresponds to a Zero-Coupon Bond (BO) and $\mathbf{1}_{\{x\geq L\}}$ is the Heaviside step function (H_L). Applying the general UO pricing theorem:

$$\begin{aligned} UOP(t, s, K) = \Bigg[& Put(t, s, L) + (K - L)\Big(BO(t, s) - H(t, s, L)\Big) \\ & - \left(\frac{L}{s}\right)^{\frac{2\tilde{r}}{\sigma^2}} \left(Put\left(t, \frac{L^2}{s}, L\right) + (K - L)\Big(BO(t, s) - H\left(t, \frac{L^2}{s}, L\right)\Big)\right) \Bigg] \mathbf{1}_{\{s<L\}} \end{aligned}$$

Note: The Zero-Coupon Bond price $BO(t, s) = e^{-r(T-t)}$ is independent of the underlying asset price s, meaning $BO(t, L^2/s) = BO(t, s)$.

2.16.3 Pricing Down-and-Out (DO) Contracts

Down-and-Out contracts pay the payoff $\phi(S_T)$ only if the underlying asset price S_t has never fallen below a certain barrier L during the life of the option. The modified payoff is $\phi_L(x) = \phi(x)\mathbf{1}_{\{x>L\}}$. By linearity, we can decompose any payoff into a linear combination of the building blocks defined above, and then apply the DO pricing formula to each component.

Down-and-Out Zero-Coupon Bond

A DO ZCB pays 1 if the barrier L is never touched. The modified payoff is $\phi_L(x) = 1 \cdot \mathbf{1}_{\{x>L\}} = H_L(x)$.

$$BO_{DO}(t, s) = \left[H(t, s, L) - \left(\frac{L}{s}\right)^{\frac{2\tilde{r}}{\sigma^2}} H\left(t, \frac{L^2}{s}, L\right) \right] \mathbf{1}_{\{s>L\}}$$

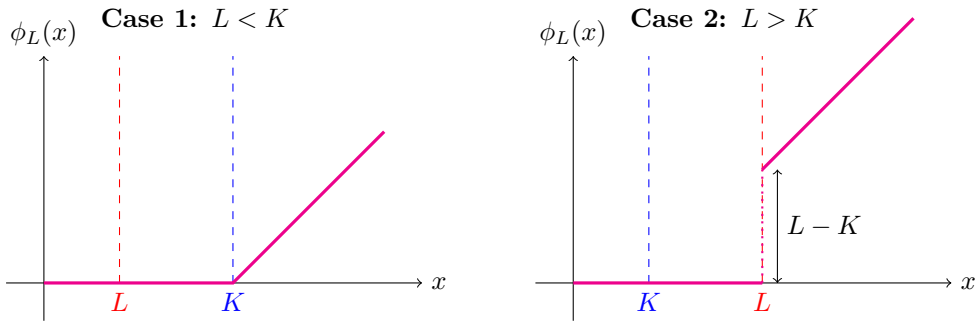
Down-and-Out Stock

Pays 1 share of the stock if the barrier is never touched. Modified payoff: $x\mathbf{1}_{\{x>L\}}$. We algebraically decompose it: $x\mathbf{1}_{\{x>L\}} = (x-L)^+ + L\mathbf{1}_{\{x>L\}} = C_L(x) + L \cdot H_L(x)$.

$$ST_{DO}(t,s) = \left[C(t,s,L) + L \cdot H(t,s,L) - \left(\frac{L}{s}\right)^{\frac{2\tilde{r}}{\sigma^2}} \left(C\left(t, \frac{L^2}{s}, L\right) + L \cdot H\left(t, \frac{L^2}{s}, L\right)\right)\right] \mathbf{1}_{\{s>L\}}$$

Down-and-Out Call (DOC)

The complexity depends on the relationship between the barrier L and the strike K.



Modified payoffs for the Down-and-Out Call. If $L < K$, the barrier condition is redundant for the payoff. If $L > K$, the payoff introduces a discontinuity (requiring a Digital option to replicate).

Case 1: $L \leq K$. The condition to finish in-the-money ($S_T > K$) strictly implies $S_T > L$. The modification $\phi_L(x) = (x-K)^+\mathbf{1}_{\{x>L\}}$ simplifies exactly to $(x-K)^+$. Therefore, the building block is just a vanilla call:

$$DOC(t,s,K) = \left[C(t,s,K) - \left(\frac{L}{s}\right)^{\frac{2\tilde{r}}{\sigma^2}} C\left(t, \frac{L^2}{s}, K\right)\right] \mathbf{1}_{\{s>L\}}$$

Case 2: $L > K$. Here, the option becomes worthless if $S_T < L$, even if it was in the money ($S_T > K$). The modified payoff has a discrete jump at L of size $L - K$. We algebraically decompose it: $(x-K)^+\mathbf{1}_{\{x>L\}} = (x-L)^+ + (L-K)\mathbf{1}_{\{x>L\}}$.

$$\begin{aligned} DOC(t,s,K) = \Bigg[&C(t,s,L) + (L-K)H(t,s,L) \\ &- \left(\frac{L}{s}\right)^{\frac{2\tilde{r}}{\sigma^2}} \left(C\left(t, \frac{L^2}{s}, L\right) + (L-K)H\left(t, \frac{L^2}{s}, L\right)\right)\Bigg] \mathbf{1}_{\{s>L\}} \end{aligned}$$

2.16.4 In-Out Parity

Any vanilla contract can be perfectly replicated by going long on an *Up-and-Out* contract and long on an *Up-and-In* contract with identical strikes and barriers.

$$Price_{Vanilla}(\phi) = Price_{UI}(\phi) + Price_{UO}(\phi)$$

$$Price_{Vanilla}(\phi) = Price_{DI}(\phi) + Price_{DO}(\phi)$$

This holds because the sum of the indicator functions perfectly covers the sample space: $\mathbf{1}_{\{\text{Barrier Hit}\}} + \mathbf{1}_{\{\text{Barrier Not Hit}\}} = 1$. This parity is universally used to price "Knock-In" options, as they are simply $Price_{Vanilla} - Price_{Knock-Out}$.

2.17 Lookback Options

Lookback options have payoffs that depend on the maximum or minimum achieved by the underlying over the entire contract life. They exhibit extreme path dependency.

- **Lookback Call (Floating Strike):** $(S_T - \min_{0\leq t\leq T} S_t)^+$ (buy at the lowest historical price).
- **Lookback Put (Floating Strike):** $(\max_{0\leq t\leq T} S_t - S_T)^+$ (sell at the highest historical price).

2.17.1 Pricing a Lookback Put

Pricing requires the joint distribution of S_T and $\max_{t\in[0,T]} S_t$. The closed-form solution for a Lookback Put evaluated at time $t = 0$ is:

$$\begin{aligned} Price_{LBP} = &-S_0\Phi(-d_1) + S_0 e^{-rT}\Phi(-d_1 + \sigma\sqrt{T}) \\ &+ S_0\frac{\sigma^2}{2r}\Phi(d_1) - S_0 e^{-rT}\frac{\sigma^2}{2r}\Phi(-d_1 + \sigma\sqrt{T}) \end{aligned}$$

where $d_1 = \frac{(r+\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$.

Distribution of the Maximum. The CDF for the maximum of an Arithmetic Brownian Motion X_t starting at α with drift μ and volatility σ is:

$$F_{\max X}(x) = \Phi\left(\frac{x-\alpha-\mu T}{\sigma\sqrt{T}}\right) - e^{\frac{2\mu(x-\alpha)}{\sigma^2}}\Phi\left(\frac{-x+\alpha-\mu T}{\sigma\sqrt{T}}\right)$$

For Geometric Brownian Motion S_t, we substitute $X \to \ln(S_t)$, $\alpha \to \ln(S_0)$, and $\mu \to \tilde{r} = r - \sigma^2/2$.

2.18 Pricing When Volatility is Stochastic

When the volatility σ_t is no longer constant but a stochastic process, the underlying asset S_t is no longer log-normally distributed. Consequently, we cannot compute the expected value of the payoff directly using the standard Black-Scholes formulas.

Under the risk-neutral measure $\mathbb{Q}$, assuming deterministic interest rates, the price of a generic payoff at maturity T is:

$$\pi_t(\text{Payoff}_T) = e^{-r(T-t)}\mathbb{E}_t^{\mathbb{Q}}[\text{Payoff}_T] = e^{-r(T-t)}\int_{-\infty}^{+\infty} g(x_T)f(x_T|x_t)dx_T$$

where $X_T = \ln S_T$ is the log-price, $g(X_T)$ is the payoff function in terms of the log-price, and $f(x_T|x_t)$ is the unknown conditional density of the log-price.

2.18.1 The Fourier Transform Approach

Since the density $f(x)$ is unknown but often its **Characteristic Function (CF)** is known analytically, we map the problem into the Fourier domain. The generalized Fourier transform of a function $f(x)$ is:

$$\hat{f}(z) = \int_{\mathbb{R}} e^{izx}f(x)dx = \mathbb{E}[e^{izX}]$$

where $z \in \mathbb{C}$ is a complex frequency parameter $z = u + iv$. The inverse transform is:

$$f(x) = \frac{1}{2\pi}\int_{\mathbb{R}-i\alpha} e^{-izx}\hat{f}(z)dz$$

By Parseval's Theorem, the pricing integral can be rewritten as the integral of the product of two Fourier transforms:

$$\pi_t = \frac{e^{-r(T-t)}}{2\pi}\int_{\mathbb{R}-i\alpha}\hat{f}(z)\hat{g}(-z)dz$$

This beautifully decouples the problem into two distinct parts:

1. $\hat{f}(z)$: The characteristic function of the log-price. This is **model-dependent** but **product-independent**.
2. $\hat{g}(-z)$: The Fourier transform of the payoff. This is **product-dependent** but **model-independent**.

2.18.2 Fourier Transforms of Standard Payoffs

When dealing with generalized Fourier transforms where $z = u + iv$, we must ensure the integral $\int e^{izx}g(x)dx$ converges. This imposes strict conditions on the imaginary part $\Im(z) = v$.

Covered Call Payoff

The first building block analyzed is a payoff that is bounded from above. Consider the function $h(X_T) = \min(e^{X_T}, K)$, which in terms of the underlying asset $S_T = e^{X_T}$ corresponds to $\min(S_T, K)$. This is financially equivalent to a **Covered Call** (holding the stock and writing a call option on it).

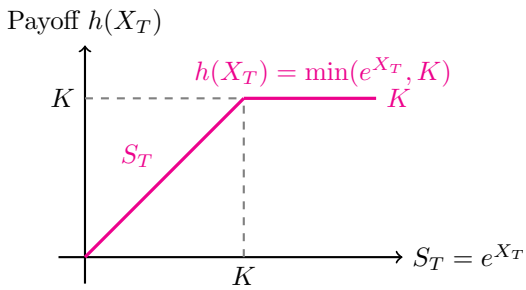


Figure 2.7: Payoff profile of the Covered Call. The boundedness of the payoff prevents explosive behavior at $+\infty$, directly affecting the domain of convergence of its Fourier transform.

To compute its generalized Fourier transform, we split the integration domain at the strike threshold $X_T = \ln K$:

$$\begin{aligned}
\hat{h}(z) &= \int_{-\infty}^{+\infty} e^{izx} \min(e^x, K) dx \\
&= \int_{-\infty}^{\ln K} e^{izx} e^x dx + \int_{\ln K}^{+\infty} e^{izx} K dx \\
&= \int_{-\infty}^{\ln K} e^{(iz+1)x} dx + K \int_{\ln K}^{+\infty} e^{izx} dx
\end{aligned}$$

Domain of Convergence. For a generalized frequency parameter $z = u + iv$ (where $v = \Im(z)$ is the imaginary part), we must guarantee that both integrals converge:

1. **First integral (lower bound $-\infty$):** The exponent is $x(i(u + iv) + 1) = x(-v + 1 + iu)$. For the integral to converge as $x \to -\infty$, the real part of the coefficient must be strictly positive:

$$1 - v > 0 \implies v < 1$$

2. **Second integral (upper bound $+\infty$):** The exponent is $x(i(u + iv)) = x(-v + iu)$. For the integral to converge as $x \to +\infty$, the real part must be strictly negative:

$$-v < 0 \implies v > 0$$

Therefore, the Fourier transform of the Covered Call payoff is well-defined only in the complex strip:

$$\mathcal{Z}_h = \{z \in \mathbb{C} : 0 < \Im(z) < 1\}$$

Evaluating the integrals within this domain yields the closed-form solution:

$$\hat{h}(z) = \left[\frac{e^{(iz+1)x}}{iz+1}\right]_{-\infty}^{\ln K} + K\left[\frac{e^{izx}}{iz}\right]_{\ln K}^{+\infty} = \frac{K^{iz+1}}{iz+1} - \frac{K^{iz+1}}{iz} = \frac{K^{iz+1}}{z^2 - iz}$$

European Call Option. The payoff is $g(x) = (e^x - K)^+$.

$$\begin{aligned}
\hat{g}_{Call}(z) &= \int_{\ln K}^{+\infty} e^{izx}(e^x - K) dx = \int_{\ln K}^{+\infty} e^{(iz+1)x} dx - K \int_{\ln K}^{+\infty} e^{izx} dx \\
&= \left[\frac{e^{(iz+1)x}}{iz+1}\right]_{\ln K}^{+\infty} - K\left[\frac{e^{izx}}{iz}\right]_{\ln K}^{+\infty}
\end{aligned}$$

For the upper limit to vanish, we require $\Re(iz+1) < 0 \implies -v+1 < 0 \implies \Im(z) > 1$. Evaluating at the lower bound yields:

$$\hat{g}_{Call}(z) = -\frac{K^{iz+1}}{z^2 - iz} \quad \text{provided } \Im(z) > 1$$

Financial Interpretation: The condition $\Im(z) > 1$ requires the model to have finite moments of order $\alpha > 1$. If the model has "moment explosion", the characteristic function might not be defined in this strip.

European Put Option. The payoff is $g(x) = (K - e^x)^+$. By similar integration from $-\infty$ to $\ln K$:

$$\hat{g}_{Put}(z) = -\frac{K^{iz+1}}{z^2 - iz} \quad \text{provided } \Im(z) < 0$$

2.19 The Heston Stochastic Volatility Model (1993)

The Heston model defines the instantaneous variance $V_t = \sigma_t^2$ as a mean-reverting Cox-Ingersoll-Ross (CIR) process.

Heston Model

Under the risk-neutral measure $\mathbb{Q}$:

$$\frac{dS_t}{S_t} = rdt + \sqrt{V_t}dW_t^S$$
$$dV_t = \kappa(\theta - V_t)dt + \xi\sqrt{V_t}dW_t^V$$

where the two Brownian motions are correlated: $d\langle W^S, W^V\rangle_t = \rho dt$.

- $\kappa > 0$: Speed of mean reversion.
- $\theta > 0$: Long-term average variance.
- $\xi > 0$: Volatility of volatility (vol-of-vol).
- $\rho \in [-1, 1]$: Correlation. In equity markets, $\rho < 0$ captures the "leverage effect" (prices drop $\implies$ variance spikes), correctly reproducing the asymmetric volatility smile (skew).

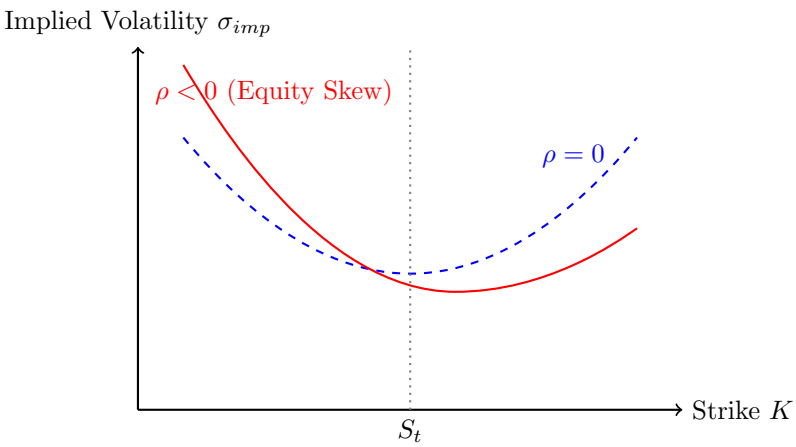


Figure 2.8: The effect of correlation ρ on the Implied Volatility Smile. A negative correlation ($\rho < 0$), standard in the Heston model for equities, skews the smile to price out-of-the-money puts higher due to crash risk.

2.19.1 Feller Condition and Squared Bessel Processes

The variance process V_t is a time-changed Squared Bessel Process (BESQ). To ensure that V_t remains strictly positive ($V_t > 0$ almost surely), the parameters must satisfy the **Feller Condition**:

$$2\kappa\theta \geq \xi^2$$

If this holds, the boundary at 0 is inaccessible.

2.19.2 The Characteristic Function PDE (Feynman-Kac)

To find the characteristic function of the log-price $G(\tau, x, v) = \mathbb{E}_t^{\mathbb{Q}}[e^{izX_T}]$, we write the dynamics of $X_t = \ln S_t$. By Itô's Lemma:

$$dX_t = \left(r - \frac{1}{2}V_t\right)dt + \sqrt{V_t}dW_t^S$$

Using the 2D Itô formula to find the infinitesimal generator $\mathcal{L}$ of the joint Markov process (X_t, V_t), we note the cross-variation $d\langle X, V\rangle_t = \rho\xi V_t dt$. The Feynman-Kac theorem states that G must satisfy:

$$\frac{\partial G}{\partial t} + \mathcal{L}G = 0$$

Switching to time-to-maturity $\tau = T - t \implies \frac{\partial G}{\partial t} = -\frac{\partial G}{\partial \tau}$, we obtain:

Heston Characteristic Function PDE

The fundamental PDE for the Heston characteristic function:

$$\begin{cases} -\frac{\partial G}{\partial \tau} + \left(r - \frac{1}{2}v\right)\frac{\partial G}{\partial x} + \kappa(\theta - v)\frac{\partial G}{\partial v} + \frac{1}{2}v\frac{\partial^2 G}{\partial x^2} + \frac{1}{2}\xi^2 v\frac{\partial^2 G}{\partial v^2} + \rho\xi v\frac{\partial^2 G}{\partial x \partial v} = 0 \\ G(0, x, v) = e^{izx} \end{cases}$$

Because the PDE coefficients are affine in the state variable v, the solution admits an exponential-affine form $G(\tau, x, v) = \exp(A(\tau) + B(\tau)v + izx)$, which reduces the PDE to a system of Riccati ODEs more tractable to solve.

2.19.3 Solving the Heston Characteristic Function

We want to find the characteristic function of the log-price $X_t = \ln S_t$ in the Heston model. The function $G(\tau, x, v) = \mathbb{E}_t^{\mathbb{Q}}[e^{izX_T}]$ must satisfy the Feynman-Kac PDE. An exponentially affine solution:

$$G(\tau, x, v) = e^{A(\tau) + B(\tau)v + C(\tau)x}$$

where $\tau = T - t$ is the time to maturity.

PDE Substitution and Coefficient Matching

We compute the partial derivatives of our guess:

$$\begin{aligned} G_\tau &= G \cdot (\dot{A} + \dot{B}v + \dot{C}x) \\ G_x &= G \cdot C, \quad G_{xx} = G \cdot C^2 \\ G_v &= G \cdot B, \quad G_{vv} = G \cdot B^2 \\ G_{xv} &= G \cdot (C \cdot B) \end{aligned}$$

Substituting these into the Heston PDE and dividing by G, we collect terms for x, v, and the constants. By the polynomial identification principle, this must hold for all x and v, yielding a system of three Ordinary Differential Equations (ODEs):

1. **Coefficient of x:**

$$-\dot{C}(\tau) = 0 \implies C(\tau) = iz$$

since the initial condition at $\tau = 0$ is $G(0, x, v) = e^{izx}$, meaning $C(0) = iz$.

2. **Coefficient of v (The Riccati Equation):**

$$\dot{B}(\tau) = \frac{1}{2}\xi^2 B^2(\tau) - (\kappa - \rho\xi iz)B(\tau) - \frac{1}{2}(iz + z^2)$$

with initial condition $B(0) = 0$. This is a nonlinear quadratic ODE with constant coefficients, known as a Riccati equation.

3. **Constant term:**

$$\dot{A}(\tau) = \kappa\theta B(\tau) + riz$$

with initial condition $A(0) = 0$. Once $B(\tau)$ is found, $A(\tau)$ is solved via direct integration.

Solving the Riccati Equation

We solve the Riccati equation by introducing the discriminant Δ:

$$\Delta = (\kappa - \rho\xi iz)^2 + \xi^2(iz + z^2)$$

We define the roots $\psi^{\pm} = -(\kappa - \rho\xi iz) \pm \sqrt{\Delta}$. Through a standard substitution $B(\tau) = -\frac{2}{\xi^2}\frac{\dot{E}(\tau)}{E(\tau)}$, the Riccati equation is transformed into a second-order linear ODE, yielding the closed-form solution:

$$B(\tau) = \frac{\kappa - \rho\xi iz - \sqrt{\Delta}}{\xi^2} \cdot \frac{1 - e^{-\sqrt{\Delta}\tau}}{1 - ge^{-\sqrt{\Delta}\tau}}, \quad \text{where } g = \frac{\kappa - \rho\xi iz - \sqrt{\Delta}}{\kappa - \rho\xi iz + \sqrt{\Delta}}$$

Integrating $B(\tau)$ to find $A(\tau)$ yields:

$$A(\tau) = riz\tau + \frac{\kappa\theta}{\xi^2}\left[(\kappa - \rho\xi iz - \sqrt{\Delta})\tau - 2\ln\left(\frac{1 - ge^{-\sqrt{\Delta}\tau}}{1 - g}\right)\right]$$

Remark: The "Little Heston Trap". In the complex plane, the logarithm is a multi-valued function. When evaluating $\ln(1 - ge^{-\sqrt{\Delta}\tau})$, improper phase tracking can cause discontinuities in the characteristic function, leading to numerical instability during pricing. Modern implementations strictly format the complex logarithm branch to ensure continuity across all maturities.

2.19.4 Alternative Approach: Matrix Exponentials

A fundamental property of the Riccati equation is its connection to the Grassmannian manifold, allowing us to linearize it. We can write $B(\tau)$ as a "ratio" of two functions (or matrices in multi-dimensional generalizations): $B(\tau) = H^{-1}(\tau)F(\tau)$. By substituting this into the Riccati ODE, we obtain a linear system:

$$\frac{d}{d\tau}\begin{pmatrix} F(\tau) \\ H(\tau) \end{pmatrix} = \begin{pmatrix} -(\kappa - \rho\xi iz) & -\frac{1}{2}(iz + z^2) \\ -\frac{1}{2}\xi^2 & 0 \end{pmatrix}\begin{pmatrix} F(\tau) \\ H(\tau) \end{pmatrix}$$

with initial conditions $F(0) = 0$ and $H(0) = 1$. The solution is given by the **Matrix Exponential** $e^{M\tau}$.

Properties of Matrix Exponentials

- $e^A = I + A + \frac{A^2}{2!} + \cdots = \sum_{n=0}^{\infty} \frac{A^n}{n!}$.
- If A is a symmetric matrix, it can be diagonalized as $A = PDP^{-1}$, where D contains the real eigenvalues. Then, $e^A = Pe^D P^{-1}$.
- For a symmetric positive semi-definite matrix ($A \in \mathcal{S}_d^+$), there exists a **unique well-defined square root** $\sqrt{A} = P\sqrt{D}P^{-1}$. This uniqueness only strictly holds for symmetric matrices; generic non-symmetric matrices may have multiple, or zero, square roots.

While the standard approach using the discriminant is sufficient for the classic 1-factor Heston model, this matrix formulation becomes an indispensable tool when extending the model to multiple dimensions. In multi-factor models, the Riccati equations evolve into complex matrix Riccati equations, where the matrix exponential offers an elegant and computationally stable resolution path.

2.19.5 Multi-Factor Stochastic Volatility (2-Heston)

To better capture the dynamics of the volatility surface—specifically disentangling the short-term skew from the long-term skew—we extend the model to a 2-factor Heston model. The variance is driven by two independent CIR processes:

$$\begin{aligned}
\frac{dS_t}{S_t} &= rdt + \sqrt{V_1}dZ_1 + \sqrt{V_2}dZ_2 \\
dV_1 &= \kappa_1(\theta_1 - V_1)dt + \xi_1\sqrt{V_1}dW_1 \\
dV_2 &= \kappa_2(\theta_2 - V_2)dt + \xi_2\sqrt{V_2}dW_2
\end{aligned}$$

Affinity Constraints and Independence

To preserve the exponentially affine structure of the characteristic function (allowing closed-form pricing), we must impose strict correlation constraints. If dW_1 and dW_2 were correlated, the cross-derivative term in the PDE would involve $\sqrt{V_1V_2}$, breaking the affine structure.

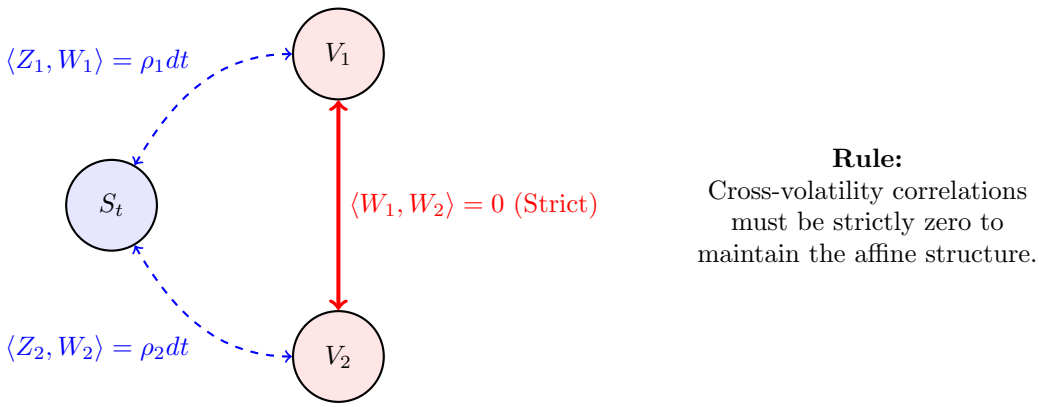


Figure 2.9: Correlation structure in the 2-Factor Heston Model. The underlying asset correlates with each volatility factor independently.

Stochastic Correlation Feature

Although the volatility factors V_1 and V_2 are independent, the *effective* instantaneous correlation between the asset's return dX_t and the total variance $d(V_1 + V_2)$ becomes **stochastic**:

$$\text{Corr}(dX_t, d(V_1 + V_2)) = \frac{\rho_1\xi_1V_1 + \rho_2\xi_2V_2}{\sqrt{V_1 + V_2}\sqrt{\xi_1^2V_1 + \xi_2^2V_2}}$$

This is a powerful feature: the skewness of the smile is no longer governed by a static parameter ρ, but is a dynamically weighted average of ρ_1 and ρ_2 dependent on the current levels of V_1 and V_2.

2.20 Limitations of the Multi-Factor Heston Model

In a multi-factor Heston model, we attempt to model an asset S_t with multiple volatility factors V_1, V_2 to capture different aspects of the volatility surface (e.g., short-term vs. long-term skew).

$$\begin{aligned}
\frac{dS_t}{S_t} &= rdt + \sqrt{V_1}dZ_1 + \sqrt{V_2}dZ_2 \\
dV_1 &= \kappa_1(\theta_1 - V_1)dt + \xi_1\sqrt{V_1}dW_1 \\
dV_2 &= \kappa_2(\theta_2 - V_2)dt + \xi_2\sqrt{V_2}dW_2
\end{aligned}$$

We allow correlation between the asset and its volatility factors: $d\langle Z_1, W_1\rangle_t = \rho_1 dt$ and $d\langle Z_2, W_2\rangle_t = \rho_2 dt$. However, to maintain the **affine structure** required for Fourier pricing, we must strictly forbid:

1. Correlation between W_1 and W_2. If $d\langle W_1, W_2\rangle_t \neq 0$, the infinitesimal generator would contain $\sqrt{V_1 V_2}$, which is not affine.

2. Cross-correlations like $d\langle Z_1, W_2\rangle_t \neq 0$.

This rigidity implies that the short-term and long-term volatility factors cannot communicate with each other.

2.21 Generating Dependence while Preserving Affinity

To model multiple assets (S_1, S_2) with non-trivial correlation while keeping the affine structure, a naive approach of correlating their respective Heston volatility factors fails. Instead, we use **Common Volatility Factors**.

Let X_1, X_2, X_3 be independent squared Bessel processes:

$$dX_i = (\dots)dt + \sqrt{X_i}dW^i, \quad i = 1, 2, 3$$

We define the asset dynamics such that X_3 acts as a shared factor:

$$\frac{dS_1}{S_1} = rdt + \sqrt{X_1}dZ^1 + \sqrt{X_3}dZ^3$$
$$\frac{dS_2}{S_2} = rdt + \sqrt{X_2}dZ^2 + \sqrt{X_3}dZ^3$$

Assuming Z^i and W^i are correlated but independent across different i's. The variances of the assets are affine:

$$\frac{1}{(S_1)^2}d\langle S_1, S_1\rangle_t = (X_1 + X_3)dt, \quad \frac{1}{(S_2)^2}d\langle S_2, S_2\rangle_t = (X_2 + X_3)dt$$

The instantaneous covariance between the two assets is generated entirely by the common factor:

$$\frac{1}{S_1 S_2}d\langle S_1, S_2\rangle_t = X_3 dt$$

Because X_3 is a stochastic process, we have successfully generated **stochastic correlation**. However, since X_3 is a squared Bessel process, it must be strictly positive. This implies the correlation between S_1 and S_2 can never be negative, which is a strong and often unrealistic restriction.

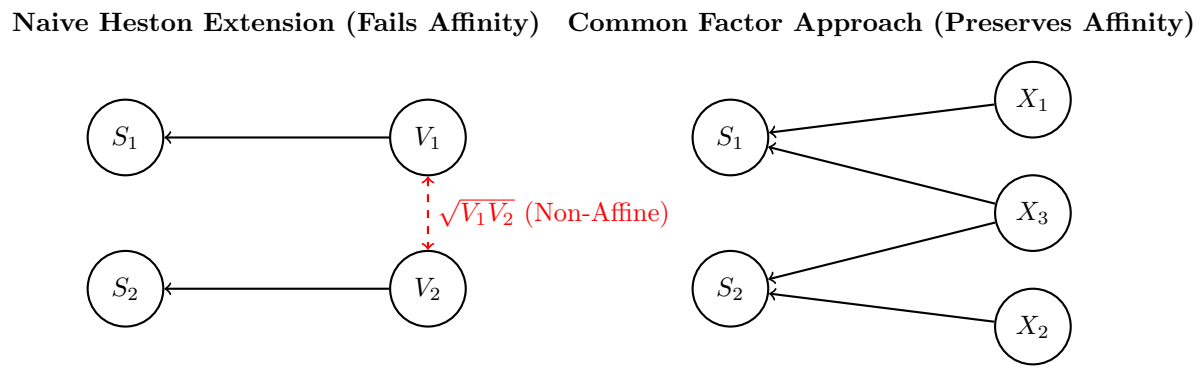


Figure 2.10: To generate dependence while preserving affinity, we cannot correlate volatility factors directly. Instead, we use a common independent volatility factor X_3. This yields stochastic correlation, but restricts it to strictly positive values.

2.22 The Wishart Process

To allow for full stochastic covariance matrices (including negative correlations) while maintaining affinity, we extend the squared Bessel process to matrix spaces.

2.22.1 Static Wishart Distribution

Let B be an $n \times d$ matrix where each entry B_{ij} is an independent standard Gaussian random variable. If $n \geq d$, the $d \times d$ symmetric matrix:

$$M = B^T B$$

follows a Wishart distribution. For $d = 1$, $M = B_{11}^2 + \cdots + B_{n1}^2 \sim \chi_n^2$, which is exactly the distribution of a squared Bessel process evaluated at a fixed time. The condition $n \geq d$ ensures the matrix is full rank and the distribution does not degenerate.

Wishart Process Definition (Bru 1991)

We define the Wishart process of dimension d and index n (with $n \geq d$), starting at a symmetric positive semi-definite matrix $S_0 = C^T C$. Let N_t be an $n \times d$ matrix whose elements are independent standard Brownian motions, starting at $N_0 = C$. The Wishart process is defined as:

$$S_t = N_t^T N_t$$

By construction, S_t is a symmetric, positive semi-definite $d \times d$ matrix.

Dynamics of the Wishart Process

Applying multi-dimensional Itô calculus to $S_t = N_t^T N_t$:

$$dS_t = d(N_t^T)N_t + N_t^T dN_t + d\langle N^T, N\rangle_t$$

Let dB_t be a $d \times d$ matrix where each entry is an independent Brownian motion. By the matrix analogue of Lévy's characterization, the cross-variations of the components of N_t allow us to represent the dynamics as:

$$dS_t = \sqrt{S_t} dB_t + dB_t^T \sqrt{S_t} + n I_{d\times d} dt$$

Notice the symmetry: we must add the transpose of the diffusion term because the product of a symmetric matrix ($\sqrt{S_t}$) and a non-symmetric matrix (dB_t) is not necessarily symmetric, but S_t must remain symmetric. The drift term $nI_{d\times d}$ is the exact matrix analogue of the Bessel drift.

Infinitesimal Generator and Affinity

The infinitesimal generator $\mathcal{A}$ of the Wishart process acting on a function $f(S)$ is given by the trace operator:

$$\mathcal{A} = \mathrm{Tr}\left[nD + 2SD^2\right]$$

where $D_{ij} = \frac{\partial}{\partial S_{ij}}$ is the matrix of differential operators.

To verify affinity, we analyze the 2×2 case ($d = 2$). Let $\Sigma = \sqrt{S_t}$.

$$\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}, \quad S_t = \Sigma^T \Sigma = \begin{pmatrix} \sigma_{11}^2 + \sigma_{21}^2 & \sigma_{11}\sigma_{12} + \sigma_{21}\sigma_{22} \\ \sigma_{11}\sigma_{12} + \sigma_{21}\sigma_{22} & \sigma_{12}^2 + \sigma_{22}^2 \end{pmatrix}$$

The diffusion term of the SDE is $dM_t = \Sigma^T dB_t + dB_t^T \Sigma$. Evaluating the top-left element:

$$d(S_{11}) = (\dots)dt + 2\sigma_{11} dB_{11} + 2\sigma_{21} dB_{21}$$

Its quadratic variation is:

$$d\langle S_{11}, S_{11}\rangle_t = 4(\sigma_{11}^2 + \sigma_{21}^2)dt = 4S_{11}dt$$

Similarly, $d\langle S_{22}, S_{22}\rangle_t = 4S_{22}dt$.

For the cross-variation between the diagonal elements:

$$d\langle S_{11}, S_{22}\rangle_t = (\dots) \text{ which initially evaluates to 0 because } dB_{ij} \text{ are independent.}$$

However, the cross-variation between S_{11} and the off-diagonal term S_{12} yields:

$$d\langle S_{11}, S_{12}\rangle_t = 2(\sigma_{11}\sigma_{12} + \sigma_{21}\sigma_{22})dt = 2S_{12}dt$$

Because the quadratic variations and covariations of the matrix elements are strictly linear combinations of the matrix elements themselves, the coefficients of the second-order differential operators in the generator $\mathcal{A}$ are **linear affine** in S. Therefore, the Wishart process preserves the affine structure and permits closed-form Fourier pricing techniques, despite modeling a fully stochastic covariance matrix.

2.22.2 The Laplace Transform of the Wishart Process

We recall the standard continuous-time Wishart process $S_t \in \mathcal{S}_d^+$ (the space of symmetric, positive semi-definite $d \times d$ matrices) driven by a $d \times d$ matrix Brownian motion B_t:

$$dS_t = \sqrt{S_t} dB_t + dB_t^\top \sqrt{S_t} + n I_{d\times d} dt$$

where $n \geq d$ is an integer dimension.

Because the infinitesimal generator $\mathcal{A}$ of this process is **linear affine** in the elements of the matrix S_t, we can compute its Laplace transform. The Laplace transform for a matrix random variable is defined using the trace operator, which acts as the matrix analogue to the scalar product of two vectors:

$$\mathbb{E}_{S_0}\left[e^{-\mathrm{Tr}[uS_t]}\right] = \det(I + 2tu)^{-\frac{n}{2}} \exp\left(-\mathrm{Tr}\left[S_0(I+2tu)^{-1}u\right]\right)$$

This closed-form solution confirms that the transform is **exponentially affine**: it is an exponential function of a linear combination of the initial matrix elements S_0.

2.22.3 Extension to Real Dimensions (α)

Boundary Classification

We can generalize the integer dimension n to a real parameter $\alpha \in \mathbb{R}$. The generalized Wishart SDE becomes:

$$dS_t = \sqrt{S_t} dB_t + dB_t^\top \sqrt{S_t} + \alpha I_{d\times d} dt$$

The parameter α strictly dictates the analytical properties of the process, particularly boundary behavior and eigenvalue collision.

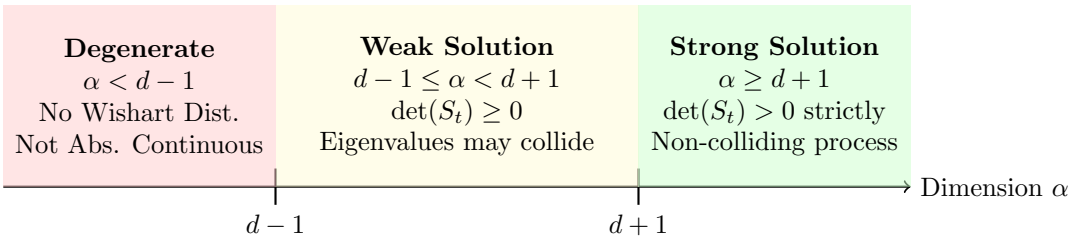


Figure 2.11: Classification of the boundary for the generalized Wishart process based on the real dimension parameter α. The threshold $d+1$ is the matrix equivalent of the Feller condition.

- $\alpha \geq d+1$: **Strong Solution**. The determinant is strictly positive ($\det(S_t) > 0$) for all t. The eigenvalues $\lambda_1 > \cdots > \lambda_d > 0$ are distinct and never collide. This is the matrix equivalent of the Feller condition.
- $\alpha \in [d-1, d+1)$: **Weak Solution**. The matrix remains positive semi-definite ($\det(S_t) \geq 0$), but eigenvalues can collide or hit zero.
- $\alpha < d-1$: The process degenerates. The distribution loses absolute continuity with respect to the Lebesgue measure, making it useless for standard financial modeling.

2.22.4 Adding Mean Reversion: The Affine Wishart Model

To use this process as a realistic stochastic covariance matrix Σ_t, we must add mean reversion. Similar to how the CIR process is generated by squaring an Ornstein-Uhlenbeck process, we introduce a matrix Ornstein-Uhlenbeck process $X_t \in \mathcal{M}_{n\times d}$:

$$dX_t = dB_t Q + M X_t dt$$

where B_t is an $n \times d$ matrix Brownian motion, Q is a diffusion parameter matrix, and M governs mean reversion.

By applying the multi-dimensional Itô formula to $\Sigma_t = X_t^\top X_t$, we obtain the fully generalized mean-reverting Wishart process:

$$d\Sigma_t = (\beta Q^\top Q + M\Sigma_t + \Sigma_t M^\top)dt + \sqrt{\Sigma_t} dW Q + Q^\top dW^\top \sqrt{\Sigma_t}$$

where β generalizes n (requiring $\beta \geq d-1$ for existence), and W is a $d \times d$ standard matrix Brownian motion.

The infinitesimal generator $\mathcal{A}_\Sigma$ for this process is:

$$\mathcal{A}_\Sigma = \mathrm{Tr}\left[(\beta Q^\top Q + M\Sigma + \Sigma M^\top)D + 2\Sigma D Q^\top Q D\right]$$

where $D_{ij} = \frac{\partial}{\partial \Sigma_{ij}}$ is the matrix of differential operators. Note that this generator remains perfectly **linear affine** in the elements of Σ.

Financial meaning: this matrix M allows us to model mean reversion in the covariance matrix, which is crucial for realistic modeling of volatility dynamics. The parameter β controls the long-term level of covariance, while Q governs the volatility of volatility. In real cases it's unrealistic to assume that the covariance matrix can explode to infinity, so mean reversion is a necessary feature.

2.22.5 Wishart's characteristics function

We wish to compute the conditional Laplace transform:

$$\psi(\tau, \Sigma_t) = \mathbb{E}_t\left[e^{-\mathrm{Tr}[\Lambda \Sigma_T]}\right]$$

Because the generator is affine, we guess an exponentially affine candidate solution:

$$\psi(\tau, \Sigma_t) = \exp\left(\mathrm{Tr}[A(\tau)\Sigma_t] + a(\tau)\right)$$

where $A(\tau)$ is a $d \times d$ symmetric deterministic matrix, and $a(\tau)$ is a scalar deterministic function, with boundary conditions $A(0) = \Lambda$ and $a(0) = 0$.

Applying the Feynman-Kac theorem:

$$\frac{\partial \psi}{\partial \tau} = \mathcal{A}_\Sigma \psi$$

Taking the derivatives of the candidate function:

$$\begin{aligned}
\frac{\partial \psi}{\partial \tau} &= \psi \cdot \left(\mathrm{Tr}[\dot{A}(\tau)\Sigma_t] + \dot{a}(\tau)\right) \\
D\psi &= \psi \cdot A(\tau) \\
D^2\psi &= \psi \cdot A^2(\tau) \quad \text{(applied carefully within the trace)}
\end{aligned}$$

Substituting these into the generator and dividing by ψ, we isolate the terms proportional to Σ_t and the constant terms. By polynomial identification, this equality must hold for all matrices Σ_t, yielding a decoupled system of ODEs:

Linearizing the Matrix Riccati ODE

To compute the Laplace transform of the Wishart process, we previously arrived at a system of ordinary differential equations. The matrix $A(\tau)$ must satisfy a Matrix Riccati ODE:

$$\begin{cases} \dot{A}(\tau) = A(\tau)M + M^\top A(\tau) - 2A(\tau)Q^\top Q A(\tau) \\ A(0) = \Lambda \end{cases}$$

Integrating this non-linear quadratic ODE numerically at every step of a calibration routine is computationally prohibitive. Instead, we linearize the system by doubling the dimension.

We propose the substitution:

$$A(\tau) = F^{-1}(\tau)G(\tau) \implies G(\tau) = F(\tau)A(\tau)$$

where $F(\tau)$ and $G(\tau)$ are square matrices of the same dimension, and F is invertible. Taking the derivative of G:

$$\dot{G} = \dot{F}A + F\dot{A}$$

Substituting the Riccati equation for $\dot{A}$:

$$\dot{G} = \dot{F}A + F(AM + M^\top A - 2AQ^\top QA)$$

We can decouple this into a linear system for F and G by grouping terms:

$$\begin{aligned}
\dot{G} &= GM \\
\dot{F} &= -FM^\top + 2FAQ^\top Q = -FM^\top + 2GQ^\top Q
\end{aligned}$$

This can be written compactly as a linear matrix ODE in a block format:

$$\frac{d}{d\tau}\begin{pmatrix} G(\tau) & F(\tau) \end{pmatrix} = \begin{pmatrix} G(\tau) & F(\tau) \end{pmatrix} \begin{pmatrix} M & 2Q^\top Q \\ 0 & -M^\top \end{pmatrix}$$

Given the initial conditions $G(0) = \Lambda$ and $F(0) = I$, the solution is given directly by the matrix exponential:

$$\begin{pmatrix} G(\tau) & F(\tau) \end{pmatrix} = \begin{pmatrix} \Lambda & I \end{pmatrix} \exp\left(\tau \begin{pmatrix} M & 2Q^\top Q \\ 0 & -M^\top \end{pmatrix}\right)$$

If we denote the block elements of the matrix exponential as $E_{ij}(\tau)$, the solution for $A(\tau)$ is recovered perfectly algebraically:

$$A(\tau) = (\Lambda E_{12}(\tau) + E_{22}(\tau))^{-1}(\Lambda E_{11}(\tau) + E_{21}(\tau))$$

Solving the Scalar Term $a(\tau)$ The scalar term $a(\tau)$ is governed by:

$$\dot{a}(\tau) = -\mathrm{Tr}[\beta Q^\top QA(\tau)]$$

Using the property of the trace operator $\mathrm{Tr}(ABC) = \mathrm{Tr}(CAB)$ and our system $\dot{F} = -FM^\top + 2GQ^\top Q$:

$$2GQ^\top Q = 2FAQ^\top Q \implies AQ^\top Q = \frac{1}{2}F^{-1}(\dot{F} + FM^\top)$$

Substituting this back into the trace:

$$\dot{a}(\tau) = -\frac{\beta}{2}\mathrm{Tr}\left[Q^\top QA\right] = -\frac{\beta}{2}\mathrm{Tr}\left[F^{-1}\dot{F} + M^\top\right]$$

Recognizing that $\mathrm{Tr}(F^{-1}\dot{F}) = \frac{d}{d\tau}\ln(\det F(\tau))$ (or analogously treating it via the matrix logarithm derivative), we integrate from 0 to τ:

$$a(\tau) = -\frac{\beta}{2}\ln(\det F(\tau)) - \frac{\beta}{2}\mathrm{Tr}(M^\top)\tau$$

This gives a completely closed-form solution for the Laplace transform of the Wishart process.

2.22.6 Feynman-Kac with State-Dependent Discounting

To price options where the discounting or running cost is a function of the state space (e.g., $e^{-\int \mathrm{Tr}(\Gamma\Sigma_s)ds}$), we apply the second version of the Feynman-Kac theorem. This modifies the PDE by introducing an absorption/discount term $-K(x)F$. In our matrix framework, $K(x) = \mathrm{Tr}(\Gamma\Sigma_t)$.

This simply adds a constant matrix Γ to the Riccati equation for $A(\tau)$:

$$\dot{A}(\tau) = A(\tau)M + M^\top A(\tau) - 2A(\tau)Q^\top QA(\tau) + \Gamma$$

The linearization procedure is identical, but the extended block matrix incorporates Γ:

$$\frac{d}{d\tau}\begin{pmatrix} G(\tau) & F(\tau) \end{pmatrix} = \begin{pmatrix} G(\tau) & F(\tau) \end{pmatrix} \begin{pmatrix} M & 2Q^\top Q \\ \Gamma & -M^\top \end{pmatrix}$$

2.23 The Multi-Factor Wishart Volatility Model

We extend stochastic volatility models to a fully multi-dimensional setting where the covariance matrix $\Sigma_t \in \mathcal{S}_d^+$ evolves as a Wishart process. The asset price dynamics under the risk-neutral measure are:

$$\frac{dS_t}{S_t} = rdt + \mathrm{Tr}\left(\sqrt{\Sigma_t} dZ_t\right)$$

where Z_t is a matrix Brownian motion that is correlated with the matrix Brownian motion W_t driving the Wishart variance process.

2.23.1 Matrix Brownian Motion Correlation

To introduce a specific correlation structure (the leverage effect/skew) while keeping Z_t a valid matrix Brownian motion, we construct Z_t using a deterministic correlation matrix R:

$$dZ_t = dW_t R^\top + dB_t \sqrt{I - RR^\top}$$

where W_t and B_t are independent $n \times d$ matrix Brownian motions.

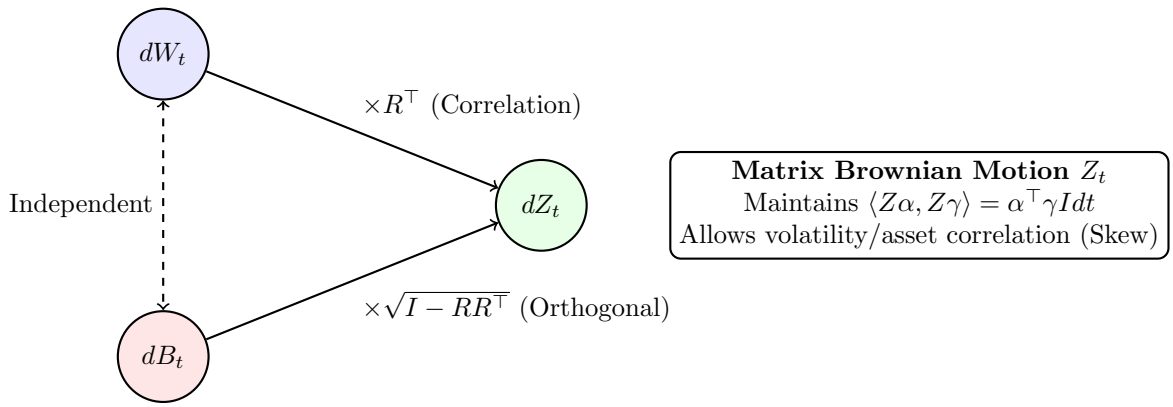


Figure 2.12: Construction of the correlated matrix Brownian motion Z_t via Cholesky-like decomposition.

To prove that Z_t is a matrix Brownian motion, we evaluate its quadratic covariation along arbitrary vectors $\alpha, \gamma \in \mathbb{R}^d$:

$$\begin{aligned}
d\langle Z\alpha, Z\gamma\rangle &= d\langle (WR^\top + B\sqrt{I - RR^\top})\alpha, (WR^\top + B\sqrt{I - RR^\top})\gamma\rangle \\
&= d\langle WR^\top\alpha, WR^\top\gamma\rangle + d\langle B\sqrt{I - RR^\top}\alpha, B\sqrt{I - RR^\top}\gamma\rangle \\
&= \alpha^\top RR^\top \gamma I dt + \alpha^\top \sqrt{I - RR^\top}\sqrt{I - RR^\top}\gamma I dt \\
&= \alpha^\top (RR^\top + I - RR^\top)\gamma I dt = \alpha^\top \gamma I dt
\end{aligned}$$

By the matrix analogue of Lévy's characterization, Z_t is confirmed to be a standard matrix Brownian motion.

2.23.2 Log-Price Dynamics and Joint Generator

For Fourier pricing, we require the dynamics of the log-price $Y_t = \ln S_t$. Applying Itô's Lemma:

$$dY_t = \left(r - \frac{1}{2}\mathrm{Tr}(\Sigma_t)\right) dt + \mathrm{Tr}\left(\sqrt{\Sigma_t} dZ_t\right)$$

To price options using the characteristic function $\hat{f}(z) = \mathbb{E}[e^{izY_T}]$, we must solve the Feynman-Kac PDE associated with the joint Markov process (Y_t, Σ_t). The joint infinitesimal generator is:

$$\begin{aligned}
\mathcal{A}_{Y,\Sigma} = &\left(r - \frac{1}{2}\mathrm{Tr}(\Sigma_t)\right)\frac{\partial}{\partial y} + \frac{1}{2}\mathrm{Tr}(\Sigma_t)\frac{\partial^2}{\partial y^2} \\
&+ \mathrm{Tr}\left[(\beta Q^\top Q + M\Sigma_t + \Sigma_t M^\top)D\right] + 2\mathrm{Tr}\left[\Sigma_t D Q^\top Q D\right] \\
&+ 2\mathrm{Tr}\left[\Sigma_t R Q D \frac{\partial}{\partial y}\right]
\end{aligned}$$

where the last term captures the cross-variation (correlation) between the asset returns and the variance matrix components. This fully specified generator maintains its affine structure, allowing the exponential affine guess to perfectly solve the multi-asset stochastic volatility pricing problem.